

Fin 340 The Politics & Legal Foundations of Finance

Numbered Transcript

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Lecture 1: Economic Development Foundations

At a glance — Calomiris introduces institutions of finance for economic development.

- 1.1 Welcome, everyone, to this course, Institutions of Finance for Economic Development. And I'm very happy to be with all of you today. As you can tell from the title of this course, it's about a three-way connection. The ultimate question, the ultimate purpose of the class, is to consider what makes economic development happen, but specifically looking at how finance is helpful for making economic development happen. And then one step back from there, finance doesn't always work well.
- 1.2 So there are institutions in societies that make finance work better, so that it can better fund economic development. And so that's the three-way linkage that we're going to be exploring, and mainly looking at what kinds of institutions we're talking about, how they matter, why they matter, and I hope that you'll travel on this journey with me. It's probably a topic that most people who aren't involved in financial sector reforms in developing countries today probably haven't thought much about.
- 1.3 But by the end of this class, I hope to convince you that this is an absolutely crucial dimension of what keeps countries from getting rich, that is, when these institutions don't exist or exist in a way that's very imperfect. So we're going to be talking about, when we're talking about finance, two different pillars. You can think of them as the two legs of the financial system. One of them has to do with capital markets and the other has to do with banks, or what we sometimes call intermediaries.
- 1.4 The literature in financial economics and economic development has shown, and we're going to summarize that literature for you in this class, has shown that both of these pillars can be extremely useful. So on the one hand, you need a banking system in your country. It collects savings from individuals and corporations, and then it disperses them in the form of loans, typically. Banks are unique. In the U.S., they're a unique source of capital, just like in developing countries, because banks are kind of local rumormongers. They keep track of what's going on in the local economy and they figure out, especially with respect to small businesses, who's worthy of credit.
- 1.5 Whereas when we talk about the securities markets, securities markets are operating on a global basis. They typically are providing credit to much larger kinds of firms or equity investment, not just credit, to much larger kinds of firms based on more firms that have a lot of public information available about them. So you can think of these two pillars as reflecting, on the one hand, smaller firms funding needs that are financed by local banks that specialize in information about the local economy. And then you can think about larger firms being funded by securities markets that are providing public information, really throughout the globe, and then getting securities market funding.
- 1.6 That's too simple a divide. Large firms also depend on the banking system. Sometimes small firms can get access to securities markets through initial public offerings after

venture capital. I'm not trying to say that there's an absolute divide, but it's helpful to think of it that way, that to have a healthy financial system you need both. You need funding opportunities through, ideally, a fully integrated international market, so that your firms that are relying on placing equity, stock, in the market could be funded by people all over the planet. And that your banks are able to build a healthy local presence based on access to local savings, which they can then use to provide for their local firms.

- 1.7 So that's kind of, in an ideal sense, that's what we hope for in a mature financial system. And if you're, if you've been living in the United States or you're familiar with the United States, you know that we do have a very large banking system and we also have a very large securities market, both equities, or stocks, and also bonds. And corporate bonds, they're debt, like bank loans, providing debt to companies. But again they're providing debt to very large companies that have a public record and tend to be very mature.
- 1.8 So, you kind of think about firms going through different life cycles and being in different circumstances. And they want to be able to rely sometimes on the banking sector when they're young firms, and then later on the equity market and on the corporate bond market. So when we're thinking about the financial system, that's sort of what we're thinking about. What are these three different kinds of funding? Two pillars I'll call securities markets, bond markets and stock markets and banks, and what are the institutions they rely on.
- 1.9 And so what we're going to see, which I think might surprise some of you, is how rare it is to have a well developed financial system. Most countries in the world have banking systems that are almost nonexistent. And that's strange because banking systems have been around for, the basic building blocks of a commercial banking system was fully developed by the middle of the 18th century, most of the ideas coming from Scotland actually. And so it's not a very difficult transaction to start up a bank, to take advantage of those ways to collect funds and then to lend funds. And yet, in most of the world, banking systems seem to be very hard to build.
- 1.10 Similarly, stock markets, stock markets, you'll see in some countries are hardly existent at all. Why would that be? You know, these are pretty essential. You can't really have a company develop without funding. And so these intermediaries and markets are absolutely essential, and yet they seem to be lacking. And what we're going to do in this course is we're going to show that this lack of finance, this, or sometimes we call it the shallowness of the financial markets or the intermediation, the banks, is traceable to public and private governance shortcomings.
- 1.11 So, just more vocabulary. When I say private governance, I'm talking about the ways that, for example, stockholders have control or don't have control over the company that they're funding. That's a private governance. Public governance has to do with the state. It could have to do with the political structure of the state. It might have to do

with the legal system that the state creates. So when I say private governance, I'm talking about the ways that corporations manage themselves, govern themselves, whether the voting rules, the participation rules for stockholders. When I talk about public governance, I'm really talking about the whole rules of the game of the society that the state determines.

- 1.12 So what we're going to do in this class is we're going to lay out the institutional underpinnings of finance in some detail. And then we're going to link those, on the one hand, link them forward to their role in promoting economic development, and then link them backward to the institutions that allow them to function. And what we'll see is, again, it's those failings of public and private governance that create institutional constraints that make finance incomplete and often very scarce.
- 1.13 How are we going to approach these questions? I'm an academic. I've been a professor for over 40 years. I believe that social science and history are factual disciplines that we approach with credible facts from reliable sources and the application of logic. What are facts? What are reliable sources? Facts for me are things that are coming from a place that is credible, meaning top-referee journal articles or top books published by academic presses. So this is a hard course, because what we're going to do is we're going to be studying cutting-edge academic work.
- 1.14 And why? Because there really isn't anything else. Newspapers, speeches by politicians, those are not reliable sources, in case you haven't noticed. And so what we're going to do is, this is an academic discussion and, of course, it's largely going to be lecture. But I will be inviting you to participate with questions, too.
- 1.15 Let's start with the most basic thing, the puzzle of persistent poverty. And also, we'll talk a little bit about not just the distant past but also about this incredible watershed decade of the 1990s and how much it transformed all of our thinking about poverty and economic policy toward development. So I think the most important question, and I wonder if you all have thought about it, when you read the news or talk to people at school, why are there so many poor people? We all are, I think, affected by poverty. Human suffering is something that's hard to witness. Human dignity is something that I think most of us feel inherently is sort of self-defining as an important topic.
- 1.16 We also, beyond that, want to see people not just rich, but also having self-determination, freedom, economic freedom, by which I mean the freedom to decide how they interact with each other in the marketplace, and also personal freedom. And poverty undermines all of that. It tends to be associated with very limited representation, very limited freedom, lots of suffering. So we'd like to get rid of it.
- 1.17 Now, from the standpoint of standard economic modeling that at least I learned when I was in college and graduate school, poverty's persistence is something of a puzzle. And the reason it's a puzzle is because it's easy to show that overcoming poverty would not be a zero-sum game. It would be something that would make everyone better off. And

you can also show that the things that it would take to overcome poverty are actually profitable. So it's profitable to invest in basic human needs. It's profitable to educate people and help them find productive employment, and it's profitable to route capital to places in the world that have low wages.

- 1.18** This is all neoclassical economic theory, because when you do that, you make your profits greater as a supplier of capital. So we would think in a globalized world that there'd be lots of people who would find it beneficial to help overcome poverty as a matter of private investment, to send capital to places where labor's cheap. And you've seen some examples of that, of course. India, for example, is a major sort of outsourcing location. You've also seen China, much more in manufacturing.
- 1.19** Notice India services, China in manufacturing. This isn't a class about sectoral allocation. But if you never thought about that, think about that for a minute. Why is it that China has been so successful as a cheap labor place to do manufacturing, whereas India has been relatively much more successful as a cheap labor place for services? And the answer, by the way, in the development literature, is that China has a lot of institutions that allow manufacturing to thrive that India doesn't have. So, if we were going to talk even about that topic, we would come back to some other kinds of institutions, not just financial institutions.
- 1.20** But you get my point. The point is, we've seen how low-wage places can attract huge amounts of capital. So why don't they attract so much capital that it overcomes poverty? The first clue of this is that, when you look at poverty in the world, you see that it's largely, not entirely, but largely a country problem. A problem—so poor people, of course, we have poor people in the United States, but they're a small fraction of the population, really deeply poor people. Poor people mainly live in poor countries. And so the problem of poverty seems to be, just from that sort of noticing that fact, that some countries seem to be incapable of developing.
- 1.21** And we know from a lot of historical literature, Simon Kuznets, Peter Lindert, David Dollar, that when you have economic development, the poor tend to be the biggest beneficiaries, or at least equal beneficiaries. And I'll show you some data about that for recent times a little bit later today. And so it's interesting to notice that poor people tend to live in poor countries because then the problem of poverty reduces to a country problem. Why are some countries seemingly unable to fix themselves economically? Let's look at a little bit of data about that.
- 1.22** So these data were created by Angus Maddison in this sort of data book that he created, *A Millennial Perspective on the Global Economy*. And he published that, the OECD published that, in 2001. Some people have complained about some of the details of these data, but we're not interested in the details. We're interested in the broad scope. And from that standpoint, the data are clearly defensible. So what I want you to notice is, if you go back to about 1000 AD, about 1000 years ago, the world looks pretty

much the same, according to Angus' data. That is, people are all pretty much at subsistence.

- 1.23** And when you get up to about 1700, you see a big divide has opened up between Europe and the rest of the world, and especially Western Europe. So in Western Europe, the United Kingdom, what's today the United Kingdom and the rest of Western Europe, you can see that, by 1700, they're at maybe three times the income per capita of the rest of the world. That period, what you're seeing there in what's sometimes called early modern Europe, you're seeing a bunch of changes that are happening from about 1600 till about 1800. And sometimes we call that era the era of mercantilism, imperialism. It's also an era that saw a lot of really important innovations in transportation, navigation, building of empires and merchant networks.
- 1.24** And sometimes people refer to the increases in per capita income that are happening there as part of what's sometimes called a Smithian revolution, which refers to Adam Smith— a Smithian revolution, meaning the building of markets, the integration of the global economy, allowing everyone to benefit, if you're not a slave or on the wrong end of invasion in the empire-building, allowing all the winners at least to benefit from integrated global markets for commodities, allowing specialization. And so these are the benefits of trade that Adam Smith talked about. Increased specialization, being able to do things that we can do best and let others do what they can do best.
- 1.25** And then after that, if you look at this, you'll see that was just the beginning. Because after 1820, we have something called the Industrial Revolution. And there are actually several industrial revolutions. People tend to refer to the first one in the mid-19th century, the second one at the end of the 19th and beginning of the 20th century. And the third one may be the information revolution that we're all still living through right now. The point of this is, this period around 1850, sometimes it's referred to as the hockey stick. And the reason it's referred to as the hockey stick is that most of global history, from a per capita income standpoint, kind of looks very flat. And then all of a sudden it takes off, and there's no stopping it.
- 1.26** So, compared to where we were in 1700, if you're looking at Western Europe or its offshoots, which is the U.S., Canada, Australia, and New Zealand, you can see that per capita income at more than—this graph stops in 1998, let's say roughly 20,000, \$2,000 per person on average. Well, that's a lot more than it was in 1820. In fact, it's about 10 times what it was. And what's interesting is, now we all assume this. We assume that this just happens perpetually. There's every reason to believe that. Growth isn't stopping. Growth is going to continue. Technological change and innovation and the spread of great ideas to make things more productive, there's every reason to believe that's permanent.
- 1.27** I do want you to realize that, as of 1850, almost no one believed that that could be possibly true. The idea that per capita income would grow, possibly grow forever, was not an idea that any of the classical economists had. Adam Smith would not have had

that idea at all. He and all of the economists around the end of the 18th and beginning of the 19th century, including, by the way, Karl Marx, sort of believed that there was a limit to how much economic wealth we could create per capita.

- 1.28** So the other thing you notice from this is not just this hockey stick phenomenon in Western Europe and places like Western Europe, the U.S., Canada, Australia, New Zealand. Do you also notice that other regions aren't sharing in that? And so this, when I was talking about how poverty is country-specific, you can see economic development, again, is a very country-specific thing. On the other hand, we see, occasionally, some countries kind of jumping out. So, for example, notice that Japan in 1870 looks pretty much very similar to the rest of underdeveloped Asia and Latin America and Africa. And yet by 1973, Japan is looking more like the United States. It's almost identical in terms of the per capita income.
- 1.29** What happened? By the way, the answer is something called the Meiji Restoration that happens in Japan in the last quarter of the 19th century. It wasn't a restoration at all. Whenever governments use words like restoration, you know they're not really telling you the truth, right? It was actually a revolution of making Japan like Western Europe. Conscious building of Western European institutions, people even dressed in Western European costume. If you go to Tokyo today, you'll see the Central Bank of Tokyo looks like the Central Bank of Belgium, as of the 1870s. Why? Because it is a copy of the Central Bank of Belgium.
- 1.30** So that's how some people took on this question of, well, how do we create economic development? And in the case of Japan and in the case of Germany, it was very much a conscious effort of the last quarter of the 19th century and beginning of the 20th century of governments to promote development by copying what worked in other countries that had already done it. And they did so very successfully.
- 1.31** What was driving them, by the way, was military conquest. So just like in the era of the early modern period, 1600-1800, the thing that was really driving the institution-building that made finance possible was the competition among the states for building empires that made them want to have effective financial systems. Because the main function of the financial systems was to fund commerce so that you could build your network of merchants to take over the world and fund the sovereign so that the sovereign could fight wars. So you needed to fund your navy and your army and you needed to have a funding channel that would work, make commerce work.
- 1.32** And so, in the 16th to 18th century, the incentives were very aligned to build, if you were, especially the Dutch, the British and the French, to try to build these institutions. Some were more successful than others. The British and the Dutch were the most successful at figuring out these institutions that helped to really make their empire strong in this competition. And similarly, when we look at Germany and Japan in the late 19th and early 20th centuries, they're building financial institutions that's underlying this record that we're seeing.

- 1.33 And they're doing it why? Because they want to fight. You know, Bismarck, what was Bismarck interested in? Well, he told us. Iron and blood. If you don't know that speech, go, go read it. It's a pretty good one. Germany wasn't subtle, neither was Japan. They wanted to make themselves important in the globe in terms of conquest, as well as economy. But they weren't interested in building per capita income for common people. They did it, and their policies did focus on it. But why? Because they saw that as the means to national importance, militarily.
- 1.34 So I just want to emphasize, when we think about the most successful countries of the distant past and why they built the institutions of finance that they did and figured out all these things we're going to be talking about in this class, they did it for reasons where the sovereign's interests were very much aligned with building these institutions. Why? For wealth creation in the form of the sovereign's expansion of territory and which also was allied with merchants' expansions of networks of trade. It wasn't about making you better off, but that was an unwitting consequence. We're all sitting here, wealthy, in our nice hockey stick lives, because of that and more, as we'll talk about.
- 1.35 I want to talk a little bit about the 1990s. So this is some research that David Dollar did when he was at the World Bank. And basically, what he did was using economic statistics to look at global inequality and to talk about how much of the inequality in per capita income put us all, all the people in the world, put us into one big bowl, and notice how unequal we are in terms of our income. And then ask, how much of the variation in that bowl of all of us thrown together is explicable in terms of what country you're located in, as opposed to within-country variation, right?
- 1.36 So there, you can think that there are two kinds of variation. We sometimes call this in economics a variance decomposition or an analysis of variance. And so what we can do is, we can say, well, first we see how much of your inequality is explicable by virtue of what country you're in. And then holding that constant, how much more is explicable in terms of variation within the countries? And so what he shows us is, most of the variation goes back to the 60s and through to the end of the millennium. Most of the variation has to do with what country you're in. That's what I mean when I say poverty is largely a country problem.
- 1.37 And what you see happening in the 1990s is, you see that green area, which is the country contribution. You see it shrinks a lot. And you see that the blue area is growing a little bit more. And the real drivers of this, in terms of the graph, what really makes the graph happen, are two countries that I've already mentioned, China and India, because they are extremely populous places. And when China and India decide to put aside state control of the economies and to liberalize, the effects on poverty reduction and overall economic growth are astounding.
- 1.38 And that is really what makes inequality across countries shrink. What makes it grow? Well, same thing. China actually becomes a place of increased inequality within

country, as does India. Why? Because some places in China are getting rich—the coastal areas, primarily—because they're the ones that are engaged in and integrated with the global economy. So, the progress that you can see, even in this graph, comes from something that we sometimes call emergence.

1.39 And I want to talk about that a little bit, because during your lifetimes, assuming that most of you were born after or around the 1980s or 1990s, maybe I'm wrong there, but during your lifetime, some kind of amazing things have happened. And the way that economists think about the question of poverty or economic development really changed a lot, too.

1.40 So let's just first ask the question, where does this national poverty come from? You might think it's a function of natural resource endowment. The old philosophy that believed that was called the physiocrats, the philosophers. And there's no known physiocrats in the world today. And the reason is because there's no evidence that natural resource abundance has anything to do with poverty.

1.41 I could give you a lot of examples, but from the very beginning, the conquest and enrichment of Rome, relative to other places, Rome was a completely resource-bereft place. It just conquered all the places that had good resources, starting with the Etruscans, that were like the Saudi Arabians of the time, because they had, in the Bronze Age, they had all the copper and the tin almost. And then the Iron Age, they had all the iron.

1.42 So you might say, well, why didn't the Etruscans conquer and take over the world? Because natural resources weren't the deciding thing. The deciding thing was, I would say, the ability to create effective political and economic institutions of governance. And that's what Rome really had going for it. I could say the same thing, when we look in the Middle Ages and at Western Europe, we see that cities are the places of bright light in terms of economic development.

1.43 And there's an old German saying that "city air breathes free." And the idea being that it's the institutions of the city, and particularly the fact that you weren't subject to all of these feudal constraints that you were subject to, if you were working in the countryside as a serf. So it was something about the institutional structure of cities that allowed them to attract people and give people a chance to operate as relatively free economic agents.

1.44 We could talk about the Dutch and the British, two places which, in contrast to the Spanish and the Portuguese, who of course had claim, early claim, to very resource-abundant areas, but it was the Dutch and the British that won out. What was it that allowed these small countries to win out? And again, we'll see that it's something about their governance structures and things that they did successfully.

1.45 The United States, you might say, well, here we are. We all know the U.S. is in North America. North America is extremely resource-abundant, except it isn't. North America

is one of the least resource-abundant continents in the world. Japan, well, if you've ever been to Japan, you know it's not known for its resource abundance. And yet, as we've already pointed out, it had a growth miracle that took place, 1870 to 1930. I could go on and on. Hong Kong, Singapore, and more recently China, all are cases where there's been big spurts of economic development that seem unrelated to natural resources.

- 1.46** Well, if it's not about the inputs, what is it about? And the answer is, we can see that per capita income is very closely related to economic efficiency in the use of inputs. In other words, what makes countries rich is not that they've got physical inputs, but that they somehow have a way to make those physical imports more productive. And we call that in economics total factor productivity, which means, whatever you take as your inputs—land, labor, and physical capital—how much output do you get? And if you have lots of high total factor productivity, that means you're getting more output for that given vector of inputs.
- 1.47** I included this quotation. I'm going to show you some data, but I like the quotation from David Landes' book, which is called *The Wealth and Poverty of Nations*. And in that book, David quotes somebody from the country Mali, talking about the failure of attempts to make economic development happen by just importing inputs. So, post-World War II, there was a complete lack of appreciation of the institutions of economic development. And people were literally just building factories in Africa and thinking, well, what don't they have in Africa? Well, they don't have our factories. Let's build some, and then they'll be rich, too.
- 1.48** And so this observed, talking about how the factories don't work and how they were a waste of money. "We built them because you said so and the way you told us, but now they don't work. You tell us we must pay for them because the World Bank lent the money, and now they want the money back, right? That's not fair. You told us to build them, you should pay for them. We didn't want them." So the point is, it wasn't the absence of the inputs, but it was something about how to make the inputs productive that was absent in Mali. And it was bad thinking.
- 1.49** I'll recommend a great book to you by William Easterly, *The Tyranny of Experts*, which was all about how bad thinking of developed countries, very mechanical thinking, wasted a huge amount of resources during this period because they weren't really thinking about fundamental institutions of economic development. This gives you a little bit of a sense of this connection between efficiency and total factor productivity. Total factor productivity is measured in the rightmost two columns and GDP per capita is measured in the leftmost two columns.
- 1.50** And this is from a study that was done a few years ago by Clark and Feenstra. And if you look, you'll see, both in 1910 and in 1990, that there's a clear association, not just at a moment in time, but throughout really human history, between if a country is ranked high in terms of GNP per capita, it also is ranked high in terms of total factor

productivity. So when we think about, physically, what is the thing that delivers total factor, economic development, per capita income, it's key that total factor productivity is essential.

- 1.51** Now that doesn't really tell you much about the mechanism. It just tells you that it was not a very smart idea to import inputs and expect that that would make countries rich. So you might say, oh, I know what drives total factor productivity. It's technology. It's knowing, having the recipe of knowing how to combine inputs. It's really just a kind of engineering thing. Well, it turns out, it's pretty clear that's not the case. And rather than show you lots of current evidence, let me just remind you that China, until very recently, was operating as a country for millennia, which was always at the subsistence level of per capita income.
- 1.52** But China had all of the technologies of the Western European industrial revolution, centuries before Western Europe had them. China had printing, iron-making, paper production, water power production, gunpowder, navigation, shipbuilding, and even water-powered textile spinning, which was sort of the quintessential early industrial revolution technology. It had that hundreds, all of those, hundreds of years before the West.
- 1.53** So if China had the technology, why didn't they have total factor productivity? And the answer that many historians have given, and you can read about this in a book by Eric Jones from the 1970s, called *The European Miracle*, is that China lacked institutions that promoted productivity. And here we're thinking about legal institutions having to do with property rights, rule of law. And you could say, well, what kinds of problems? Well, you're prohibited to transfer or inherit land. And so, your incentives to improve land, which is a very important part of productivity growth in early economies, are kind of nonexistent.
- 1.54** You're not allowed to engage in international trade. By the way, I think it was in the Ming dynasty, they had merchant enclaves with very advanced shipbuilding, and they started to experiment with international trade. The Chinese government decided to burn the ships and destroy all of the progress of the technology in how to do shipbuilding. Why? It wasn't a stupid thing. The emperor figured creating a power center of merchants on the coast of China who would be involved in international trade might not be good for the emperor.
- 1.55** Why? Because it could be that those kinds of enclaves could create revolutions. They create locations of people who are now linked to the rest of the world, are learning from the rest of the world, are making themselves wealthy, and before you know it, they're gonna come after you. And so, the point that I'm making is really a very central point in this course about all the institutions we're going to talk about, which is, there usually is a political or political economy story about why economic institutions or legal ones are not being developed. It's in someone's political interest not to have them.

- 1.56** I'll tell you a funny story. When I was a young professor and I did a lot of work in Latin America, I was working in Mexico right after the Mexican crisis of 1994-95. And I was down there consulting with the Mexican regulatory agencies, the Mexican government and World Bank, on how you build financial institutions, because Mexico didn't have very good ones. And I said, "Well, you know, the main thing you need is, you need to have—and we'll talk about this later in the class— a registry for your capital goods so that creditors know that the collateral is registered." We'll talk about that. It's little simple things like this that make finance work or not work.
- 1.57** And so, they're all sitting listening to me talk about the way we do it in the United States and with smiles on their faces. And I said, you know, they said, "Well, professor, you think we don't know about this?" And I said, "What do you mean? You know about this and you don't have it?" They said, "Yeah, we don't have it." I said, "Why?" They said, "The big banks don't want it." And I said, "That's impossible." I said, "The main beneficiaries of this set of institutions I'm talking about would be the big banks." He said, "They don't think they need it and they don't want it."
- 1.58** So that was the most puzzling thing. And then I realized what's going on. If you're running banks that are lending to themselves, that is, the bankers are, let's say, the six families of Mexico and their friends, and they're the ones getting all the credit— so if you're lending money to yourself, you don't need the law to enforce paying back, right? And by having no law, it means that you don't have competition, because nobody else can borrow because there's no law that makes them pay back. So not having a financial system can be a political decision that serves the existing locus of power in the economy.
- 1.59** Because having arm's length credit, as you'll see in just a minute—and Adam Smith and John Stuart Mill and others talked about this—once you have markets and you have arm's length credit available, all of a sudden you can contest power. People with good ideas who are willing to work hard can disrupt the power of the preexisting group. And so, if you're in the six families, you don't want to have a legal system that makes arm's length credit work better, because you don't like arm's length credit. You like insider credit because that's what you depend on. And so, this story that we're telling is one not just about China and not just about Mexico, but really about almost all of these things.
- 1.60** Now, another thing people have pointed to in trying to answer cross-country poverty questions is human capital. So is it really just about human capital? That is, poor countries are poor because people there aren't educated. OK, so first let's just look at some data. The first thing to notice if you look at, so this is from a famous paper by my friend Robert Barro, written in 2010.
- 1.61** And so, what this shows is, if you look at average years of schooling, as of 1950, and then you look at per capita income in 2006, because you have to look at two different dates because the schooling doesn't immediately cause the income; it causes it with a

lag, right? You see a very high positive relationship. So, voilà, eureka, eureka, I found it. This is a very important variable for understanding poverty.

- 1.62** But now we get into another sort of important concept I want you to think about. And that is, what is causation? What is causality? So, yes, it's true that rich countries have well-educated people, but when we look at that historically, what we find is that that gets back to institutions, too. So the decision to educate poor people is a political decision. Many societies choose not to educate people, just as those societies also choose not to build institutions of finance.
- 1.63** Well, why not? It's a positive. Again, it's not a zero-sum game. If you educate people, it makes the whole economy more productive. We all benefit from each other's education in terms of lifting productivity and making the marketplace work better for all of us. Well, it costs something to educate people. So Engerman and Sokoloff, in a classic paper in 2002, they compared and contrasted the colonial systems in Latin America and North America. And they noticed that one of the key differences from the very beginning in U.S. colonization was investment in education.
- 1.64** So that every one of the North American colonies had, as part of its foundational structure, investing in public school education for the whole population, whereas in Latin America, it was not true. There was no interest of the ruling class in investing in public education. OK, so let's think about why that might be. Of course, education is costly, so you have to tax people to provide for education. And then once you have education, that allows people to have opportunities to also be able to, by the way, challenge authority because they're more educated, right?
- 1.65** So, in the U.S., how were we getting people to come to the U.S.? In the North, prior to slavery becoming important in the South— so really, for the first hundred years or so, the primary labor that North America is bringing in is indentured servitude, which is voluntary contractual labor, which is immigration coming from Western Europe. What are those people coming here for? To work for four, five, six years under indenture, then be liberated from their work contract, given a grant of land, and being able to go out and start their own lives as entrepreneurs, farmers, really.
- 1.66** Education was an important part of attracting people. Also, all of the colonies were democratic. They all had bicameral legislatures. And so actually, the good institutions of the North reflected in part the absence of gold and silver and spices, and the big one, sugar. So this is just a bereft place. The British wanted it to be a place where they could bring people so that they could lay claim to it and build an empire location here. How were they going to do it? They had to attract people who wanted to live there because it was going to be a democracy. They were going to be free. They were going to be educated. They were going to have private property.
- 1.67** And so all those good institutions that I just listed were in the interest of the empire to create. How about Latin America? Well, in— let's take the coast of Brazil. They had

sugar. That was the big one, right? They had sugar. And sugar is best from the standpoint of profitability. It's very hard to get people to want to labor in Brazil in sugar. So you solve that problem with slavery. And the people who are setting up the colonies are basically a military encampment that is bringing slave labor.

- 1.68** Now, when you, and, and it's not giving people rights to themselves, much less other property rights. It's a very narrow, autocratic military government. So are they going to educate the slaves? Why would they? It doesn't make any sense. And so, even after slavery is abolished, for example, in Brazil in the 1880s, you still have a kind of continuation of that same class divide in Brazil. And there's just, if you're part of the minority that's controlling the country, creating opportunities and education for the masses is not high on your list. It's not going to help you.
- 1.69** So what I'm trying to get you to see is, we keep coming back to political choices about institutions, whether they're institutions of education or institutions of property rights, in the Chinese case, or in the cases more pertinent to what we're going to be talking about, in Mexico, institutions having to do with building a good financial system and banking system. There are conscious choices not to have what we would regard as good institutions. And that actually leads us to another kind of recognition, which is, what's good is in the eyes of the beholder.
- 1.70** So if you're running an autocratic slave labor system, good institutions, from your standpoint, it's not the same as what we're thinking, right? So you have to understand, there's a certain political economy logic to institutions and economic development. So we shouldn't just look at countries that have bad institutions and think that they're just dumb. The people in charge are not dumb. They're making conscious choices.
- 1.71** That also is a little depressing, because what it tells you is, it's not a matter of educating them. And it may even be that it's incompatible with their existent political structure to try to even import good institutions. That's a little depressing. But you did, I hope you didn't come here for feel-good. We're here to learn about where these institutions come from or don't.
- 1.72** OK, finally. Well, is poverty maybe about participation in market transactions with economic freedom based on having property rights, living in a system under rule of law, which means we all are living under similar rules? It's not about who I am that gets to decide. The law doesn't decide based on who I am. It decides based on what I do. And markets and then intermediaries that are both markets and intermediaries able to fund investments. So, yeah, it is about that, but that this is not a mechanical choice. This is a political choice.
- 1.73** When you look back at what people were writing, the great philosophers of the 18th century, of liberalism, liberalism means, what does it mean to be liberal? It means that you're not a slave. That's what—you're free. What does free mean? You're not a slave. You can speak as you like, you can worship as you like, but you can also do as you like in

the market. Don't take that for granted. In most places in the world, that was not the case and it still isn't.

- 1.74** So, and when Adam Smith and David Hume and John Stuart Mill were making the arguments for liberalism, for creating freedom of markets for people to interact in, they were very conscious that this was a political choice that would have, not just a political choice, but would have political consequences. One of my favorite quotes from Adam Smith is, he said that eliminating the servile dependency of individuals on their superiors was, quote, "the least observed advantage of commerce and by far the most important of all its effects."
- 1.75** So notice, the most famous economist in all of history also understood that man does not live by bread alone, that the point of economic progress and freedom to act in the market was to give you control over your life, to allow you to compete. We're going to talk about free trade in the current context, but there's an economics history study of free trade going back to the 19th and early 20th centuries. And what you find is that deciding to allow people to engage in free trade, that is lowering barriers on competition from other countries.
- 1.76** So this is from this study by Lopez-Cordova and Meissner, quote, "A one standard deviation increase in trade," so allowing countries to trade among themselves, "tends to result in countries like Indonesia, Russia, or Venezuela," which are among the least free places on the planet, "to be as democratic as the U.S." Democratic is referring to something political scientists use, called a polity score, which measures on a continuum how autocratic you are, how much there's self-governance and representation.
- 1.77** So, the point of this is, it turns out that, in the political science literature, there's actually a lot to confirm what Adam Smith thought, which is, if you allow trade to happen, economic freedom, to characterize your economy, there will be political consequences. Now, of course, that is exactly why people don't want free trade. Put this together, right? This shows you, when you think autocrats are stupid, you're wrong. They think like this. They understand this, and they've often lived through it.
- 1.78** OK, what do we mean by institutions? Well, we're going to use Douglass North's definition. So, Douglass North got the Nobel Prize in economics for his work in defining institutions and showing that they mattered for economic development. And he wrote in his book in 1990, defining institutions as "the humanly devised constraints that shape human interaction." That's a very broad definition. I love it. And the thing I love most about it is that he uses and understands the word constraint to also mean "possibility creator."
- 1.79** So you might think that constraint has a negative connotation. Igor Stravinsky said, "The more constrained I am, the more creative I am." And if you know his music, you know he's one of the most creative minds in all of the history of music. So what's interesting about institutions is they are constraints. They define whose property this

is. They constrain you from coming up and taking it. That's the sense in which they're constraints. But every constraint tells you what's an illegal action that you'll go to prison for. Tells you how you may vote or not. Tells you how you can operate your business or not. Tells you if you can be a bank, licensed or not.

1.80 These are all things that constrain you, but of course, they also make it possible for you to be acting in the capacity of an empowered institution or individual. So corporations, banks, institutional investors like pension funds or mutual funds, public equity contracts, when you go and buy stock in a company, you're using that contract. Debt contracts, when you borrow money, you're entering into a contract, and all of the legal, political, and economic entities that create those and oversee them, and including also, not just the laws, but also the norms that shape how we all interact. Those are all the institutions that we're talking about.

1.81 OK, is the importance of institutions for development controversial? Yes and no. So first of all, what's the sense in which they're not controversial? If you look at the question, are institutions obviously a key part of any economic development strategy? The answer, uncontroversially, is yes. By the way, there's a great article to read by Daron Acemoglu and James Robinson, which was published in the Journal of Economic Perspectives, which is all about the decline of general laws for explaining economic development.

1.82 And what does he mean by general laws? Well, he started, they mean the kind of thinking that Karl Marx had, or Thomas Piketty had more recently, which believes that you can think about development or inequality or growth as having some kind of mechanical, inevitable trajectory. So for Marx, it was class struggle that was defining human history. For Piketty, it was whether the interest rate is lower than the growth rate or higher than the growth rate. These were kind of mechanical ideas. They're clearly, demonstrably wrong. And that's what the point of the Acemoglu and Robinson article is.

1.83 But there's not just a negative point to their article. The other point is to show that, if you really want to understand where development comes from, you have to look at institutional history. But I want to emphasize. So why do I say it's somewhat controversial? Because then the question becomes, OK, we understand that institutions are chosen. Sometimes they're chosen to be good, sometimes chosen to not work so well from the standpoint of economic development. But where does that political choice come from? And that's where it gets controversial.

1.84 So I'll give you two examples. So, my close friend Deirdre McCloskey has written a wonderful thing called the Bourgeois Trilogy, which is three long books about the history of economic development, in which she's arguing that it's the culture of freedom in the Netherlands circa 1600, and in Britain circa 1700, that culture of freedom that really makes the difference for this whole hockey stick thing and for building the institutions. So, she regards the institutions, that it would be wrong to say

institutions—she would kill me if I said, "Institutions cause development." She'd say, "No, no, freedom caused institutions and development." Right?

- 1.85** And there's some basis of, I think she's right, actually. And then Steve Haber has a series of papers and now a book that he's working on that's about geography and geology really being the more important. And he's asking a different question. He's asking, why are some political systems, call them autocratic systems, why did they develop with the combination of institutions that they had, that makes them resistant to these imports of favorable or good institutions for development?
- 1.86** And his argument is, it has to do with where you are located, has to do with rainfall. So it's a lot of geography and geology. So, for example, let me give you a taste of that. What is the storable wealth at the beginning of humanity development? It's grain. That's what storable wealth is. Or livestock, or oil. So these are things that you can store. Olive oil can be stored, but grain can be stored for hundreds or maybe thousands of years. And so granaries were absolutely crucial for the creation of anything like wealth.
- 1.87** Once you have some wealth created, all of a sudden you need laws about protecting it, about incentivizing the creation of it. It does matter that the locations where grain can be produced depend on rainfall. If you have too much rainfall, you can't produce grain. If you have too little, you can't. So, for example, if you were in the Democratic Republic of Congo, you'd say, well, it's pretty much bananas. How long can you store a banana? It's like, what is the sound of one hand clapping, right? You can't store a banana. How do you store bananas? You get people to eat them and then you enslave them. That's how you create wealth out of bananas.
- 1.88** So that gives you a little bit of a sense of when people are thinking about natural, climate kinds of influences, they're thinking about, wow, you know, Western Europe and Eastern Europe, by the way, and certain parts of the world, it kind of mattered that they were creating grains, whereas other parts of the world either were too dry, the Stans, or too wet. And that tends to lead to— and so, this book, which is going to come out, and I know, I wish it were out already, I would be talking about it more, has a lot to say for it, because what it sort of does is, it shows you how the combination of natural circumstances can actually favor autocracy.
- 1.89** That under some—I'm talking about pre-1800, where agriculture is pretty much the whole economy—that under pre-1800 economies, autocracy was kind of a necessary institutional complement to bananas. So anyway, you get the idea. So there's a literature out there that's talking about where institutions come from, with kind of very deep causality of thinking about nature or culture. We're going to be much more modest. We're going to use the fact that those kinds of stories don't determine everything.

- 1.90** And then we're going to look at, on the margin, when countries make choices that are within their choice set, we're going to look at how that matters. So, for example, when a country decides at a certain point in time to create a certain institution on the margin, does that make a difference? So, even though I recognize that it is true that, generally, the bundle of institutions and their quality is determined by something beyond the institutions in the political sphere, we're going to take that for granted and we're just going to look at more local choices that aren't already predetermined by those deep causal parameters.
- 1.91** And that's going to be the content of our course. We're going to look at things like what happens when Brazil decides to create something called the Novo Mercado, which is a revolution in private governance of publicly traded stock companies. Or when Korea decides to create a government version of that. In Brazil, that's done privately. In Korea, that's done with something like strengthening minority shareholders' rights as a matter of law. Whereas in Brazil they're doing that not as a matter of law, but as a matter of a private company's decision to belong to a club. Very interesting.
- 1.92** Or when countries decide to improve the registration. I already mentioned the registration of collateral interests in Mexico that they weren't interested in improving. Well, some countries are, and I think my friends Steve Haber and Deirdre McCloskey would agree with me that those decisions weren't determined by the culture and they weren't determined by the rainfall pre-1800. And so, we're not trying to explain everything. We're trying to explain the things that, on the margin, we see countries deciding to do and, of course, will expand from that list to more general understanding, as we've already done, of the role of institutions for development.
- 1.93** But what's great is, because we're looking at this very focused question, which is financial institutions, we can make a lot of progress. If we were trying to understand where does democracy come from, that would be very hard. I'm not sure we could do it, not with the tools we're going to use here. Where does freedom come from? I'll leave that to Deirdre and to Steve.
- 1.94** What kind of evidence are we going to be looking at? We're going to be looking at lots of evidence, econometric studies. So, yes, you're going to be seeing a lot of regressions. I promise you, this is not a math course. I will help you understand what the regressions mean. So we're going to use econometric evidence. We're also going to use narratives, as we have been doing so already—narrative evidence telling you stories about what countries did, why it mattered.
- 1.95** One of the readings on the course list is by Peter Rousseau and Richard Sylla, and it's discussing the narrative about where the building of financial institutions happened in that early period, 1600 to 1800, as I've already described. But even in their article, once they've spelled that out, they turn to econometric evidence to show how the building of a system of banking, successful building of a system of banking, in what are now today's developed countries, how that successful building of banking systems in the

19th century created immediate consequences for economic development in Western Europe.

- 1.96** So it's not true now, by the way, that once you've built a banking system, you won't see the same kind of consequences of building, of a little bit more credit growth, because you've created the institutions. So, what they're looking at is at the building period, where you went from having no banking system to having a banking system and showing how much that mattered for the economy's ability to grow.
- 1.97** I want to just make sure that we have a clear— I think I've been clear about it, but I want to be very clear. When we talk about the consequences of building institutions, we're not believing as policymakers that, therefore, we could solve a country's problems by importing that institution to the country. I want to make that clear, because as I've already pointed out, countries have their own political logic which is resistant to importations of factories, which aren't very useful because they don't have the total factor productivity, or of other ideas about institutions.
- 1.98** And that makes our study here a little bit depressing, because what we're really saying is, we can tell you what works. We can tell you also that the ability to get it is very dependent on the political environment. And so, therefore, we can't tell you that we can make you rich. But I only tell that to my clients after they hire me. No, that's not true. I tell them upfront, but back in the 1990s, I didn't know that. And so I went around thinking that, as a bright young professor, that I could make everyone rich. And that was not true. I couldn't.
- 1.99** Now, when we talk about the 1990s, I've already mentioned this, we get to talk about something happy, which is the period of what's called emergence. And if you didn't live through this, as I did, you wouldn't even know that this is a thing. But I'm going to tell you, it's a big thing. So when I was in college, I went to Yale. It was considered pretty much, I won't say it was considered a fact, but it was pretty strongly believed that a strategy like emergence wouldn't really make a difference. I'll talk about what that is in a minute.
- 1.100** And part of it was that people believed in something called dependency theory. So, dependency theory was a theory. It was initially invented by Raúl Prebisch in Brazil. But the basic idea was, the system is rigged. The whole global system is rigged. Poor countries, if they just tie themselves to the little engine called global trade, it's not going to end well because the system's rigged. The wealthy countries are in charge. That will just give them more ways to exploit you. You can't, by hooking your trailer to the engine of global growth and global trade and global capital flows, by integrating more into the global economy, that won't make you rich.
- 1.101** And then it turned out that when this was actually tried— that's why you don't hear people talking much about dependency theory anymore. Because after the 1990s, dependency theory disappeared because it turned out, it was demonstrably false. The

way that that came about was that this emergence phenomenon started off in a few countries, notably Chile and Latin America and South Korea, that experimented with emergence. And now, of course, they're the richest countries in their regional spheres. And when their success became apparent by roughly 1990, then other developing countries said, "Well, maybe we should try emergence."

- 1.102** So, what is this emergence that emerging market countries are doing? What are they emerging from, first of all? And the answer is, they're emerging from state control over the economy. And that state control over the economy is manifested in four dimensions. So it's state control over the actual governance of industry. So, emergence has—part of emergence is privatization of state-controlled enterprises. So, you take your banks or your oil company or your transportation company or your steel company that are owned by the government, and you privatize them.
- 1.103** Second thing you do is liberalize the banking system. So again, this is something that is no longer as fashionable as it used to be. But when I was an undergraduate, governments pretty much decided who banks lent to through most of the world. By the way, that was even true in Great Britain. So which sectors you lent to and often which firms you lent to was part of government planning, believe it or not.
- 1.104** And so the banking system was very much controlled by, and it, it gave rise to an expression that my professor in graduate school, Ron McKinnon, called financial repression, which is basically a banking system whose function is to lend money to the state on favorable terms or to lend money to the private sector that the state identifies for you as the recipient of your money.
- 1.105** So, liberalization of banking means stopping that, allowing interest rates to be freely determined in the market, allowing a lending market to exist where the banks look for the best customers and lend accordingly, and where the government doesn't keep poaching the savings in the banking system. What's the best way to do that? Something called reserve requirements. So, I'm the government, I tell the banks—and by the way, in Britain even, the banks had, I think it was more than 30% of their deposits had to be in either government bonds or reserves held at the central bank.
- 1.106** And if the reserves are held at the central bank, they bear zero interest. So the government says, "Oh, by the way, as part of the privilege of being a banker, please give me an interest-free loan equal to 20% of your deposits." So you get the idea. Liberalization means, let the banks have the money they raise to lend out, let them decide who to lend it to. Liberalization of trade. Allowing your country to participate in international trade through reductions in tariffs and other quotas.
- 1.107** Now reducing the limits on trade has many benefits. We already talked about Adam Smith's benefit of creating competition, so that it reduces the servile dependency of laborers in the country. Secondly, by promoting comparative advantage, it boosts income. Your country produces what it's best at producing in the global economy. But

there's a third advantage, which is, if you don't do global trade, then how could you also have the fourth liberalization, which is the liberalization of capital flows?

- 1.108** So if I'm going to get money from you, suppose I'm in Chile and I want you Americans to send money my way, how are you going to expect to be paid back? In dollars, right? You're going to send me dollars. That's what you've got, and you're going to expect dollars back. Well, great. How do I pay you back? I have to actually be able to earn dollars. How do I earn dollars? Trade.
- 1.109** So, liberalization of trade not only created direct economic benefits, but it also created a big indirect benefit of allowing your country to also get funding from other countries, not just through the World Bank deciding to give it to you, but through private sources. So these are the four liberalizations that are happening, that are part of the phenomenon called emerging markets.
- 1.110** So these graphs are just as convincing as any regressions I could show you. And David Dollar in 2002 was kind of pointing this out. So most of this choice after—so Chile and Korea are doing this emergence in the 1970s. Other countries are copying them in the '80s and the early '90s. That's sort of the heyday of emerging markets, is the '80s and early '90s. And so he's saying, well, what happens to countries that choose to go in the direction of free trade? What happens to poor countries that choose to do that?
- 1.111** And he finds that, well, if you choose to go in the direction of free trade, look how much, by the 1990s, your GDP is growing. It went from under 2% in the 1970s to 2.5 in the 1980s and to 5.1 in the 1990s. What if you were in the group of countries, developing countries, that didn't choose to open up to free trade? And so, notice that your experience in the 1970s was better. You were growing at 2.6% instead of 1.8%. So don't think it's just that you were already rich and that's why you did better in the 1980s and 1990s.
- 1.112** And actually, these countries, which had been doing better, marginally better, immediately were doing much, much worse because the world becomes much more competitive. And so what's going on is, countries are learning very quickly by the end of the 1990s that the strategy of emergence is very successful. Again, the initial example of countries like South Korea and Chile in the 1970s gets copied massively.
- 1.113** And you might also wonder, well, OK, fine, it causes very much an acceleration in growth, but is that good for the poor? And so this, from Dollar's same paper, he looks at the relationship between the poorest quintile, that's the lowest fifth of the population, and the average growth rate. And what he finds is that the growth of incomes of the poorest quintile in emerging market countries is growing at the same rate. If you look at the red line, you can see that line has a slope of about 1.
- 1.114** So what that means is that the average growth rate of the country, which is on the horizontal line, and the growth rate of the lowest quintile, which is the vertical line, are tracking each other on average, one for one. So if you ask, does growth tend, average

growth tend to be also favorable for the poor, the answer is yes. But of course, we knew that also from that first graph I showed you, also from his work about how much global inequality had changed, because, by the 1990s, two of these countries are massive in terms of having hundreds of millions of people are coming out of poverty. And that's a pretty amazing thing.

- 1.115** And there's also a lesson here. Remember in China, when hundreds of millions of people came out of poverty, that also caused inequality to go up. Was that a good thing or a bad thing? I would say clearly a good thing because the interior area didn't change. It just had people still at sort of subsistence level. And the people on the coast all got better off, inequality got worse, poverty got substantially reduced. So, be very careful when you hear people talk about inequality. Look especially at not just average growth, but at poverty. Because often stories about poverty and inequality go in opposite directions.
- 1.116** The last thing before we take a break, I want to talk about is convergence. So the idea of convergence is another way to think about, for developing economies, the consequences of emergence for their growth. And so all these diagrams about convergence, what they do is, they start with a moment on the horizontal axis of GDP per capita, that is, how rich a country are you, as of the 1970s? And then the horizontal, the vertical axis is the growth rate since then.
- 1.117** And so, what convergence means is a negative slope, because that means that, if you're poor, you're growing faster, and if you're rich, you're growing slower. So, if you want to know, is the world on a convergent trajectory, where poor are getting richer, and the rich are not widening the difference between themselves and the rest of the world, then you expect, when you put countries into this two-dimensional space, you should get a negative relationship. So, when you put the open economies into this diagram, you see convergence taking place, right?
- 1.118** If I had put the closed economies, they'd all be, like, down around the origin, making those kind of a square, with no relationship. But the open economies show clearly a convergence going on. And what's interesting is, if you see the slope there of convergence, negative 0.01, if I then make the sample more selective to say open economies that also had high scores of something called rule of law. Rule of law means, it's survey evidence to say, how much is the law administered in a non-corrupt and impartial basis? So if you have high rule of law and you opened your economy during this time, notice that the regression coefficient is negative 0.02.
- 1.119** What does that mean? It means that convergence is happening faster. So that, if you're a country that had other good helping institutions like lack of corruption and better rule of law, then when you opened up to free trade, you got even faster growth. We'll come back to that. I want to stick to our timetable fairly strictly. So we're out of time for our first session and we'll come back and pick up where we left off. Let's take a break.

Lecture 2: Governance and Growth

At a glance — Governance and growth, and the emerging-markets phenomenon of the 1980s-90s.

- 2.1 Welcome back. We were in the middle of a discussion of the emerging markets phenomenon of the 1980s and 1990s, and of this graph that shows that countries that opened their economies up to international trade in the context of legal systems that earned high scores from the standpoint of adherence to rule of law, that is, having good rule of law as opposed to rule of man, autocracies, tended to perform relatively well and converge to higher growth a little bit faster.
- 2.2 That same group of high-performing growth economies that were adhering to rule of law and openness also exhibited some other features. They had, relative to the average, they had 4% more, that is, those who had the good rule of law scores, compared to those that didn't, but that were also opening up, the ones with the good rule of law scores had 4% more trade, too, relative to GDP. That's a very large amount.
- 2.3 So what does that tell you? It tells you that opening up to free trade doesn't necessarily create as much free trade as if you open up to free trade in the context of a good set of domestic institutions. You'll also get more of any good thing by having good domestic institutions, including adherence to rule of law. Other measures that are complementary to this are more foreign direct investment, 1% of GDP more, which is, again, a huge amount.
- 2.4 If you're not, and this has been shown in a lot of studies, if you're not as corrupt, that tends to be very highly correlated with good rule of law. That is, less corruption and higher rule of law are almost the same thing, statistically. You get more foreign direct investment. Companies are willing to come to your country and establish branches of their company in your country, if they feel comfortable with your administrative law enforcement.
- 2.5 Finally, here's a negative, fewer worker remittances. Now, that may sound like, what's going on there? Why are there fewer worker remittances? Well, there are fewer worker remittances because people don't have to leave your country. Worker remittances are what bad rule of law countries, that's their ticket to trying to get out of poverty. That is, send their population to another country and have them send it back to mom and dad, remittances. So when you actually improve your domestic institutions, you have less outmigration and fewer remittances.
- 2.6 This is one of my favorite graphs because this is from a completely different study. It's a study of individual U.S. states, and the horizontal axis is the state's per capita income as of 1880, and the vertical axis is from 1880 to about a century later. So what you can see there is, these states in the U.S. are all almost exactly on the line. It's really a remarkable fit, right? And what it's telling you is, if you have a common rule of law, very high rule of law regime, and you have completely free trade, because trade is not,

cannot be restricted across the states, so among the states, you're in the highest level of rule of law, at least on planet Earth.

- 2.7 It may not be perfect, but it's pretty good. And completely free trade, convergence happens. So if you were having a conversation with someone who said, "I don't understand. In neoclassical economics, they tell us that convergence should happen, because if you're in a poor place, capital should migrate there and help people in a low-wage, poor place to grow faster." So you expect convergence. And I mentioned, we don't always see convergence, at the very beginning of this class. My point is, institutional failures can circumvent the process.
- 2.8 But if you know, like in the case of the U.S., that you have absolutely free trade, free mobility of people, too, by the way, and rule of law operating similarly in all the states, you see convergence works really well. So, the principle of convergence is not wrong when applied to an environment in which the institutions are functioning. We're going to now look at institutions in more depth. And we're going to divide our analysis. We're going to do mainly statistical analysis. You'll see some country-specific data, too, but mainly statistical analysis, looking at emerging market countries broadly.
- 2.9 We're going to be analyzing both equity markets and public debt markets, as well as banks. The sequence we're going to do it is, first we're going to look at equity markets, stock market liberalizations, and how they affected things. Then we're going to look at banking system liberalizations, and then public debt liberalizations. We have a lot of data because the emerging market phenomenon occurred, and so there's a lot of countries that did liberalize, and we track from the moment they liberalize, what happened afterwards, and compare them to the countries that had liberalized long before or that never liberalized. That's basically the strategy of most of these studies.
- 2.10 Let's first note that, roughly, the timing and the way this worked. I'm going to show you the components of this, but this is an overall index of liberalization divided into emerging countries and mature markets. So the bottom line is basically the U.S. and developed countries in Western Europe and elsewhere. The top line are poorer countries that are choosing emergence.
- 2.11 And what you see is that, as you get—so this is like, how many impediments are there to emergence? Remember the list that we talked about, but especially with respect to the financial emergence, it has to do with allowing international capital to come into your country. It has to do with liberalizing your banking system, to allow your banking system to provide private credit as the market dictates. And it also has to do with reducing equity market limitations on foreign ownership. Those are the main things we're going to be exploring.
- 2.12 There are other equity liberalizations we're going to be talking about, equity market liberalizations, but those are the three main ones. Those are the three that are factored into this index. And so what you can see is, so lower values are better, less restraints,

less limitations. So you can see that the mature markets start off pretty similar to the emerging markets, a little bit better, that is, a little lower. You can see that they liberalize a little sooner.

- 2.13** In the early 1980s, you're starting to see a drop, and then again, in the late 1980s, further drop. By the late 1990s, developed countries have pretty much eliminated almost all of the barriers on both the domestic banking system structure, as well as capital flows coming to their country and foreign ownership of equity. You can see that, for the emerging markets, they also are liberalizing. They start off and end up in similar places, but it's a much more protracted process.
- 2.14** What this is showing you is that there's also learning from other countries' experience. I mentioned that there's a lot of learning from the early experience of Korea and Chile, but also as more countries adopt liberalizations of financial systems and the successes of those become apparent, other countries are willing to imitate them. It becomes standard practice by the end of the 1990s.
- 2.15** I mentioned the three dimensions. The three dimensions are liberalization of the capital account that is basically allowing foreign capital to come in; liberalizing the domestic financial sector, that's pretty much allowing banks to provide private credit; and liberalizing the stock market, which has mainly to do with allowing foreign ownership. You can see that there are three different graphs, that there was less of a change in stock market ownership rules as the developed countries started off allowing ownership of stock across borders, which is why that lower diagram already starts off pretty low. But the basic trajectory is similar among all three.
- 2.16** And, of course, the point of this course is, to get the finance reforms to work, it's not going to turn out to be just a matter of allowing them to happen. It's not just a matter of saying, "OK, we're going to allow banks to operate legally. We're going to allow it, or legally, we're going to allow foreign capital to come in, or legally, we're going to allow foreigners to own equity."
- 2.17** That's a necessary condition. If you don't allow it, it can't happen. But our point is going to be, just allowing it doesn't solve all the problems. You have to also build institutions that make it happen. So there's removing the negative barrier, but then you also have to provide causes. And so that's what we're going to be exploring one market at a time, starting with the equity market and moving on, at what kinds of things need to be done.
- 2.18** I think it's also worth mentioning, although we won't be talking much about this, that—we will at the very end, though—there's a lot of errors. Those errors tend to take the form of crises, where countries liberalize and find out that they didn't do it quite right. So I'm advertising now for my other course here at the Peterson Academy, which is the history of financial crises, where we talk a lot about those crises. But as we'll see, there's a positive aspect to these crises.

- 2.19** You can imagine that, if you're sitting there governing the country and you decide to do all four of these financial liberalizations at once, and then try to build all of the institutional changes that are going to make them work well, you're basically changing everything, right? That's a lot to change at once. Are you likely to make a mistake? Are you likely to not know how to do everything perfectly the first time? Yes. And then the question is, how do you learn? What's the learning process?
- 2.20** And we'll see some evidence that, for the typical emerging market experiences, as they're liberalizing and experimenting with these new policies, they're making some mistakes. There are some financial crises, and the crises help them highlight the errors that need further improvement. And so, liberalization is not a static one-time decision. It's more of a commitment to a process where you learn over time and things get better.
- 2.21** It doesn't always work out that way, of course. Some countries give up on liberalization. We've seen, unfortunately, that's the case in Argentina. That is, they went through major liberalization in the early 1990s, and then after their crisis, 2002, there was a backlash against liberalization, and we're still waiting to see if they can restore a commitment to emergence as part of their development strategy.
- 2.22** And I would also mention in Mexico right now, we're seeing that AMLO does not really see much value in preserving all of the institutions that have been part of building the success of Mexico since 1995, preferring one-party rule and state control as a model of economic development. I think that's a very unwise decision, but I'm very worried that Mexico is going through that right now, too. But the overall pattern is much more optimistic, of countries learning from their mistakes and continuing on a path of improvement, wealth creation, and higher income.
- 2.23** When we talk about governance problems, they divide into public and private governance problems. So public governance problems that the institutions of the state are related to have to do with everything from the legal system— is this a rule of law country? How well are creditors' rights articulated and enforced? Do stockholders have rights to fire the managers effectively, to work as a group in their own best interest, or are managers or a certain cadre of insiders controlling the companies for their own personal interests at the cost of the shareholders?
- 2.24** So the institutions are designed to fix these kinds of problems. And, of course, legal institutions are essential, if you're a creditor, if you're a bank. How do you collect if the law doesn't back you up and your right to collect from a borrower? But there's more to it than that. There are political risks. Banks face political risks of expropriation. Mexican banks, for example, were expropriated in the 1980s. That is, the government took them over. The banks were then privatized several years later.
- 2.25** But there are many ways to expropriate with regulation, rather than the formal act of taking the bank. You can just make the bank have to give you an interest-free loan by

having reserves at the central bank at zero interest. Pass laws that require the bank to give credit to your friends, your political friends. So there's a lot of ways in which just creating a liberalization by itself doesn't mean you've really followed through on all of the institutions that make it successful.

- 2.26** In a frictionless world—let's talk about equity markets for a minute— in a frictionless world, what we finance professors imagine, when I say frictionless world, I mean one in which governance issues aren't relevant. Governance issues are solved. There are no barriers coming from governance. We'll see that that is not true, and we'll pinpoint those failures.
- 2.27** But in a world that doesn't have such problems, what we expect is, under something called the capital asset pricing model, which we find very useful to teach in business schools, about how investors should allocate their wealth in forming portfolios, what we teach is that you should pretty much have, all have the same, or more or less the same, stock portfolios because there's going to be an efficiency frontier of portfolios. I'm not expecting you to learn what I'm saying, but I'm just telling you how we do it in finance. There's a point of tangency between that frontier and a riskless rate that's available to you.
- 2.28** You can all invest in dollar, U.S. government securities, let's say. In a frictionless world, you'd all hold some combination. Insofar as you're holding a stock portfolio, you'd hold the same stock portfolio and then some other assets like government bonds or cash or something like that, but your stock portfolio would all be the same. In fact, we gave a Nobel Prize to someone for showing that. The problem, of course, is that is not, as we'll see, that is not the case. People don't all own a diversified stock portfolio of all the world's stocks.
- 2.29** We have to try to figure out, why is that? Why is it that it looks like, in emerging markets, it's actually pretty few firms that are participating in the global market, in the sense of having investors invest in them? Typically, it's a handful, certainly not more than, if you put China aside, not more than 20 or 30 firms in any emerging market country that have substantial global investor participation. That doesn't mean it's unimportant, as I'll show you. Those firms get a lot of equity invested in them internationally.
- 2.30** It's not the case that all the firms that participate, let's say, in the stock market in Brazil or in Mexico, are participating in the global aspect of the stock market. Not because there's a law against it. There's no law against it. It's just that, it turns out that, de facto, the firms that are participating in the global market are a very select number. That's something for us to try to figure out, why that would be. We'll also see that, in terms of risk-sharing, measures of risk-sharing, that our hypothetical notion of a global stock portfolio with global risk-sharing, we're very far away from that, maybe 80% away from that, if you were, and I'll show you how we try to capture that notion statistically.

- 2.31** We'll see when we talk about René Stulz's work that he's going to point to institutions of private governance being mainly to blame for that. That is, that in many countries, minority shareholders' rights are not adequately protected. And so, all outsiders, including foreigners who are outsiders, but also domestic outsiders. So what's an insider? Someone who's a powerful person sitting on the board of directors or managers of that company in that country. Outsiders are all the other people in the world, including people in that country and people outside.
- 2.32** And so, René Stulz is going to show us that the limitations on foreign ownership seem to reflect largely private governance shortcomings that make outsiders discouraged from buying shares. It's not to say that removing the legal limits wasn't helpful. It was. But by itself, it didn't create the perfectly integrated common global portfolio because it didn't fix all problems. But before getting into all the limitations, let's just revel in some pretty impressive accomplishments.
- 2.33** The early papers in this literature, looking at how much it mattered to remove limitations on foreign ownership of equity in emerging markets, found results that were so astounding. Remember, I've been teaching this material for about 30 years. So astounding that it used to be that, when I taught these studies, I said I didn't believe them, but I now do believe them. I'll explain why it's reasonable to have doubts about these kinds of studies. But now, as we go on past this to the micro-analysis studies that I'll present to you, we can see these effects are real.
- 2.34** So what's so astounding? Well, the macroeconomic study. So what that means is, you're just looking at the whole country as a whole. So a country year is a data point in your study. So you're saying, "Oh, I allowed liberalization." What happened to the GDP in that country? What happened to the prices of the equity in the index in that country? What happened to employment in that country? What happened to productivity? So it's really big-picture stuff, right?
- 2.35** So they found that permitting cross-border ownership increased the market value of equity on average by about 3%. That's a pretty big number. For the securities that the foreigners were eligible to purchase because the rules didn't say across the board that all securities could be purchased; they may have limited which securities could be purchased. For the ones that were specifically eligible for foreign purchase, their price went up by 10%. So that's a really big effect.
- 2.36** They also found, as you'd expect from risk-sharing, of constructing a global portfolio, their risk premia fell. So let me explain. The required return that investors are expecting on any stock consists of two components. One, the risk-free return. So, think of that as the treasury bill return. Plus, a premium that you earn because stock is riskier than a treasury bill. And that's called the risk premium. So when you globally diversify, it's allowing investors across different countries to hold a broader stock portfolio. What you expect is that the risk premia for stock all over the world should come down, because now we're all able to diversify.

- 2.37** Think about it this way. You're in Chile. Huge risk in your stock market is the price of copper because Chile is one of the world's top suppliers of copper. And copper market affects a lot of things within Chile. So even if you looked at the Chilean—if you were a stockholder who could only buy your stock in Chile, you would have a very big risk associated with the movement of the global copper price. Now, if you can hold stocks outside of Chile, global copper prices are just one component in a global portfolio and not a very important one.
- 2.38** And so the risk premium, once you remove the barrier, the risk premium of allowing foreigners to be in your country and you to be in countries outside yours should be a big part of the adjustment. Why does this matter? Because when the risk premium falls, that means that firms can raise funds more cheaply, because the outside investor is not expecting a return that's as high. That means the cost of raising funds is lower, which means, I can raise more funds, invest more, and be more profitable.
- 2.39** The whole point of these equity liberalizations is not just to give finance professors a job to explain about equity markets and portfolio frontiers. The whole point is to reduce the cost of funding of firms through reductions in the risk premium that allow the firms to attract equity investments more easily, fund more investments, and grow. It better be the case that we find a big reduction in the risk premium, and we do.
- 2.40** How about the real economy? Well, these same authors find that fixed capital investment, that means building things, so fixed capital investment is plants and machinery investment, structures, too, increased by 22% as a result of these liberalizations. GDP growth increased by 1%, and worker productivity by 2.3%. Now, these are huge numbers. 1% of GDP growth is a very big number for an equity market liberalization to have achieved. And worker productivity growing by 2.3% is astounding.
- 2.41** The other thing that all these studies found, I've just given you the average numbers, but just like in that study of trade that we went through by David Dollar, it turns out that if countries had good institutional quality, better rule of law, lower corruption, the effects were even larger. These are just the average effects. So I used to say, look, I don't really believe time series evidence very much, even panel of time series. In other words, country-year observations, there's a lot that can be going on that's not captured. Now, all of these studies tried to control for every imaginable thing. To really pinpoint and hone in on equity market liberalizations.
- 2.42** But to be honest, the effects are so large that I was suspicious. Then why didn't I do something about it for all those years? My bad, as they say. So I would teach these studies and say, well, I believe there was some advantage, but this is, like, unreasonably large. And I didn't do anything about it for, I don't know, more than a decade. And then, with some friends of mine, we thought of a way to use some microeconomics to see whether we could actually see this mechanism happening.

- 2.43 If you're interested in being an economist and doing empirical work, this is one of the most rewarding things I've done in my life. I can tell you, it is very rewarding to find out the answer to something and know that you really have it. The way that you get that, in my opinion, is never from a time series study. You get a lot of interesting facts from those. But if you actually look at firms and you see what they're doing, and you see the actual mechanism underlying all of this happening, and you can identify it very clearly, that's very satisfying.
- 2.44 What I'm going to now describe to you is a paper that we published in 2021, my friends Mauricio Larraín, who's from Chile, and Sergio Schmuckler, who's from Argentina. And what we did was, we said, well, why don't we look at, when we see capital flowing into these countries, does an exogenous shock, and that's hard to identify. Remember, what we're looking for is something on the supply side. When there's a supply-side decision of foreigners to send more capital to my country, what do I see firms doing in the country? And what we expect, if those macro studies are right, what we expect to see is that those firms should issue more equity.
- 2.45 If they're not issuing more equity, then I don't believe it. The whole point is, we made the cost of equity cheaper. We know that there's dollars flowing in from outside to buy that company's stock. What is the company doing that's creating all this increased investment, increased productivity? What's the company doing? It must be using this foreign supply to increase investment. We said, it's got to be the case, if these studies are credible, it's got to be the case that we see a big link between capital flows and equity issuance.
- 2.46 The thing that was shocking to me, and I say this because I think if you're a student, sometimes you think everything's been done. Here I was teaching a class. I guess I had been teaching it for, I don't know, 15 years at that time. I reported these studies, but it never occurred to me that I could, if I wanted to, collect the data to do the micro-study. Then when Sergio and Mauricio and I sat down and looked at what was available, in truth, the data had become available not that long before, but it was really interesting that we saw a way to do it.
- 2.47 What we found was that the effects were big, that the capital inflow shocks, increased supply of foreign capital to the equity market, are associated with very big equity issuances. We did tend to find that this was concentrated in a few firms, and those are the firms that are themselves the ones who are participating in the global market, not surprisingly. So let's take a look at that.
- 2.48 First of all, let's just look at the aggregate evidence. And this was like, to me, I wondered how we could discover this when it was so easy to discover it. So the first thing we did was we just plotted global flows from developed countries, equity investors into emerging markets, which is the red line, and then the equity issuances of these same emerging market countries. So what do you think? Is there a relationship? Obviously, right? They're obviously very positively related. That's just looking globally.

Well, what if you look individually? Now, what are you expecting to see? You're expecting to see that inflows, capital inflows, should be coincident with or preceding major change in equity issuance.

- 2.49** Now let's look at a few individual countries. This is Brazil and Bulgaria. You can see—so the issuances are the red line, and the equity inflows are the black line. So you can see that the equity inflows, like if you look down at Bulgaria, equity inflows is clearly preceding the spike in issuances that's happening immediately in the following year. What I'm going to show you in all these graphs are that the black line typically is either preceding or coincident with the red line, and that there are very big spikes going on.
- 2.50** So what's happening is, countries open themselves up. They might also have some major achievements in terms of improving the rights of minority shareholders. That's happening a lot during this period. We're going to talk about it. So it could be either the changes— they passed the law that said that you could invest, or they had a major improvement of minority shareholder rights. Then also, of course, these capital inflows can also reflect the fact that, during some years, global institutional investors, that is US pensions and mutual funds, stock funds, might have more money that they want to invest in emerging markets or not. So it's all those three factors are determining the time variation of capital flows into these equity markets.
- 2.51** But if you look at them, here's now Chile and China, you see very similar pictures that the black line and the red line are correlated, and usually the black line, when it's spiking, is preceding a similar spike in the red line. It's great to look at regressions, but another thing to learn when you're doing statistical analysis is there's no shame in actually looking at the data. Then you kind of wonder, how come ours was the first study, published in 2021, to talk about this? Isn't that unbelievable? To me, it is. I don't have any excuse, except I just didn't think to do it.
- 2.52** You can see, I could take you through many countries. You see a similar picture. It's a pretty convincing story that, at least visually, big inflows of foreign capital into domestic equity markets are associated with more issuance. I just showed you what almost constitutes, and certainly on a weighted basis, constitutes the emerging market portfolios. It's not just a little isolated statistical anomaly that we picked up in some regression. It's like screaming out from plotting the data, even plotting the aggregate data. So again, we're very surprised that it works so clearly, and yet wasn't really discovered until pretty late.
- 2.53** So, of course, that's not enough. If you're a professor, you can't take a clear answer and leave it at that. You have to actually make sure that what's hitting you right between the eyes is real. The way we do that is with statistical analysis, to really pinpoint that link and to look at individual firms. That's where the key is, not just at the nations, but at individual firms, to see how the firms are behaving.

- 2.54 Then furthermore, guess what? We know firms also report how they use the funds they raise. What we can do is we can look at not just what drives the inflows, and in particular, we want to identify exogenous inflows. That means inflows that are related to the supply side, which I'll explain more. We want to show that this is linked to equity issuance; for which firms is that happening? Then we want to look and see, what do the firms do with the funds? So do they invest the funds, as the macro studies would suggest they do?
- 2.55 You can see it's a big number. We have about a trillion dollars of capital inflows, total capital inflows, at about 93 billion of equity inflows during this sample period that we're looking at prior to 2014, or I'm sorry, prior to 2018. We're going to do two kinds of things here. First, we're going to just show you some country-level analysis, just looking at the relationship between aggregate country-level inflows and aggregate country-level issuances. What we're going to do is show that, at the country level, those graphs, I already showed you country-specific graphs, but those aren't the same as regressions because you have to also control for other things.
- 2.56 We want to make sure that, in a regression analysis, we're also controlling for other influences when we measure the effect of the inflows. What we find is big effects, and we find that those effects are similar for the domestic and foreign-placed equity. Let me also note that each one of these emerging markets, some of the firms that are in their equity markets are also placing their equity in foreign markets, so that you can buy some companies in the New York Stock Exchange from Brazil, and you can also buy them in Brazil. But then there's some companies that you can buy only in Brazil.
- 2.57 So I want you to be thinking about three kinds of companies, OK? Large globally participating companies that are listed both in the home market and the New York Stock Exchange or the London Stock Exchange. Second category, other large globally participating companies that aren't listed abroad, only listed in their home country. The third category are smaller companies that are not listed abroad and really aren't participating very much in the global equity market. What I mean by "aren't participating" is that foreign investors haven't identified them as firms to include in their portfolios.
- 2.58 Those are three categories. And what I'm going to show you is that we get similar results for the larger firms that are participating in the global market, whether they have foreign listings or not. And we're finding very large effects on stock issuance on average coming out of that. Then when we look at the firm level analysis, what's interesting is, there is no relationship on average between capital inflows and domestic stock issuance. The reason for that is, the third category of firms is by far the largest number of firms, the small firms that aren't participating.
- 2.59 So for most publicly listed firms in Brazil, capital inflows don't affect their issuance because their stocks aren't being bought by global investors. But for the firms whose stocks are bought by global investors, whether those firms have foreign listings or only

domestic listings, there's a very big effect on their stock issuances. They raise a lot of new money when there's a big shock of foreigners' interest in buying their stock. And not only that, but we find that they tend to invest that money very much in new plant and equipment.

- 2.60** Because we're doing econometrics, we also can deal with what's called the identification problem, so that we can make sure that the findings that we're telling you about are really reflecting supply-side shocks. This becomes especially possible when you are looking at many countries at once. We develop a strategy for identification. OK, let's step back a minute and think, what is this identification problem coming from? Suppose it's just the case that something good happens in Chile, like the price of copper goes up, and everybody says, wohoo, we're doing great.
- 2.61** And then the firms say it's a good time to do issue equity because we have great opportunities to expand. And you also see foreigners coming and investing more. We'd call that more, not the supply of capital side, but just the demand side. That is, there's increased productivity because the price of copper went up. Firms have decided to invest more. And then really, there's just a passive mirror image of that coming from foreign investors. But that doesn't prove very much. It's still interesting. It shows that foreigners are facilitating the investment process, but it doesn't show that they're causing the investment process.
- 2.62** What would it mean to show that they cause it? Suppose you can identify supply-side shocks of foreign capital. So, foreigners are unpredictably coming into your country, and when they come in, that's causing you to issue more equity. And then you use that equity, because the cost of equity is lower, there are projects that you've wanted to fund. Now, when that cost of equity goes down, you fund them and you build them. That's a much stronger argument.
- 2.63** How do we get a supply-side shock story? Well, the way we do it is, we know that our data are driven by institutional investors' interests. We look to see, if institutional investors are investing in other emerging market countries, we use that. That's not correlated. We make sure that's not correlated with your current economy. We look to see when institutional investors are increasing their investment in other emerging market countries in ways that aren't linked to your own domestic productivity, and then we were going to say, well, then that, when your country is affected by that shock, we know it's not driven by your own country having an improvement in productivity.
- 2.64** So I won't go into the econometrics too much on that, but we're going to use an instrumental variables approach where we're using the investors' interest in other countries as our way to create exogenous supply shocks that are also relevant for your country. That's how we're going to do it. When we do that, we find that the relationship is very strong. But again, it's only strong of capital inflow shocks. It's only strong for equity issuances for large firms because those large firms are part of

investors' interest. We'll get into a little bit more of the understanding what's driving that in just a minute.

- 2.65** Just to give you a little bit of background, when we talk about firms issuing, in each of these countries, the average annual value of issuance is given as the third column of numbers here. What we're doing is, we're also looking at the number of firms that are in the market. A very large number of firms may be in the market, the first column. But as we'll see, it's only a fraction of those that are accounting for the issuance.
- 2.66** The frequency of issuance is pretty high. If you're coming from the US market, you'd be very surprised to think that, on average, firms issue equity one out of every six years. That would be a really unusually high rate of issuance. But that's what this emerging market data set is telling you. Firms are really making use of this. Equity issuance is a very important source of funds in emerging markets.
- 2.67** OK, let's start with the country-level analysis. All we're doing in the country-level analysis is we're just looking at equity issuance as the dependent variable, the responding variable, and equity inflows are treated as the exogenous variable. Again, we're not doing, strictly speaking, instrumental variables or other kinds of exogeneity here. We're just looking at covariance. But what we're seeing is a remarkably large covariance. So what this is telling us is, when equity flows in, equity investments by institutional investors flow into your country, equity issuance goes up by a lot.
- 2.68** Now, this is in logarithms, so you have to translate these numbers, so that these coefficients, about 0.5, they translate into—and notice they're the same in columns two and three. They're the same for those who are doing foreign participation, for those who are large ones who are, I'm sorry—they're the same for foreign participation, equity issuance, and for domestic. So foreign investors can either buy your stock on the New York Stock Exchange, or they can buy your stock in the Bovespa. It turns out, it doesn't matter. The coefficients are pretty much the same.
- 2.69** What does a coefficient of 0.5 mean? It means in levels, not in logarithms, a coefficient of about 0.2, 0.16, to be more exact. What that means is, for every dollar of, or let's say, for every \$100 of equity inflows, there's \$16 more equity issuance. That is a very big number. OK, now, when you're looking at the country data, you can also look at it other ways. You can divide by GDP when measuring the effect. It doesn't matter.
- 2.70** One thing I want to point out that's very beneficial about equity markets and emerging markets is this asymmetry. If you look at the last column here, you'll see that when inflows are negative—OK, so notice, inflows don't always have to be positive. Foreign investors could be cashing out their investments in your country, too. The measure of total inflows is on net, whether it's positive, if foreigners are buying more equity than they're selling. But they can be negative inflows if foreigners are selling more equity than they're buying.

- 2.71** Notice that there's no effect on issuances when foreigners are selling more equity than they're buying. That's illustrating an important positive effect of equity markets. Notice that it's very different from debt markets. In debt markets, if foreigners want to sell, one way that they sell is that they don't renew the debt, so that you actually have a contraction of foreign investment in your country. In equity markets, when the foreigners sell their stock, that doesn't cause the firm to have to pay a dividend to the equity investors.
- 2.72** See how different that is from debt? In debt, if foreigners decide they want to withdraw from your country, you got to pay back the principle. In equity, that's not true. Equity has a very nice stabilizing role because of this asymmetry between issuances and capital flows. When capital flows are positive, issuances are very positive. When capital flows are negative, issuances are zero, or the consequent issuance is zero. That's a positive. From a stabilization of your economy perspective, a very nice feature of equity investment. Foreigners can't demand their money back. They can in debt markets, but not in equity.
- 2.73** OK, now let's look at the firm level, and we're going to have similar, looking at issuance responding to inflows. First, we're going to do this without the instrumental variables, without this identification of the supply side. Interestingly, identifying the supply side is going to make the effect even larger. But when we just do the overall covariance as a simple regression, it's what I told you. First of all, notice, if we look at all the firms together, the first two columns, we get zero. There's no effect. There's a slight positive coefficient, but it's not statistically significant.
- 2.74** But when we look at either large firms or at firms that list abroad, you'll see that firms that list abroad have a highly statistically significant coefficient in the third column of 0.09. And large firms that are listing domestically have a very high and statistically significant coefficient of 0.106, or in the last column, 0.09. So the effect is identical. If you're a large firm, whether you list abroad or just list domestically, when foreigners decide they want to buy more of your country's equity, there's a very significant effect on your issuance, but not for small firms. Why? Because the foreigners want to buy the large firms.
- 2.75** What is it about large firms that makes them more attractive to foreigners? Well, first of all, they tend to have much more liquidity. When you look at volume trading, so of course, some of them have foreign listings, that's true, but that doesn't seem to be very important. When foreigners get into a stock, they want to know that there's a fairly liquid investment, so that if they want to get out, it's not that they have to find somebody to buy their stock. So large firms are going to often have better corporate governance, more understandable public information, so that you're more comfortable investing in them.
- 2.76** And they also are going to have more liquidity, which matters hugely if you're a pension or mutual. When you decide to buy or sell, you're typically doing pretty large

increments, and you want to know that the investments you're making can be made in a liquid market. And so I think the best argument explaining this is, it's really two things. One, larger firms tend to have more transparency, globally. We know more about them. Secondly, larger firms are more liquid, and so institutional investors are more comfortable investing in them.

- 2.77** I mentioned that we're going to not just look at the covariance, not just at the simple statistical relationship, but also going to instrument so that we can identify the inflow that's related to inflows that are coming from a broader emerging market interest of institutional investors that's not specific to your own country. And so that's what we do in the rest of the paper. And the key thing to notice is, it's the second stage regression, the columns on the right that are the relevant ones. The coefficients now are twice or even more as large. What it's telling us is that, when we have a properly identified supply side story, it's actually—the coefficient is even larger.
- 2.78** Now, let's go to the next thing. Now, we want to know, well, is this affecting investment? We have two different approaches for this. One is to run a separate regression for each category of investment. So firms can invest in plants and equipment that's called CAPEX. They can invest in acquisitions. They can buy other firms, which is a form of CAPEX. They can invest more in research and development. They can invest more in accumulating inventory. They can invest more in accumulating cash assets, or they could pay down their debt. Those are all alternative uses of the funds.
- 2.79** The first approach that we do is, we just went to see, can we find responses across all these dimensions, or some of them, coming from issuing equity? When firms issue equity, do they tend to deploy it to all these and in what proportion? What you can see from this table, and this is true whether you're a foreign issuer, a foreign placing firm, or just domestic market placing, that you're putting some of the proceeds from your equity issuance into a little bit of everything, but substantially into R&D, acquisitions, and CAPEX. We're getting very large responses from those, as well as accumulating inventory and cash.
- 2.80** What we're seeing is firms are definitely increasing their investment in productive resources using the proceeds of this. So far, proceeds' response, the response to capital inflows, is a lot of issuance, and the issuance is also being used a lot for investment, not just for accumulating cash. By the way, if we were looking at bonds, and we will be looking at them later in the class, not so much. Firms that were taking advantage of some opportunities in the bond market were using those proceeds to accumulate cash for the most part. That's something that we'll come back to and discuss, why those should be different. It's a very interesting question.
- 2.81** There's another methodology that was developed by professors Kim and Weisbach in an article they published in the Journal of Financial Economics in 2008 that has a way to come up with a simultaneous penny-for-penny allocation, of how this particular funding source is being allocated across all the uses. And so, the two different

approaches are quite similar. But what's nice about this Kim-Weisbach approach is it gives you a cleaner way of talking about, what do you do with the dollar of issuance? How do you allocate it across all the different proceeds?

- 2.82** Just to give you a sense of what it aggregates to, when you use this approach, it turns out that, if you take CAPEX, research and development and acquisitions, all things that you would think of as highly productive uses of growing the firm, that exhausts 90 cents on the dollar of the issuance. So just to remind you, let's do a back-of-the-envelope calculation now. If there's a million dollars of foreigners that decide they're suddenly interested in buying your shares, that will lead you to issue \$160,000 worth of equity, and 90% of that will go into investment, which means basically \$130,000 of investment. So for every million dollars of equity supply shock, there's \$130,000 of what we would call this fixed investment.
- 2.83** So these numbers are very much coincident with and supportive of huge macroeconomic effects that the macro literature had been identifying. What they're telling us is, we who have been studying emerging market finance for all these years, of course, have believed this, that there are lots of firms that are very productive in emerging markets but are facing financing constraints from less well-developed securities markets or banking systems. And so they are very productive, but also they're in riskier places. I mean that as a matter of politics and institutional risks. Emerging markets are riskier, and bad things happen in emerging markets.
- 2.84** Right now, Mexico is about to throw away two decades of progress in building institutions. That's a big risk. If you invested in Mexican firms, you knew that there was that risk. That's not going to happen in the Netherlands. That's not going to happen in the United States. Emerging markets are places where institutional innovation is much more up for grabs. They are riskier places, and that translates into higher required returns for investors investing in equity, which means that there are lots of investments. Firms, if they were in the United States, they would have invested a lot more.
- 2.85** But they're in a capital-scarce environment because of higher risk and because of limitations on participation in the financial markets, and that when those participation limits are relaxed and when institutions are improved, risk premia fall, investment in those buying portfolio investors goes up, and the firms use that to do a lot of investment that they would have been doing all along if they'd been located in the United States.
- 2.86** So that's what we think of when we talk about financial constraints. That's what we're talking about. We're talking about the fact that positive net present value investments that we tend to think of firms wanting to do are not being done as much in emerging markets because of a politically and institutionally riskier environment. What we're showing is, if you're an emerging market country and you can do something to increase

the supply of foreign funds coming to your country, that would have a major effect on investment and growth in your country.

- 2.87** Now, in our study, we didn't talk a lot about, what are those things that you could do? Now we're going to turn to that question with a study that we've already referenced by René Stulz. Stulz, in 2005, in his presidential address to the American Finance Association, it starts with this kind of fact that we sort of knew, but no one really knew it until René looked up the fact. I mentioned that because, again, I think it's very important for people who are interested in this area to realize, there are very few people who look under the hood at some very basic facts. There's a lot there to be done. This literature is still in its infancy.
- 2.88** He decided to ask the question—remember, this study is written right after those other studies that we already referenced, that had been written showing huge advantages from reducing the legal barriers to ownership of equity markets. And so, Stulz is then asking the question, "OK, I guess, since we got such big effects, it must be the case that the finance professors' dream that we have an integrated global equity market, we must be pretty close to that to have gotten such big effects." And. Of course, what he finds is we're not anywhere near that.
- 2.89** If we were to ask the question, "How close are we to an integrated global capital market?", we'll see the answer is, "We're not even 10% of the way," in terms of stock ownership on a diversified basis all over the world. In other words, people pretty much still own, 90% just own the stock in their own country. So only on the margin do people invest in foreign stocks. If you're in Chile, how much should Chilean firms be in your portfolio? A trivial portion, right? And if you're in the US, the US should be maybe a quarter of your portfolio, but not 90%.
- 2.90** So what Stulz is doing is, he's looking to see, "What's going on? Where are we?" And the answer is, we're pretty much still where we were. Yes, there's some foreign investment on the margin, but it's still small. People are still pretty much holding the portfolio of their own country. And he says, "So, why is that? Why is it that, once the legal barriers are removed, we're still seeing these limitations?" And of course, you know the answer because we've said it already five times, which is, there are other things that keep you as an investor from wanting to invest.
- 2.91** But Stulz is going to talk about this in terms of two expropriation risks. So if you're an equity investor, there are two kinds of things that can happen to you that are bad. One is public expropriation. The government says, "We have decided to take over this firm. And of course, we're going to pay the equity investors what we regard as fair compensation," which is generally not what the equity investors think of as fair compensation. That's what we call expropriation risk. So, I allow you to come into my country, but if you come into my country, you know there's a pretty good chance I'm going to take your money. Not such an attractive investment, right?

- 2.92** But then he says, but there's also private expropriation risk. That is, if the insiders running the firm don't work in the interest of the shareholders but rather work in the interests of themselves, and there's 30 ways to Sunday for them to rip off the shareholders. What are some of them? Pay myself a very high salary, even though I'm not worth it. Protect myself from being fired, even though I'm a completely feckless manager.
- 2.93** Tunnel. So what is tunneling? Tunneling is, well, I'm in the position of being CEO of this company. It just so happens that my family owns another company. Oh, by the way, they're in the same sector. And I'm going to license the technology that this company has to this other company. And of course, I'm going to charge this other company a fair price. It's just, in my opinion, the fair price is almost nothing. What have I done? I've taken the technology that's very valuable in this country, and I've dug a little tunnel and sent it to the other company.
- 2.94** Maybe in this company, I own 5% of the shares, the company that you all own. And in this other company, my family and I own 100% of the shares. So when I send something that's worth a dollar over here and they only pay a cent for it, this company makes 99 cents profit, which all comes to me. And I may lose a little bit here from having underpriced that, but I only own 5% of this, I own 100% of this. That's called tunneling. There are many ways to do this. You can sell shares on a selective basis to your actual global company, or you can, as I said, you can sell technology or other kinds of resources, or sell whole factories to them on the cheap.
- 2.95** Stulz's point is, it shouldn't be surprising that emerging markets have risks, both relating to public expropriation and private expropriation. He's going to call those the twin expropriation risks. Then, what is this statistical modeling going to do? It's going to look at measures of those two risks and show that foreigner participation is reduced when those risks are high. A lot of our understanding of why do foreigners participate more in some economies in equity market holding and not as much in others is going to be explicable once we quantify the twin expropriation risks.
- 2.96** We're going to come back to this in a more systematic way with a study that was done a decade later that tried to be much more comprehensive in its analysis of all the institutional features that matter for equity market investors. But what Stulz did was, he came up with a small list of things that would measure public expropriation and private expropriation and showed that this mattered very much.
- 2.97** Now, let me come back to the graphs. So first thing to notice is, when you just look at foreign asset holdings relative to GDP worldwide, obviously, it looks like we're in an unprecedented place. And prior to World War I, this was also high, as you can see in this graph. Then the combination of World War I, the Great Depression, and the protectionism that came with that, and then World War II, completely destroyed cross-border holdings of wealth. It's only after World War II that that started to be rebuilt.

- 2.98** But as you can see, as we noticed with the financial liberalizations, first in the mature economies and later in the emerging markets, it's really not until the 1990s that we get back to where we were, prior to World War I. You could look at this and say we've made enormous progress since this period of emergence, and that's true. However, that's not his point. His point is, cross-border flows, while they've gone up—this is another graph that shows lots of cross-border flows. The prior graph was looking at the stock. But then he says, what about this?
- 2.99** So this is looking at a measure of how globally diversified is the economy. So I'll just stick to the black line here. So if you have an absolute national segmentation, if individuals only hold shares from their own economy, that black line should take the value one. If they all share in holding a common global portfolio, that number should be zero. Where is it? He says, well, it's about 0.75. So we're only, let's say, we're only 25% or 20-25% of the way to what we think of as a fully integrated global market. That a very insightful way to proceed.
- 2.100** Now he's going to get at this question of, how do I think about these twin expropriation risks? The first thing he's going to do for Americans and British and a few other countries, if you're from the United States, you're used to thinking that insiders only hold a very small percentage of public company stock. Maybe on average, 10%, and of large companies, even a little less because they're very large. It's hard for a few people to own a very large percentage of something that's very big.
- 2.101** But as he points out, if you look at the investable world, by the way, including France and Germany, the typical number is more like 50%. This is as of 2002. In some countries, it's more like 80 or 90%. And so his point is, when we're thinking about expropriation risks, he develops this model that says, "We expect the percentage of insiders holding your shares to be higher when the twin expropriation risks are high."
- 2.102** Let's think why that would be. There are two reasons. First, let's look at the private expropriation risk. So if private, if it's very likely that managers who are the insiders in the firm can expropriate from the minority shareholders, then that's going to limit minority shareholder participation. Another way to put it is, if you're going to steal from this firm, you're going to have to own most of it. That's one way to limit the amount you'll steal from it. So that's an equilibrium.
- 2.103** If there's private expropriation risk because of poor minority shareholder protection, we expect the insider holdings' share to be greater. Notice that has nothing to do with allowing foreigners into your country. That's going to be true even if it's just domestics. Secondly, what about public expropriation? This is the more creative point, because he's going to say, this is also going to be true, that when we have more public expropriation risk, we're also going to still see more insider percentage holding. As somebody who has studied this for my whole career, I've seen the validity of what he's modeling here.

- 2.104** So if you are the CEO and main shareholder of a firm in a corrupt centralized autocracy, you spend a lot of your time working as a member of the network of the government. You're making sure that maybe your son marries the daughter of a prominent general. You're spending a lot of effort managing, believe it or not, managing the expropriation risk. The point is, I need you to own a large amount of the share, not just because it will reduce the amount of private expropriation you'll do, but it will also help insulate the firm from public expropriation.
- 2.105** By the way, there's a whole fund that specializes just in emerging markets, and I won't name the fund. But the business model of the fund is to make sure that a highly politically influential family from each of its investment countries is participating in the fund, so that they can not just have good information, but that they can actually manage their political risks. Imagine if you had a member of the top six families of Mexico, a member of the top 300 families of China. By the way, you're probably sitting there, saying, "Is there such a thing?" Yes, there is such a thing. You have members of these families participating in your fund, and that is a risk control mechanism.
- 2.106** And so similarly, in these kinds of situations, you want the manager to have a big incentive to manage that public expropriation risk, too, by owning a large amount of the firm. So Stulz's, I think, really creative insight here is, it's not just about we want insiders to own big shares because of private expropriation risk, we also want them to own big shares because it mitigates public expropriation risk. Just to give you a sense, we're going to talk about stock market depth more generally, but there's very big differences across countries in how much depth there is in the market that's related to these kinds of risks.
- 2.107** So, you're looking at countries right next each other. Total market value of equity divided by GDP in Malaysia is 10 times, as of 1995, what it was in Indonesia. This is during the Suharto regime in Indonesia. Then in Argentina and Chile, Chile, market value of equity relative to GDP, 10 times what it was in Argentina in 1995. The point is, don't think of—the magnitude of equity markets is going to be constrained by public and private expropriation risk. Of course, the higher those risks are, the more the market is going to be restricted to insiders' ownership because that mitigates the incentives of stealing, both by the government and by the managers.
- 2.108** When we're looking at these statistics and we're trying to understand what in the world can explain why countries right next to each other that seem to have a lot in common have such different equity markets, that's where we have to learn to think. Stulz is then just going to show us in these regressions, and I'm not going to spend much time on this, just to tell you, you can read his presidential address, which is published in the American Journal called the Journal of Finance in 2005. What he does is, he takes all of these measures.
- 2.109** We have survey evidence from business people about how high they think expropriation risks are around the world. We also have measures of not just survey

measures, but also we know things about minority shareholder protection. When we get to Andrew Karolyi's book that I referenced earlier, we're going to talk about a full list of what those things are. But they are things like whether you have anti-dilution protection, whether you call meetings at a place where people can attend them, what kinds of practices are required, what kinds are prohibited.

2.110 And so, in teaching corporate governance, we have a sense of a list of things that are very helpful to protect minority shareholders, and we also have a list of things that are not. What Stulz finds is, when public expropriation risk is higher, you have more concentration on insider holdings. Also, when private expropriation risk is higher, you have that same concentration. That also explains the lower participation of foreigners in these markets.

2.111 Now, how do you fix these problems? There's a whole raft of things that you can do that have been identified in the corporate governance literature to deal with private governance problems. Public governance problems are much harder. How do you turn an autocracy into a democracy? Because democracies have much lower expropriation risk than autocracies. It's not just whether you have a Suharto, it's also what powers the president has versus the legislatures. There are big literatures in political science on, what are the key dimensions that matter of public governance?

2.112 We can look at happy stories and sad stories. What is harder, and we can also look at transitions. There's a wonderful book that my friend Lee Alston and several others wrote about the transitions, the transition that happened in Brazil in the 1990s toward better public governance. What we'll see, too, is that was also associated with better private governance, as we'll talk about. But it's very hard to predict them, I will tell you, very hard to predict them and to try to model them. We're going to talk a little bit about that.

2.113 But on the private governance side, we can certainly point to things that you can do to improve your disclosure, so that your managers of these firms are being held to high standards of disclosure of facts to their minority shareholders, and also to prevent them from doing certain things that are contrary to the interests of minority shareholders. We'll talk about those when we come back from the break, and we'll talk about some other extreme cases.

2.114 But I want to emphasize, our list of solutions is a list of solutions that people are well aware of, but that is easier said than done. So even though we'll look at countries that have implemented these, there are lots of countries that haven't implemented them. And so then we have to ask the question, why would it be that countries wouldn't implement positive changes that protect minority shareholders and that increase investment and development? Of course, the answer is, some people don't want that.

2.115 If you're, if you're running one of the powerful domestic firms, some of those firms might want to see an improvement in minority shareholders' rights, and they're free to

do it, as we'll see when we talk about the Novo Mercado in Brazil. But other firms don't want that improvement because they like being able to earn what we call rents from control at the expense of minority shareholders. And especially in environments where public and private expropriation risks are high, autocracies with small networks of families that are basically both running the government and running the business, they are very resistant to these kinds of reforms because things are going very well for them as it is.

2.116 These kinds of reforms would create more competition. They don't necessarily want more competition. In fact, they generally don't like competition. You have to nest this whole discussion of reforms back into the political economy of what makes countries want them. But we're still going down our list of what would favorable reforms look like. We'll come back and look at some examples of how much they mattered when they were done in different countries after the break. Thank you.

Lecture 3: Emerging Market Reforms

At a glance — Emerging market reforms and recent improvements.

- 3.1** Welcome back. Where we left off, we were talking about improvements that have happened in recent years in emerging market countries' private governance systems to protect minority shareholders. And this has been a really interesting topic since about 1997. One of the things that, again, I'm always selling here, so I'm selling my other course on history of financial crises. But one of the things that came out of the 1997 Asian financial crises that happened in Korea, in Indonesia and Thailand, was a recognition that these countries had actually been doing very well, in many respects.
- 3.2** They weren't spendthrift governments, so it wasn't a case of unsustainable monitoring of fiscal policies that were driving their difficulties, but that it was largely the funding of uncompetitive firms through a combination of bad public choices and bad private governance. And what was interesting then about this crisis was that it led to a recognition in emerging markets that to be able to have highly productive firms getting funding, even in countries we already mentioned—Korea was a leading example, going back to the '70s, of emergence.
- 3.3** But what had happened was that somehow the largest Korean manufacturing firms, the ones linked to the so-called chaebol, which are groups, industrial groups, that were very much tied in with the governments and with people who controlled the banks, that it was a little bit too cozy, a little partnership of the largest manufacturing firms, the government, and the banks, and that what they had done—you can hear all about this more in my other course—but what they did was they continued to fund enterprises that really were not continuing to be productive and competitive.
- 3.4** And then when you peeled back the onion and tried to figure out, well, what had tolerated this, it was exactly those cozy relationships that allowed these firms to continue having access to funding, even though the management and the government that was assisting them was not really working in shareholders' interests. The reason that's worth emphasizing is that, if you're paying attention, if you're not asleep yet, you will notice that all the reforms I'm going to talk about happened in the same year.
- 3.5** 1999 was a big year of reform. And that is not a coincidence. It's because it's a direct reaction to a global new sensibility about the importance of private governance. And so that's a happy story in the sense that things really did change in a lot of countries that were listed here— Korea, India, Bulgaria, and we'll see Brazil, too, where the difference is that, in Brazil, this happens through a non-governmental initiative, which is called the Novo Mercado. It happens slightly later, but still the same sort of picture. So this window of time, 1999 to 2001, just a huge amount of conversation among international investors about the need for this kind of reform.

- 3.6 But as much as we're optimistic and see that, I just keep wanting to point you toward two deeper problems that prevent this from being so easily solved. One is the fact that in many countries, the people who benefit from bad corporate governance like to keep things the way they are because they personally benefit and they have political power that can prevent governance reform. Secondly, in some cases, it's really hard to even draw the line between public governance problems and private governance problems. We're going to talk about two extreme cases, pre-1998 Indonesia under Suharto and China. China in the early 2000s, we're going to have some discussion, but it's still true today.
- 3.7 What I'm pointing to in those cases is that these two concepts that René Stulz taught us about, which are private expropriation risk and public expropriation risk, there's a sense in which they can become one, because private governance and the actions of the state can be so integrated that the private governance apparatus can be part of a public policy, just as the government can help to assist in private, self-interested, bad corporate governance. So you'll see when we get into it that it's very hard to even separate the public governance expropriation risk from the private expropriation risk when the relationship between the management of the firms and the government is so intertwined.
- 3.8 They really almost become one risk. And that kind of thing is very hard to fix because the only way you can really fix it is with a revolution. Just to be very clear about it, if you asked me, what would it take to improve corporate governance in China? What would be my corporate governance reforms that I would be proposing? You'll see, based on the paper I'm going to be presenting to you, that the answer would be, toppling the Communist Party in China would be the beginning. So, violent revolution, in other words, would be pretty much the only way because there's such a mix in intertwining of the public governance problem with the private governance problem that I don't see how you fix it.
- 3.9 They have done some things in China to try to improve the situation, and we're going to talk about those. But it's very telling what we're going to say about them. OK, let's start with an example of that that's not coming from China, but that's coming from Suharto's reign in Indonesia. So, one of my former colleagues and my co-author, Ray Fisman, when he was doing his dissertation at Harvard Business School, he had a great idea, which ended up getting published in the American Economic Review, one of the top journals in economics, and was his job market paper. It's what got him hired at Columbia.
- 3.10 He wanted to try to quantify how much this private expropriation risk, but really kind of public-private, because how much that was intertwined with the health of the dictator. So, what he did was he collected information to try to quantify the crony network of Indonesian capitalism. What he did was, he constructed from a measure that was

available, the connectedness between Suharto and his family and close business associates in each of the large corporations in Indonesia that was publicly listed.

- 3.11** Some of them were closely connected. Let's just do it in a discrete way so we can discuss it clearly. Some of them are closely connected to Suharto, some are not closely connected to Suharto. And then it turned out that there were several times during Suharto's reign where health rumors about Suharto circulated. "Suharto's sick. Oh, Suharto is recovered. Suharto is sick. Suharto is recovered." And what he did was he looked at, how did the stock prices of the companies react to the rumors about Suharto being sick?
- 3.12** And what he found was, I bet you could guess this, the companies that were closely connected to Suharto declined in value whenever he was sick, which showed that a lot of their value was political, was value that came from what we, since Anne Krueger's work in 1974, we call the rent-seeking society, which I alluded to before, that part of what these companies are trying to do is rent-seek from the government to have a sharing of rent with the autocracy that they do in the interest of all the shareholders, too, including the people who are running the firm, who are shareholders. That's why we want to incentivize them to marry the right people, to get the right special deals.
- 3.13** So you probably would have guessed that, when Suharto gets sick, the companies that are identifiably close to him through this intermarriage or interlocking other economic relationships decline in value substantially, and they did. But you might not have guessed that the companies that aren't connected to him rose in value, because the prospect of getting rid of Suharto was a prospect of increased value through their ability to compete because they were on the outs, because he had his favored groups.
- 3.14** But if he were gone, then those who weren't in his favorite groups were also going to be worth more. And so that was really all that the paper showed, but did it in a very careful way, and showed that you could quantify crony networks, that you could show how dependent they were on their membership in the autocracy and the use of the government in promoting them. Final optimistic point about Indonesia. Notice that if Suharto hadn't liberalized Indonesia and gotten them involved in the global economy, then sub-Saharan African countries didn't participate in any major emerging market crises in the 1990s. Only emerging market liberalizers got the crises.
- 3.15** Suharto would never have been toppled if he hadn't actually liberalized. So one of the things that liberalization does is, it puts at risk some of these autocratic arrangements. In Mexico, the PRI had controlled Mexican politics for five decades. After the 1995 crisis, the PRI lost power, which it never regained. Now, something else called Morena has taken over and by somebody who is like what we might call a PRI dinosaur, meaning old-fashioned autocrat.
- 3.16** But my point is, liberalization can actually have very major positive disruptive effects. But notice that that's one of the things that keeps people from liberalizing because they

know that they're placing their bets on something a little bit risky. OK, now, I want you to think about that for a minute, this notion of the downside risk of liberalization, because the next paper is going to be all about that. This was a big surprise paper, another thing that really surprised me.

- 3.17** This is a paper I wrote with my colleague at that time, Ray Fisman, who was the one who did the Indonesian study, and also our PhD student, Yongxiang Wang, who now is a professor, and after getting tenure at, what is that? USC, then returned to China. And what we wanted to do was look at a very interesting case study of what happened to share prices of Chinese publicly traded firms around the time of Chinese discussions of privatization.
- 3.18** Now, by the way, the Chinese government doesn't like the word privatization. I don't know why, but it's very common. That's what we use, when you're getting rid of government ownership of stock and selling it to the private sector. So it's privatization. What happened in the examples that we're going to talk about is that there was a discussion of privatization, an intention, formally speaking, an announcement on July 24th, 2001, of an intention to sell all of the so-called non-tradable shares of publicly traded firms into the market.
- 3.19** So I'll explain that better. But that meant all the government shares that had been withheld. So when publicly traded Chinese companies began, they had a bunch of different types of shares. Everything's very complicated. And a lot of the shares were retained by the government. So they were partially privatized in the '90s, but not fully privatized. And then come around July 24, 2001, the government, which is holding a third of the public company stock itself, and then some other quasi-government agencies are holding almost 30% additionally in shares that were called non-tradable shares, they say, "Oh, maybe it would be a good idea to privatize the remaining component, to sell all the non-tradable shares into the tradable market."
- 3.20** A year later, June 23rd, 2002, the government said, "We're sorry we ever said this, forget it." So it's a really interesting case where we have the possibility of privatization being mentioned, announced, and then being withdrawn. How does the story end? Even though the government says in 2002, "We're never going to raise this topic again," in 2005, they institute a new plan that does effectively liquidate the non-tradable shares into the market. That is another study. How that happened is quite interesting, and what we learned from that is quite interesting and fully consistent with what I'm about to tell you. But for our purposes, this one study will be good enough.
- 3.21** What we're going to do is, we're going to look at what happened to share prices of the firms in China that had government ownership, which is all of them. And when the announcement comes out in 2001 that they're going to be privatized, that the remaining shares will be sold, and then in 2002, when that gets reversed. Now, the reason this is such an interesting case is that it's going to be an illustration of the ways,

just like in that Suharto case, you saw that Suharto is creating value for firms that had influence with him. That's the idea, right?

- 3.22** But in China, if the government owns all the firms, it's really much more systemic, right? The government has influence. The government basically is the largest single shareholder. And of course, you're not in a rule-of-law country. So the government has control over you through every which way to Sunday. They can formally regulate you or they can just have your person taken away. There's no limit on the autocratic power. There is no rule of law.
- 3.23** But the government has an ownership interest in your firm. And so what's interesting there is, the government has, of course, unlimited power to decide about your firm any way they want, but they also have an interest in your firm. So this is a little different, right? And they have an interest in all these firms. So when the government decides to get rid of its interest, is that a positive or a negative thing for the publicly traded firms? Now, the entire history of privatization, coming from all of the experiences that happened in emergence in the 1980s and 1990s, the estimated effects of privatization are always very positive on average.
- 3.24** Why? Well, because privatization in an economy that's typically somewhat rule-of-law, not crazy like, not the extreme on either end, but let's say average rule-of-law, emerging market country, when it's privatized, that means that it's under private ownership, which means the incentives to invest profitably and do things are better than under government ownership. Government bureaucracies don't tend to make great decisions in terms of profitability of investments.
- 3.25** I'm not saying that privatization is perfect. We've been talking about shortcomings of emerging market countries in terms of their governance systems. Nonetheless, privatization, generally, because it changes the incentives toward managing efficiently, tends to produce value increase. Of course, the biggest examples of that were in Eastern Europe after 1990, when the fall of the Soviet Union leads to mass privatizations, and we see huge value creation that's economy-wide.
- 3.26** But notice, that's a case where the privatization is happening and the dictatorships disappear. OK, so the Czech Republic is establishing, like it's pre-World War II, kind of, you know, highly developed economy with a good governance system and a democratic system. Very much rule of law. Hungary, very much a rule-of-law country. So there's a lot of variety there. But when we look at those cases, at least, we see that we're having privatization associated with a complete change in public governance toward having a rule of law and not having a high autocratic control situation where there's certainly not a lot of possibility for—because it's a democratic government, we don't expect a high probability of expropriation by the government.
- 3.27** So, notice that in China, mass privatization was, we're going to sell all of our government shares, but we're still here. And we're still doing things the way we always

did. Lack of rule of law, people can be sent away to death camps. Of course, you know there's still an ongoing genocide of the Uyghurs in China. So it's like, you have to always remember, despite the Chinese growth that's happened, you have to always remember we're dealing with that country when we're thinking about investing. So there aren't really any limits on what might happen to you as an investor.

- 3.28** Now, that said, privatization sounds like a good thing. So which is it? When the government sells its shares, is that going to be a good thing or is it going to be something different? Now, of course, we weren't sure what the answer would be. But as we looked at the case, the thing that was really interesting to us is that this is a very different context, where the privatization would mean that the government has no really continuing interest in the firm.
- 3.29** And so maybe in a country with this system, you get an advantage as a company if the government has a large interest in you, because that means that when you need a favor, which might just be not being sent away to die, would be one favor that you would like, or just having them regulate your competitor in a certain way, they have a vested interest. These shares were owned. It's very corrupt, by the way. So some of the shares were owned directly by the national government, some by the regional government, some by individuals who only can buy them from the national government.
- 3.30** So you shouldn't think of this as all just national government. It's not. And it is a very corrupt situation. So what you're really getting is a vested interest by the government or quasi-government agencies and people who are part of this corrupt control environment of the Communist Party that have a vested interest in you. It's like having Mafia protection, right? So obviously, that's going to matter a lot. And in fact, it did matter, which is why, on average, the announcement in 2001 caused a dramatic decline in stock prices of all the firms. And a year later, it was reversed.
- 3.31** And what's interesting, as I'll show you, is the price decline, which was very large, was exactly reversed. The same amount of price gain happened when they announced that they would not be doing it. OK, so what we're going to do is, we're going to look at the response of the individual firms and how the market responds. And what's really useful, because one of the co-authors reads Chinese and grew up in China, is that we already knew where to look for interesting cross-sectional covariates.
- 3.32** So what that means is, we're not just going to look at the average effect for all the firms. Because as I've already told you, when you're just looking at a time series of what happened on this date, there are other explanations for that. Now, it is very telling that it's an exact reversal on the two dates, negative 10%, positive 10%. But nonetheless, what's really interesting is to see whether variables of each of the firms can help you understand why it's sometimes greater than average and sometimes less than average.

- 3.33** Let me get to the point here. First, we're going to look at B shares because it's a little complicated. Chinese firms had multiple types of shares because they were being regulated on where they could raise their money and from whom and in what source. So B shares were shares that were being offered only to foreign investors. The reason that they're useful is that there's some concern that when the non-tradable A shares owned by the government got converted into tradable A shares, that that might have a direct mechanical effect through a dilution on the value of the A shares.
- 3.34** I never believed that story, but some people did. But by using the B shares, it's clear the total number of A shares hasn't changed. And so it can't be that it's had any—there's no dilution type of effect on the B shares. So we're just going to stick with analyzing the B shares for the companies that had B shares. We have 107 firms that we're analyzing. The average effect of the two different announcements was negative 10.5% and then positive 12.7. Notice that when you fall by 10.5 and you rise by 12, that takes you back to the same price, because a percentage decline and a percentage increase from two different starting points are two different things. So they're quite similar.
- 3.35** But what's even more interesting is to ask, how does the effect vary with attributes of the firm that are cross-sectionally different? So you believe the effect. The reason it got published in the Journal of Financial Economics, which is one of the top finance journals, is because of the cross-sectional evidence. If all we had done was say negative 10.5%, positive 12.7%, it's like, well, what else happened on those days?
- 3.36** But by showing that the effects that we're measuring for individual firms deviate from the average in predictable systematic ways, depending on characteristics of the firm, that should matter from the standpoint of this autocratic model that we're talking about, that is really interesting. OK, so what are those things? Well, first of all, not all firms had the same reliance on that government protection idea. Some firms, the government, or the quasi-government agencies, had a much higher stake in the firm. So they should have, when this stake was going to go away, it should have had a bigger effect on them.
- 3.37** If the government only owned 1% of me, then I wasn't getting much protection to begin with. So when the government was going to privatize that, that wasn't really mattering much for me. So does the effect vary with the magnitude of the government's stake? And the answer is yes. Secondly, can we mitigate the effect if there are other ways for the firm to have its political connectedness? What are those other ways? Well, most of the power that was mattering for these firms had to do with regional or municipal governments. Did you have a governor or high municipal official who was serving on your board of directors, former official who was serving your board of directors?
- 3.38** Because that's well known in the Chinese literature that that person's job is to be a political fixer for you and has the political connections to be able to do that. Well, if you have two ways to skin the cat, then when you lose one way to skin the cat, you've still

got the other one, right? So if you're one of the firms that had an alternative way to fix your problems, then the loss of your government ownership of your firm wouldn't have as big an effect on your stock price, right?

- 3.39** Furthermore, in special enterprise zones, the latitude of action by the government toward the firm is greater. We expect all the effects to be larger in special enterprise zones than they are for companies that are operating in other locations in China. Finally, I'll mention, suppose the company itself has a large welfare payment that the local government depends on coming from the company. Some of these companies have commitments to make either pension payments or welfare payments to the population. That's like an automatic inoculation against anybody trying to hurt you.
- 3.40** Because if you owe the thousands and thousands of laborers a pension or welfare payment, then that's like automatic inoculation against political risk. So you might say, can it possibly be the case that these things show up in the market, that on the day this announcement is made, all these nuances of the identity of the firm having to do with how politically connected it is, how politically important it is through its welfare payments, whether it's in a special enterprise, that all this would show up in predictable systematic variation in stock price reactions?
- 3.41** And the answer is yes; it does work that way. Markets are very smart. These attributes of these firms are not incidental to their identity. They're core and key to their identity. And so that's what we find. If you look at the government share coefficient, the first two columns are the 2001 announcement, the last two columns are the 2002 announcement. And you can see the coefficients on the percentage of ownership are identical. In other words, if you had a high amount of ownership or a low amount of ownership, it gets multiplied negatively by 0.05 in 2001 and positively by 0.05 in 2002. It's a direct mirror image.
- 3.42** How about political connections? Well, they helped you positively in 2001. They mitigated the effect, and then when that was removed, it was part of the mirror image. Then we also looked into the pooled regression. So this is if you take the two events and say one is just the mirror image of the other, then you get more statistical significance because you have more power by constraining the regressions and you get the same results.
- 3.43** How about if you're in a special enterprise zone? Well, if you're in a special enterprise zone, the government shares, again, it's a more pronounced negative. That's the negative 0.06, and in the pool of regressions. So what that's telling you is that the initial negative 0.05 or negative 0.06 coefficient is doubled if you're in a special enterprise zone because the dependence on government discretionary actions is making the government ownership state twice as important for you.
- 3.44** How about if you are running very influential welfare payments or pension payments? That acts as a mitigator of your exposure to the political risk. That's what we found.

What's really interesting about that is it tells you, that's why when I said before, how would you fix private governance problems in a system like this? I don't know how you do, because there's a complete integration of the public governance problem with the private governance problem. There's a complete absence of rule of law.

- 3.45** And so I'm not saying it's not possible. I'm sure some China expert could come up here and tell me something that might be helpful. But my point is, absent a complete change of regime, it's hard to see how you get to a favorable outcome. I also want to note that things haven't been going very well for China in terms of growth. The way that we have been teaching it for a long time would put it as, initially, China was able to grow very rapidly despite the fact that it had a terribly constructed banking system, complete absence of creditors' rights for the banking system, and a complete absence of minority shareholders' protections or rule of law, which it still suffers from today.
- 3.46** And the reason is that, when you are first opening up your economy to be able to participate in transacting with the global economy, there's a lot of low-hanging fruit. There's a lot of productive things you can do. And you're making earnings that are so high that just by plowing the earnings back into the firm, you can maintain very high growth. But as you get to more of a middle situation, after about a decade or two of that, you're more reliant on the financial sector and on outside investors to fund your growth. And that's when these institutional attributes of the financial system become very important constraints.
- 3.47** I would say the absence of a banking sector and a properly constructed stock market were pretty much irrelevant to China's growth in the first decade or so. But now, what are we witnessing? A long-term slowdown of Chinese growth. And the political scientist, Minxin Pei, wrote a book about 20 years ago called China's Trap Transition, where he was forecasting exactly this. And all of us who have been teaching this for some time, I think, are pretty much on the same page, which is, these institutions, this institutional weakness, maybe isn't a problem at the very early stages of deciding to allow a market to exist in goods and services.
- 3.48** But once you start becoming more dependent on the financial sector to allocate resources as a watchdog of efficient allocation of resources, then the absence of these institutions is going to be a big contributor toward a growth decline. And so Minxin Pei was really the first one to say, absent a political transition in China, there will be a significant and permanent growth slowdown, and that China would be trapped, as he put it in his book, between having to accept either permanent slow growth or a political revolution.
- 3.49** And that's where I think we are right now. So my prognosis is, China is not going to see a substantial increase or improvement in its growth because the Communist Party is not willing to give up power in order to get more growth. That makes them, in my view, by the way, even more belligerent and dangerous, geopolitically. So it's not good news for any of us.

- 3.50** OK, how about some happy news? Korea, we've talked about, was a great success story going back to the '70s, had a growth slowdown due to this failure of governance. And that is, the funding system was giving established incumbent large firms, parts of these chaebol, access to funding despite the fact that they were declining in their profitability. And so after the crisis hit Korea, which was already a wealthy country in 1997, Korea was one of the countries that instituted a law that changed the way governance practices worked at firms.
- 3.51** Now, when this law was announced, I have to admit, I was skeptical that just instituting things that require audit committees to be established in the board of directors, that require 50% outside directors. Outside director means someone who's not intimately involved in the firm's management and an outside director nominating committee. So what they basically require is that the management of the firm doesn't control who its board of directors is, doesn't control its accounting and disclosures, and that it has to have a certain percentage of outsiders in all of these key governance roles.
- 3.52** I wasn't sure that was going to be really very important, to be honest. What's interesting is that, because this was implemented only for large firms with assets greater than 2 trillion won, you have an ability to do an analysis that says, do we see a discontinuity? That is, is there evidence of a corporate governance improvement, an evaluation effect, positive valuation effect, that happens at the announcement of this, just for firms of more than 2 trillion? And that's what made it possible to do this analysis.
- 3.53** And it was done in two ways—one, by looking at, longer term or medium term, what happened to the stock price, stock value relative to reproduction cost, of firms that were subject to this? That's sometimes called Tobin's q. It went up substantially. Furthermore, you can look at it through an announcement effect, which is what the second study did, to look at whether stock prices jumped and only jumped for those greater than 2 trillion firms. And both are true. There's clear evidence that this had important value production for Korean firms.
- 3.54** In Bulgaria, there was mass privatization, like the rest of Eastern Europe, 1998. But then these fraudsters were able to, who managed to seize control, coming generally from their previous positions in the Bulgarian government, the oligarch types, seized control and then realized that once they had control of the private companies, they could do things that would benefit themselves at the expense of minority shareholders.
- 3.55** So a typical thing to do, dilution or freeze out, is that you basically sell shares at cheap prices to your friends, so that you owned, let's say, 20% of the company. Now, the 80% you didn't own is going to be worth nothing because you've diluted it by selling 100 times that to yourself for a penny. In other words, you took the preexisting minority shareholders and you reduced their interest in the firm by giving away the firm, by giving away in new share issues, new shares of the firm to yourself and your friends.

- 3.56** You might look at that and say, "That's fraud. That should be illegal." And of course, in the US, that would be illegal. In Bulgaria, it wasn't. In 2002, they passed new rules that would make this behavior illegal. What happened? Well, so there were two kinds of firms trading, some that were not subject to this rule because they were basically government-owned firms. That's the orange hashed line, and some that were subject to the rule. That's the green solid line. So the ones that were subject to the rule, when this 2002 rule was announced, the price-to-earnings ratio of those firms tripled and stayed higher, continued to grow.
- 3.57** So this is some evidence that these kinds of governance reforms can have a pretty major effect on shareholder value. All these country cases are a little bit different, though. So, now we're going to talk about Russia. Russia is a case where you don't really get political regime reform. You do get a lot of range about what's bad. So what was happening there is, there's a lot of Russian firms that, prior to the Russian crisis of 1998, so there was a lot of investor interest there—there were a lot of questions about what the governance was going to be—Russia passed rules that look like SEC kinds of regulations to protect minority shareholders.
- 3.58** But then the question is, are they going to enforce the rules? So you've met Vladimir Putin. What do you think? So how will the rules be enforced in that kind of autocracy? And by the way, welcome to emerging markets. This is an economics course, but all the time, what am I doing? I'm talking about politics, right? Because guess what? That is often what matters for understanding how the economics is going to play out. So, in the case of Russia, because there was a lot of early interest by foreign investors, because some of these firms are very big, Gazprom, Lukoil, some of these firms are potentially of great interest to investors.
- 3.59** There was a question of, let's just start tracking the governance of Russian firms. There were companies that were set up that were doing scoring of corporate governance for Russian firms. I'm going to use those scores and just show you some things about those scores. The point of this paper is not so much looking at a reform. There was no reform. There had been an SEC after, in the early '90s, SEC creation that was never really implemented. What would I mean by not really implemented?
- 3.60** Suppose you want to attend a stockholders' meeting under the rules that were set to vote your proxy. But there's a guy with a machine gun at the door to the room, telling you you may not come in. That would be an example of not-enforced rules. There was a fellow who ran a company called Hermitage Capital, which was an American investor who made a big deal out of exposing violations of the rules, which were constant. By the way, he traveled in an entourage of bulletproof vehicles throughout his time in Russia. He was ultimately forced to leave Russia. He was thrown out of Russia by Vladimir Putin.
- 3.61** So he was trying, and successfully, he thought, "When I expose things, the stock price of that firm will fall." He was trying to expose accounting disclosure, frauds, and other

kinds of practices that were inconsistent with Russian law, but that were ongoing in common. What we're doing here is, we're just looking at governance score, how it varies across firms, and how it's associated with valuations. So the point here is, from the worst to the best in terms of governance, there are going to be some firms that we're going to say, they get the worst governance scores— Gazprom, Lukoil.

- 3.62** By the way, notice something. If you're a company operating in a corrupt autocracy, which sector do you think is going to be more troubled by the autocracy? More, I'm sorry to use the word raped by the autocracy, which one is going to be troubled by the autocracy more? A firm whose operations are physically present throughout the country, like Gazprom, or a firm that's doing software development for global markets? So it's going to turn out that these governance score, governance achievements, that are partly voluntary by the firm—now, of course, if the firm is controlled by the government, that's not voluntary. Gazprom is Putin's firm. It's not making independent decisions.
- 3.63** But if you are running a firm that's doing software development out of your garage, it's less opportunity for the government to hold you up. Whereas if you're transmitting gas throughout the country, you basically are operating with the constant permission of the government. And so, before we even get to the slide, which sector do you think has better corporate governance and the scores? And the answer is, of course, it's the sectors that aren't so reliant on the government. By the way, mobile phones are going to be another example of a sector that's not as physically reliant on, through its presence everywhere, on the government, and it's going to have a kind of intermediate governance score.
- 3.64** So one of the lessons you get from this is, if you're thinking about these public-private governance problems, they can be highly sector-specific in how much value destruction is going on as a result of the governance problem. What Black and his co-authors did was, they first constructed this thing called the value ratio. What they tried to do here was to imagine, if this firm were operating in the United States, looking at the firm's sales and expenses and looking at what comparable firms are worth in the United States, they would run the same account through—if we had credible rule of law and US governance, how much more would the firm be worth?
- 3.65** In some cases, the answer to that is, it would be worth twice as much. It would be worth four times as much. They're constructing this data themselves by taking the financial reports of the firm and imagining what that firm's stock price would be based on those financial data if it were located in the US, and then dividing that by its actual value in Russia. So that the value ratio is the hypothetical value it would have if it weren't located in Russia, if it were located in US, divided by what its true value is, being located in Russia. You with me?
- 3.66** Then the governance ranking is a score dependent on what the firm has committed to in terms of protections of minority shareholders. It's a measure of both its effective

practices and its de jure practices. You see there's a downward sloping line here. What does that say? It says that when your governance ranking is higher, you wouldn't see such a big improvement in value by being located in the US. So the point is, governance is having a big effect on value.

- 3.67** But now what I want to show you is which sectors are what, because that's much more fun. Notice, Gazprom and Lukoil, the ratio for Gazprom is, stock price would be worth, let's see if I've got this right. Gazprom's ratio valuation is 0.002. I think that means it would be worth 500 times what it's worth in Russia if it were located in the US. Is that right? I think that's right. 500 times 0.002 is 1, isn't that right? So 500. Whereas VimpelCom, or Rostelecom, it would be worth maybe twice as much or five times as much. So notice that the sector, that multiple value ratio, is very sector-specific.
- 3.68** And that, of course, we already know, the governance rankings are very closely related to this. So Gazprom has the worst governance rating, and it also has the worst counterfactual, that it would be worth 500 times as much if we could only move it to the US. The point of all these examples are that the literature in emerging markets has been exploring, as part of what your job is as an investor, is to explore what kinds of investment are appealing from the standpoint of exactly this public expropriation and private expropriation risk, and which in some countries are very closely connected, like Russia and China.
- 3.69** India also passes something called a 1999 law. Again, not a coincidence on the timing, a 1999 law that changes governance standards, again, for the largest firms first, and then it will apply to smaller firms. So this is, in some ways, similar to the Korean experience. And so this study of India looks to see what's the—in finance, we call this the idiosyncratic returns. That is, after taking account of the firm's normal relationship to the market, we're not looking at the total market returns, but of what's the average effect for large firms that were subject to this rule? And what about medium-sized firms that would be subject to the rule eventually?
- 3.70** And the answer is, the announcement effect was a stock price increase, depending on which window you look at, of somewhere between 4 and 10% for the largest firms. And for the smaller firms, about half of those magnitudes for the medium-sized firms because there was a delay of when those firms would be. So if you think that there's value destruction from bad corporate governance, corporate governance is going to improve the value of the firm, then for the medium-sized firms, it's a delayed improvement, which is why the effect on the stock price is lower.
- 3.71** OK, now in Brazil, we didn't have government policies. All of these examples, India, Bulgaria, and Korea, that we've looked at, are government policies. But in Brazil, that's not the case. So one of my former PhD students, Gledson Carvalho, went back to Brazil to become a professor, and he got this idea in 2001 to see if the Bovespa, which is the Brazilian Stock Exchange, would be willing to consider a voluntary action that would allow firms to choose to adhere to, voluntarily, high-quality accounting, disclosure, and

corporate governance standards in terms of these same kinds of things—outside directors, participation, anti-dilution and freeze-out laws, all of this thing, as well as disclosure standards.

- 3.72** It's voluntary. And by the way, there are different levels. You can be a member of the Bovespa that doesn't want to participate in Novo Mercado, or you can choose minimal participation in Novo Mercado, or you can choose higher levels of participation. So what was interesting is that the firms were told, "You decide." OK, now, first question, why is this credible at all? And there you have to ask the question, why is the Bovespa going to be a credible monitor, affirm compliance? Because any rule is only as good as the institution that will verify its compliance. Are they actually doing the things that they say they're doing?
- 3.73** So you have to ask, why would the Bovespa be as credible as, let's say, a government agency or maybe more credible? What does the Bovespa get out of this? And the answer is money. If you have more trading going on, so when firms are not, when minority shareholders are not happy about firms, the firms have less stock outstanding and less trading in the stock. You're more worried about whether, if you trade, you might be taken advantage of by an insider. Float, as we say, or turnover, tends to be very low in insider-controlled, poorly governed stocks.
- 3.74** So Bovespa knows we're going to get more firms issuing stock, which means we'll have more trading going on, and we'll have more liquidity, better information, therefore more trading. And guess what? We make our cut based on how much stock trading there is. So what's interesting about this is a model. To my knowledge, it hasn't been imitated, which is interesting, but it was a path-breaking model because it said, look, some firms, OK, some firms just don't want discipline. They're politically connected, incumbents. They like their rent-seeking equilibrium. They're going to block the creation of SEC rules or other such things to force compliance.
- 3.75** But there are going to be some firms that want to have better corporate governance because that's a credible commitment that will allow more people to want to buy their shares. So did it work? And the answer is, yes, it did. So firms that committed saw a stock price rise of more than 2%. They also saw, so there were different kinds of shares that were traded by these firms, some with control discounts because they weren't for voting and all the rest of it. And what happened was, when the firms committed to this Novo Mercado, there was a compression in the pricing of their non-voting and voting shares, which is like saying, even if I'm a non-voting minority shareholder who's used to being taken advantage of, that's going to be happening less now.
- 3.76** They saw greater likelihood of the firms that were choosing the higher disclosure and governance standards, those firms were much more likely to do an offering. In fact, you might guess that part of the reason that those firms signed on to the Novo Mercado standards was because they were anticipating wanting to do an offering, and and they

wanted to get a higher price for it. And the price went up, and they did a lot more offerings.

- 3.77** Then the authors also did a placebo test. What if your firm already had a foreign listing at the New York Stock Exchange? Well, if that's so, you already had to comply because of the New York Stock Exchange's requirements for disclosure and governance with things that were pretty similar to the Novo Mercado. So for those firms, signing on to the Novo Mercado had no effects. We call that a placebo test, or it shouldn't be that it has an effect and it doesn't.
- 3.78** OK, so the point of all this, of course, is that corporate governance has real value. It's an ongoing problem for stocks in emerging markets, but there are some positive experiences, too. Now, we talked about cross-listing a little bit, so you might have wondered, is cross-listing another solution? In other words, maybe if I'm a firm, even though I'm not in Brazil with a Novo Mercado, maybe if I'm in, I don't know where, Venezuela, I could just buy cross-listing in the New York Stock Exchange, could get a lot of the advantages.
- 3.79** Now, there are a couple of problems with that. You're still subject to Venezuelan expropriation risk, which is pretty big. The other problem with that is that what we find is that you can't really cross-list your way to good governance. Because good governance requires certain things locally in terms of enforcement, I guess. So that if you're in a really bad private governance regime, you can't just cross-list in the best one and improve. And what we see typically with cross-listing is countries cross-list to a slightly better country.
- 3.80** So if you're in the tank, if you're the worst kind of minority shareholder protection and disclosure rules, what we saw is those countries would typically cross-list, not to New York or London, but to Japan, which is not as good. So there was a sense in which cross-listing can only improve you incrementally a little bit. You can't just cross-list your way to great corporate governance. That's what the literature has shown. This paper by Fernandes and Giannetti is the one I'm thinking of.
- 3.81** So really, you can't fix your governance without domestic fixes, is what it comes down to. You can maybe improve things a little on the margin by cross-listing, but not much, because so much of what people are worried about is the domestic political economy that is related to your governance problem. OK. In 2015, Cornell professor Andrew Karolyi writes this really interesting book called *Cracking the Emerging Markets Enigma*. What he's trying to do is try to explain why emerging market stocks are riskier, how they vary, why some stocks are viewed as risky and others not.
- 3.82** And so what he does—now, you guys are ready for this. You could have guessed, based on all of our discussion today. You could have guessed what he would do. He's going to come up with this list of all these institutional characteristics. He's going to try to organize them in an intelligent way, and he's going to then see whether he can show

that developed country investors' interest in emerging markets is limited by institutional factors that affect risk. This book, all he's going to do is, start with a very large list of hundreds and hundreds of institutional attributes related to exactly the things we've talked about, and I'm going to show them to you momentarily.

- 3.83** Then find a way, statistically, to boil them down to composite measures of six dimensions of institutional quality. How well or badly you think he did that? I don't think it's perfect, but it's definitely good. And then see whether those help us understand differences in international investors' interests in particular countries. The answer is, they do. He also shows not only that, but he also shows which dimensions, therefore, are the most important. It's going to be expropriation risk and corporate governance, just like René Stulz's study would have led you to expect.
- 3.84** Furthermore, he then looks at something that happened in 2013 called the taper tantrum, which is when there was a general exodus of concern coming from Federal Reserve, making it look like they were going to start increasing interest rates. Some of the interest in emerging market investments has come from low interest rates in the US, as a place where you can get higher yields if you're willing to take on some more risk. But as US interest rates were expected to go up, there was an expectation that some of the investors would be withdrawing.
- 3.85** Then the question is, is there a cross-sectional difference in which countries the investors withdrew from? The answer is yes. It looks like it was a kind of last in, first out. That is, the countries that were considered the weakest, which had been slow to get a lot of international interest, were also the first place that people went for the exits when they started thinking US interest rates were coming back. That's what his study found. Now, I'm just going to quickly go through it a little bit in our remaining time, so you can get a sense of what he did and what some possible problems are in what he did.
- 3.86** OK, so he took all of his hundreds of measurable institutional features of emerging markets, and he put them into one of these six categories, which he labeled. Again, these labels are his. Market capacity constraints. He's thinking just, I don't know how to describe this. It's like, how much market is there in this country? Now, you might say, well, what does that measure exactly? That's an outcome. That's a criticism I would make. There's market value of equity to GDP is an indication of a lack of foreign interest. It's highly correlated with lack of foreign interest. But I don't know that that's an explanation. It's just a measure of it. So I didn't love the market capacity things.
- 3.87** Operational inefficiency has to do with attributes of rules of trading. I like that category. Foreign accessibility restrictions. That's looking at de jure and de facto—does this country have restrictions that make it harder for foreigners to participate in the market? Corporate opacity. Well, that's this private governance stuff that we've been talking about. Limits on legal protection. That has to do with actual legal protections for

minority shareholders, but also for creditors. And political instability. That has to do with things like expropriation risk.

- 3.88** So this is how he categorized it. And then for each of these six categories, he then has lots of variables. So how do you compress them? He uses something called principal components. He tries to identify the common component in a long list of variables. He'll use the first one or two principal components to construct an index. I'm not going to explain the principal components. It's not the only way to go. There are other ways to try to distill information from a large list of variables, but it's a very respectable way to go.
- 3.89** And so what he does is, he then has a common index for each of these six that tries to distill all the information out of hundreds of variables into just six scores. And then he wants to see whether, in a regression, those distilled variables can help explain why investors are interested in your country or not. So what I love about this book is, first of all, it allows you to— suppose you had an assignment if you were on a trading desk and somebody said, "Well, what do you think of Indonesia?" You could look at his spider graph, which I'll show you in a minute, and it would immediately identify for you the category of things that's most weak in Indonesia, because all these countries are not the same.
- 3.90** Across these six categories, the institutional weaknesses are different. You can really get very quickly to a picture of how Indonesia is different from Chile, different from Mexico, and that's very important. Emerging market countries are all very different from each other. So his methodology allows you to summarize how those countries are different in a picture, which is really nice. The other thing is, he shows that these things matter, and he also shows which things matter. So this is the spider diagram I was referring to. So if you have six dimensions, you can think of that as a hexagon of each of these dimensions.
- 3.91** And if you look at the solid line is the average for developed economies like the US and the UK. So the point is, political instability, you can see it's virtually zero, which is why the black square, at political instability, the black square is right at the edge, right? So the point is, these are very long, mature, long-lived and mature democracies; they're not going to have a revolution. They're not going to have expropriations of private investors. It's unthinkable that the US would expropriate companies. That's not what, we're not going to do that. And so that's zero.
- 3.92** Now, if you look at the dashed interior line, it's very close to the middle of the diagram, you can see that for emerging markets, they're very close to the middle, which is the extreme value, the worst it can be. If you were going to say, where are developed economies and emerging markets most dissimilar, you would say political instability. If you were going to say, where are developed economies and emerging markets on average most similar, it would be legal protections. That's, what is the legal codes say on average. Emerging markets are still inferior, but they're closer now. This is, again,

pretty recent. If this graph had been done in 1990, they would be very much farther apart.

- 3.93** I hope this is making sense. Then you can look at them for individual countries. This is a composite score, and you can kind of see a ranking. Venezuela is the worst institutional quality, taken as a composite score across the six categories, and Taiwan is maybe the best. You can see that, if the average of developed economies, which is the solid black thing at the bottom of this graph, is pretty similar to Taiwan, South Korea. I can't remember if Israel is on this graph here. I'm not reading it.
- 3.94** But you get the idea that some, Taiwan, Malaysia, Slovenia, South Korea, are close to developed economies, whereas Venezuela, Egypt, Pakistan, Russia, Nigeria, Vietnam, Argentina, Jordan, are the opposite extreme. But then having the spider graphs for each of the countries is interesting because, as you can see, these are very different. So if you were looking at Brazil, you would say, well, let's take, just for a comparison, let's take corporate opacity in China versus Brazil. So China gets a zero for corporate opacity. Brazil gets a very high score for corporate opacity, as we've seen. They've had some important institutional improvements, right?
- 3.95** So, I think it's a good resource. If you're in this area, if this is something that you want to do, you can read the book. As I already said, I'm not in love with the market capacity constraints category because all it's really doing is looking at the outcome variables. How much are we seeing in domestic credit? How much are we seeing in stock market value, all relative to GDP? How much turnover is in the stock market? These are outcomes that we wouldn't be—of course, these outcomes should be positively correlated with foreign investors' interest, but they're not really telling us anything about the mechanism.
- 3.96** So if I had written a book, these would not have been included. They would have been included as alternative measures of outcomes, not causes. Operational efficiencies, this has to do with how the markets are run, and I very much like these. Foreign accessibility is measuring how hard as a foreigner it is to access the markets. Corporate opacity, again, it's all about accounting disclosure and governance. Limits on legal protection. What kinds of explicit legal limits exist for minority shareholders and for creditors, and how well are they enforced?
- 3.97** We have surveys in all these countries telling us whether rights are enforced. It turns out, as you'll see in some things I'm going to show you tomorrow, that rights matter interactively with how well they're enforced. That is, how much you get from the right depends on what the judicial rules are for enforcing them, and are you in a place that enforces laws or not? Those kinds of things. Political instability. We have survey measures that are done by the Fraser Institute, the Cato Institute, Heritage Foundation, the Wall Street Journal, on economic freedom.

- 3.98** Corruption is measured by Transparency International, as well as rule of law, which would be in the prior group. Governance quality measures and structure of government. So if you have a very powerful executive branch, that tends to be viewed as a negative. These are all things coming from the political science literature and various kinds of survey measurements, mainly survey measurements, but some are explicit. Some are objective, I mean.
- 3.99** So, does all this work? And the answer is, yes, it does. So let's look at this. This is taken as a page copied from his book. So, the first column does a weighted average of the six indicators statistically significantly predict foreign interest in a country. Now, there are two different databases, so we're going to have two tables, but that's all. One is more focused on US investors, the other all developed country investors. The US investors' database is considered a little better, so it's data quality versus data coverage trade-off, and what he finds is similar results for both.
- 3.100** But then he looks, what about if you just take one at a time? Market capacity, yes, it's statistically significant. I put little weight on that because, again, it's an outcome variable. But then when he looks at operational efficiencies, he finds not a big deal. No, that doesn't seem to be very important. Foreign investability restriction is not very important. The way I read that is, most of the world has relaxed those investability restrictions, and they're not really the binding constraint on what foreigners are doing. It's more the corporate governance, that is the private governance problems, as well as the public governance problems that matter.
- 3.101** Then we find that corporate opacity, that has a very large and statistically significant effect, limits to legal protection also, and political instability. My reading of this evidence is, the most important things are the legal limitations on your rights, the corporate governance practices, and the political instability, which is pretty much what we've been stressing all along and in our discussion also of René Stulz's work.
- 3.102** This is the second table just finding similar results coming from a different database. Here, the foreign investability and the operational efficiencies are both showing up. They're having smaller effects, but they are showing up as statistically significant. In this one, all the categories are showing up as statistically significant. And then he says, well, that's looking at the cross-section of the stock of foreign investment. What if we ask, how vulnerable were you to the events of 2013 relating to the Fed's taper tantrum? And there again, what we're finding is lots of different influences.
- 3.103** Here, operational efficiencies, it turns out, were more important. I can understand that as, if you're in a situation where you're worried about being able to get out of a market, maybe you get out faster. And then it comes down to corporate opacity, the three that matter the most still. Corporate opacity limits the legal protections and political instability. So that completes my discussion of the Karolyi book. I want to talk now more generally about how investors and academics that are trying to approach this topic, how the literature is changing, because I think it's quite interesting. One of the

big improvements has been the availability of a new analysis based on textual data, that is text flow.

- 3.104** For example, Hassan et al have an approach to using earnings calls. All US public companies have quarterly earnings calls where they have discussions. They present their current performance, and then they have discussions, Q&A with their investor groups. And in those discussions, you can track discussions of emerging markets. So what they did was, they thought, well, if a company and its investors think discussing Turkey is important enough to take the precious time of their quarterly earnings report, to discuss risks in Turkey, then maybe something's going on in Turkey.
- 3.105** So they aggregate across all of the earnings calls for all 3,000-some companies, how much are people talking about Turkey this quarter? How much are people talking about and in what context? Because you can also look at the words that modify that. Is it positive tone or negative tone? Are they talking about certain topics? We can do topical analysis. So textual analysis, I do a lot of work in textual analysis now. What you can do is take the context and you can say, try to tell from the words being used what the implicit topic is. And there are different ways to do that.
- 3.106** So you can have topic-specific and country-specific analysis of the frequency a country is mentioned, whether it's mentioned in a positive light or a negative light, with respect to topic one or topic two. So this allows you to think about high frequency, being able to track aspects of countries, political risks and other relevant risks for these companies, to track those on a continuing basis. And so that's what Hassan et al are doing, is they're looking at measures of country risk and trying to see whether they can measure through this textual way, construct graphs. These graphs are constructed from their analysis of country risk.
- 3.107** So, one of these is a measure of country risk based on mentions, the other is related to sentiment. And I have done some studies where I've used similar kinds of measures to see whether they also are reflected in exchange rate changes and equity price index changes for that country. Without going into this too much, this is from that Hassan et al paper. But the point is that, if you're trading in these markets, this is a new tool. What this tool allows you to do, so it is very much without— we don't have time to go through the details. It doesn't matter. The point is that these textual measures can be constructed in real-time.
- 3.108** And by seeing which countries are measured and in what light and with respect to what topic, you actually have a pretty powerful predictor of short-term influences on international investors' interest in the country, relating to risk perceptions, which are reflected in equity prices, in capital outflows or inflows, and also in exchange rates for the country. There's a bunch of studies also that use this kind of thing to track capital outflows and show that social unrest measures coming from these kinds of things are useful for seeing stock price changes, exchange rate changes.

- 3.109** This is opening up a whole new toolkit to thinking about these qualitative institutional factors and being able to map them into not just a long-term understanding of equity performance and international interest, but also predictive tools that you might be able to use actually as a trader or as an investor for trying to identify troubling moments in emerging markets.
- 3.110** You mentioned politics greatly impact emerging markets. Roughly two years ago, Russia was cut off from the global economy, no longer able to access the US dollar. What do you think Russia's five-year economic future looks like?
- 3.111** Well, it's a good question. I mean, it hasn't been cut off from the global economy because it's still been very elevated prices for oil and gas that it can still sell, although it has a more restricted list of countries that will buy from it. Of course, there have been some enforcement actions also taken against Russian individuals and companies. But that's still a bit of a Swiss cheese. Actually, Russia has benefited hugely from the increase in the oil price.
- 3.112** One of the things that some of my friends who were looking at this, I was involved also in the sanctions part of an informal working group of experts on how you would structure sanctions. If you really wanted to structure sanctions so that they weren't a Swiss cheese, you can do it. You can do it by sanctioning any insurance company that insures a Russian vessel, sanctioning China for its assistance to Russia. But if Russia can still export 20% of the physical volume, it's doing so at a very inflated price. So from a revenue standpoint, it actually might be said better off because it's getting similar revenue with losing less of its physical stock.
- 3.113** I don't think we've really had very effective sanctions against Russia. I think it was something that could have been improved if we had been more serious about it. My own view is the best policy, by the way, that would have really had an effect on both Russia and Iran would have been more serious sanctions, especially against Iran, by the way. That would be a related case. But increasing our own supply. There's no reason for the price of oil to have gone up so much or the price of gas to have gone up so much.
- 3.114** If you look at Putin's various adventures in the former Soviet Union, you will see that all three cases happened at times when oil prices were temporarily very high. If we had a policy in the US of exploring oil and gas more liberally and being more willing to increase our supply, I don't think Russia would be in the same position that it's in now. But so the problem is, they haven't been cut off from the global market, and they are still making a lot of money on their exports.
- 3.115** Five years is a long time to project things into the future. I would say that I don't have any reason to believe that everything's going to be solved in Russia, the deeper problems of public governance. The key problem there is, Putin is a strong-arm autocrat. He's probably the richest person in the world. He has one problem. He can't go anywhere. So, given his history of mass murder and individual murders, invading

another country, all the things that he's done, if he left Russia, I can't think of any—countries did take Idi Amin.

- 3.116** They'll take these kinds of individuals sometimes to give them an exit so that the world can move on. But I'm not sure that he would trust any country that would offer him asylum. And that's a problem. He's got nowhere to go. And I don't think he'll be signing up for this course, so I'm not that worried about my accusations. Another question.
- 3.117** So you discussed how, in markets like China, when there is a reversion in their control over the market, what happens with equity. Could you go through what would happen in a reverse environment, where you're going from a free market with well-established rule of law, reverting on that, going more towards a command economy, how that might affect the equity?
- 3.118** So if we're going from a situation where there's no government ownership of firms, is that the scenario? So, no government ownership to one where we're now establishing an autocracy that's going to also nationalize firms?
- 3.119** Sure, or maybe the rule of law is decreasing?
- 3.120** Yeah. It's what I'm guessing you would expect, which is, there is going to be a decline in the value. What's so interesting is these institutions are priced in to equity prices. If you get rid of your institution, whether it's democracy. And what's so interesting is, it's a cluster. These things tend to co-occur. Democracy, personal freedom, economic freedom, but by the way, limited democracy. There is a contradiction between tyrannically populist democracy and individual freedom, because as the foundational thinkers on freedom recognize, democracy could be a threat to freedom, too.
- 3.121** But limited democracy, freedom, property rights, rule of law, those all seem to coexist. So when we take the scores for those, from either Cato or Heritage or whoever is doing them, we find that countries are highly correlated. If you got low on one, you probably have low on others. And that is all difficult to disentangle. If you said to me, what would I think would happen if we got rid of democracy in the US, as opposed to getting rid of the first 10 amendments to the Constitution, as opposed— I don't know how we would try to figure that out.
- 3.122** But we know that what's so important is that all of these institutional attributes are impounded, compounded, whatever the right word is, and present into prices. At the beginning, I mentioned, it's one thing to talk about individual institutional changes that are not always just baked into culture and deep politics, which is what we're doing with these financial institutional changes. It's much harder to talk about the effect on development or aggregate stock prices, of changing your whole system of government, because it happens very, very rarely. So you don't really have a sample from which to make scientific inferences.

- 3.123** We can say, we're going to talk tomorrow when we talk about governance of banks and legal systems, that we're going to see that there's some interesting things that we can tell about institutional changes that come out of the former Soviet Union in East Europe, where that is a big regime change. And spoiler alert, it's a good thing for the banking system, right? When we could end the Soviet Union control over the banking system, which under the Soviet Union in Eastern Europe, the banking system was just an accounting apparatus for state planning.
- 3.124** There was no allocative role of banks. When we change that and then get rule of law coming back, we have a very big effect. The reason we can do that is because we have this simultaneous, many countries changing. What's so interesting is, we have heterogeneity there. That heterogeneity is reflecting deep histories where countries like Hungary and the Czech Republic, prior to the Soviet Union, already were very much participating as democracies and rule-of-law countries in a global order.
- 3.125** And so for them, coming out of the Soviet Union was return to home, whereas other countries, like Bulgaria, never really had developed those kinds of institutions. And so the end of the Soviet Union has a different effect on institutions for countries like Bulgaria or the Czech Republic, because they were different countries institutionally in their deep history. So one of the things that you learn from that episode is that there's, as Deirdre McCloskey would point out, a cultural marker that's even from 50 or 60 or 70 years ago that still matters in terms of what countries can achieve easily in terms of institutional reform. It's one of the interesting things.
- 3.126** But it's very hard to make counterfactual kinds of claims about ending one of these things. At the beginning, that's why I said, I think that those are very controversial because we don't have a lot of good data about them. We do have some studies that, like the Haber and the McCloskey studies, which I think make really valid and interesting points. But also, when I convened them and some other people to have a discussion of this, you wouldn't be surprised to hear that they disagreed.
- 3.127** Whereas in this literature, because we're talking about small ball, things that are plausibly exogenous and that we can identify and then see the causal effects of, we're much more confident. And then we can aggregate it up and say, well, the more positive institutional changes, the more we'd see improvements in governance, stock price increases, more global diversification on the equity side, we'll see on the banking side similar things. But we don't get to, for better or worse, we don't get to observe too many big changes like that.

Lecture 4: Debt and Risk

At a glance — Debt and risk — from equity markets to the institutions of debt.

- 4.1 Welcome back. We're turning now from a discussion of the institutional foundations of equity markets to the institutional foundations of debt markets. When we do that, we turn naturally from a focus from before on both public and private governance problems, especially private governance problems related to the firm, to now talk about almost exclusively public governance problems. Because from the standpoint of debt holders, the things that matter for them, they have fixed claims; they don't have stock holding.
- 4.2 So the things that matter to them are the institutional environment that relates to mainly legal risks and policies that affect the ability of debt to be issued. And so those public governance failures that we talked about, again, are going to reflect the fact that, in most underdeveloped countries, the political structure is fairly authoritarian, autocratic. In these autocracies, there tend to be powerful networks of people who are involved both in the public sphere and the private sphere, and that those networks exercise a lot of control.
- 4.3 One of the themes that we're going to talk about a little bit more in detail this morning is, unfortunately, the stability of those kinds of networks and the way that they tend to eschew things that we might think of as positive institutional reforms, because what's good for the development of the economy and the financial sector is not necessarily good for them. When you look at public governance and you think about the way those networks might not favor, for example, establishing a rule of law that treats everyone equally, well, why not?
- 4.4 Because they don't want to be treated equally, right? They want to be treated favorably. Or wanting to have a system of corruption. Why? Because they're the ones who can reliably get the favors from the corrupt relationships. Or maybe not wanting to have a lot of taxation. Why? Because they're the ones in the unequal environment that have all the money. Then you think, OK, if you don't have effective laws and you have a lot of corruption and you don't have a kind of adequately supplied fisc, from a taxation standpoint, then how is that going to affect debt markets? Well, it's going to affect them in some pretty obvious ways.
- 4.5 First of all, debt capacity of the sovereign is going to be affected. You're going to have, tend to have weak fiscal situations. Those weak fiscal situations manifest themselves in two important ways. One is limited debt capacity, so you can't borrow as much if you don't have as much in the way of fiscal taxation resources. Then also because you have a weak fiscal situation, your borrowing may be limited to borrowing from foreigners in debts denominated in foreign currency. It's sometimes in the literature called original sin.

- 4.6 Why is it called original sin? Because it's whether you like it or not, you have to borrow in someone else's currency. In the United States, we don't borrow in anyone else's currency. In Europe, they don't borrow, in the Eurozone countries, they borrow in euros. Now, those are a currency of a group rather than a single country. And it does, as Greece found out, that can create some problems in terms of default. But when you think about emerging market countries, they're all borrowing huge amounts.
- 4.7 Because they want to access that supply of funds in the global capital market, and because of their own fiscal weakness, foreigners don't necessarily want to hold domestic denominated debt from the emerging markets. Now, they do now more than they used to. Things are changing a little bit in that regard. But still, bond issues by both governments and corporations in emerging markets tend to be denominated in hard currencies. So this limited fiscal situation, again, is manifesting itself in the terms of limited debt capacity overall and also this requirement, this original sin problem, of having to borrow your money in a hard currency often that you don't control.
- 4.8 And that's a big problem, because when we have a fiscal shock in the US, and we did have one recently relating to COVID, and we passed this huge spending bill, I would say it was not warranted, but we passed it. And what happened? We had a big inflation. If you had been holding US government debt, that would have felt like a default because all of a sudden, you're holding this debt, it's denominated in dollars, and the dollar isn't worth as much. The government surprised you by making the dollar worth less than you thought it was going to be worth. And so you got paid back less in real terms.
- 4.9 That sounds like a default, right? It feels like a default if you're holding the debt. But it doesn't have the associated problems of having to actually default on your debt. Whereas if you had borrowed the debt in a hard currency, and you had had a similar kind of fiscal problem, you wouldn't just be able to inflate your way out of it. You'd have to actually default on the debt. So one of the things we're going to talk about is defaults, because part of original sin is creating that. When you have these limitations on public governance, they're going to affect sovereign debt through limited debt capacity and a reliance on dollar-denominated debt issues.
- 4.10 There's another thing that you may not have thought of, which is because the sovereigns are often impecunious, that is lacking resources, they use regulation of the banking system as a way to expand their fiscal hold. I mentioned before, if you can get banks, if you could simply regulate banks to have to lend you money at zero interest, which emerging market countries are still doing, and you make that a significant amount of deposits—let's say you say, you say, "Well, we're going to require banks to have 20% of their deposits in the form of reserves at the central bank, and we're not going to pay any interest on that," what you're doing is you're telling banks they must give you an interest-free loan on a principal amount equal to 20% of their deposits.
- 4.11 That's a tax. In fact, we call that the reserve tax in economics. You don't have to pay them zero interest, if you want to; and by the way, in the US now, banks earn pretty

much market interest, sometimes even a little above market interest, on their reserves. But the typical reality, including in US history, was when you had reserve requirements on banks, you didn't pay interest. And of course, you can do other things in the banking system, too. You can require the banks to lend to your friends. That's also a tax and transfer. You don't have the fiscal resources, so what you do is you tell the bank, who are you taxing?

4.12 Well, you could say, I'm taxing the banks. I'm really taxing the depositors. I'm telling the depositors that they have to contribute. I might limit the interest rate. They can be charged, tell you who to lend to. And of course, what I'm doing effectively is a tax and transfer, taxing the unfavoured group, the depositors, the general citizenry, and transferring to the favored group, the subsidized borrowers. And that system of forcing banks to have to give you interest-free loans, controlling what they can do with their money, we call financial repression, which I talked about briefly yesterday. We're going to talk about it in more detail today.

4.13 But what I want to emphasize is that this financial repression is a fiscal consequence. It's coming from the limited fiscal resources of governments. I'll give you an example of this again from Mexico. We're going to talk also about the way that the safety net, that is, deposit insurance, evolves from a system that almost no one is using in the 1960s to almost everyone is using by the end of the 1990s. One of the things that deposit insurance is sometimes used for is to expand the sovereign's use of the banking system for its own purposes.

4.14 A great example of this is that when the Mexican government decided, after nationalizing the banks in the early 1980s, to reprivatize them in the early 1990s, you could say, well, wait a minute. They took the banks over. They destroyed the balance sheets of the banks. The banks are either worth nothing or actually worth negative in some of the cases, very severely negative amounts of net worth. How are they now, after destroying the banks, going to sell them back to the private sector?

4.15 So first of all, the banks aren't worth anything. Secondly, you just stole the banks from me. You want to sell them back to me. Why would I buy them from you? Because you don't seem like a good partner, right? And the answer is, they got the banks to buy themselves—they got the wealthy people in Mexico to buy the banks again. What was going on there? Well, that was when the government instituted 100% guarantees on all the banks' liabilities. So if the bank is insolvent, but the government guarantees all of its liabilities, all of a sudden, you're willing to buy the bank.

4.16 The other thing they did was they said, "And you don't have to actually bring any of your own capital to invest in the bank. You can have the bank that's insolvent lend you the money to buy it with all of its liabilities guaranteed." And you might say, well, what is the point of that? Why did the Mexican government do that? Because it was already in a very financially constrained situation. It wanted to get a little bit more money. How? By selling the banks. So it needed money. If you go back to the late '70s and the

early '80s, it gets the money by increasing the reserve tax, forcing the banks to give it a very large interest-free loan.

- 4.17** Then when it still needs more money, it grabs the banks. Then when it still needs more money, it privatizes the banks. And you might say, wait, this doesn't make any sense. It does because when they privatize the banks, they use deposit insurance as the way that they got the bankers to be willing to buy them. All three of those steps had to do with the fiscal needs of the government. And then you say, well, this seems like a smart thing. Was there any cost? Well, of course, there was a cost. It was called the crisis of 1995.
- 4.18** That crisis, though, had to be paid for by a different government. One of the other things that you can think about is, when governments are really strapped for cash, they tend to be very myopic. And, of course, in a system where you have, in the Mexican case, elections every six years for president, well, maybe you only care and you're only allowed to have one term. Maybe you only care about the next few years and you're willing to create problems that the future will have to solve.
- 4.19** This is a new way maybe for you to think about when we're talking about institutional limits and problems, we're going to also talk about safety nets. We're going to talk about how they're often created in very vulnerable ways without the proper kinds of regulatory apparatus accompanying the safety net. But instead of finding that puzzling, you should find that completely understandable. Because if the point is to maximize how much resources you're taking and to sell something to someone so that you can get some resources, if you want to create a subsidy, you don't want to create regulations that then undo the subsidy.
- 4.20** As we're trying to puzzle through institutional failures in emerging markets, and we're thinking about banking systems and safety nets, on top of lack of legal systems that make functioning as a bank viable, on top of financial repression, forcing banks to give you their money for free and forcing them to lend to favorite customers, you have safety nets that are being created by governments as part of that partnership with banks to create more resources that they can share with the banks. They're not necessarily trying to create a great financial system. They're trying to create fiscal resources.
- 4.21** So a lot of puzzles about why there's a systematic long-term lack of learning about good financial regulation is understandable if you realize that the political equilibrium may be that they want to use the banking system and its regulations purely for a fiscal purpose. And then you're not so puzzled. I'm not saying that's the case in all cases; it's not. But my point is, you have to have in your thinking mindset this openness to, as you're reading through, if you're getting involved in emerging markets, you have to learn how to read the newspaper. You have to learn what kinds of types there are of governments, what kind of regime you're working with. What is its orientation toward the financial system? What is it trying to achieve?

- 4.22 Or else you'll be completely puzzled by various failures that will seem to you outrageous and dumb. But I can tell you from having been involved in this now for 40 years, I haven't really, looking back on it, I haven't really experienced too many dumb people. People sort of understand what they're doing. This is something for us to take very seriously, that there often are reasons that things are happening that have nothing to do with the stated discussions. Everything, when you see a bank regulation discussion, everything seems to be about, "Oh, we want to create a safe and sound banking system, and we have some technical discussions about that." That's real in some countries. That's not real in other countries. Fiscal matters often trump regulatory policy.
- 4.23 A great book about that recently was written by Viral Acharya, talking about the Indian case. He was a deputy governor of the Reserve Bank of India and really saw—the book is all about that all of the regulatory problems he faced, all of the problems in trying to get the banks to be healthier, ultimately, all of them were fiscal problems, where he couldn't make progress because the government was relying on the banks as fiscal tools. That's still true in India. So if you're interested in the Indian case, Viral's book, I think it came out in 2022, is really an excellent discussion of that.
- 4.24 Now, what I want to do is, having talked a lot about this, this political foundation, I want to kind of show you that this is a very old idea. Then I want to talk about a couple of recent books about the nature of what I'm going to call, following John Wallis, who coined this term, systematic corruption. The point of systematic corruption is that there's often a predictable and long-term unchanging failure to provide what we would consider proper legal and economic institutions for the financial system and for the general economy, and a very strong willingness to provide improper institutions, lacking in good property, well-defined property rights, lacking in rule of law, with lots of corruption, and also with a lot of favoritism.
- 4.25 The literature in this really goes back to Aristotle and Machiavelli in the late medieval period. The term corruption actually, which we now use to define a small thing, that is, when a bureaucrat does a favor for you for, let's say, money, you think of that as corruption. But in this older literature, John Wallis argues that the notion of corruption was much more broad than that. It was about how powerful political actors manipulated the rules of the economic system to favor themselves. And that's what we're talking about as the root of institutional failure.
- 4.26 And so when people, I mentioned Doug North yesterday. There's a group of economic historians, some of the most prominent ones. Doug North has passed, but John Wallis at University of Maryland, Lee Alston, who's recently retired—they really have taken the whole study of this area to a new level of trying to understand how political systems that are employing this strategy of systemic corruption, how those tend to change or can they change. How do they get reform?

- 4.27 And so first, I want to show you some basic facts that I've already mentioned, and that is that how the legal system operates, how much freedom there is to do things in the economy, how much corruption there is, how much rule of law is enforced, they're all very correlated. Let me just define these again for you. Rule of law is the legal system enforcing rules similarly across people, irrespective of who they are, in a predictable way. So non-discriminatory and predictable. That's what rule of law is. Corruption. This is a narrow definition of whether the bureaucracy is corrupt, whether you can pay people for favors.
- 4.28 Judicial efficiency. This is a measure of how well the courts are functioning to adjudicate disputes. It's not just a matter of corruption, it's also a matter of legal favoritism to some interests, and it's also a matter of delay of adjudication. Economic freedom. Are you free to have a go? You may be familiar with de Soto's work coming from Peru, of talking about how licensing of economic activity is often used as a barrier to preserve the favored special group. So lack of economic freedom.
- 4.29 So what do you notice here? What you notice is that, if you're in Zimbabwe, you have the worst rule of law on this list. You have one of the worst corruption scores. You have one of the worst judicial efficiencies, and you have one of the worst positions of economic freedom. Those are all different things. I want to emphasize, they're not the same, but they're all highly correlated because, really, what they're all telling you is that systematic corruption is driving the bus.
- 4.30 If we don't think this way, we're not going to understand why it is that the legal foundations and other foundations of the financial system are often lacking. I want to really emphasize, it's purposeful. And John Wallis has a book called *Leviathan Denied*, and he says something in this book that I think is really interesting. He says, "If you're thinking about who has all the wealth in these societies, it sometimes makes you think the case for reform should be stronger."
- 4.31 That is, if the people holding all the wealth agreed to a reform that would change the regime from systematic corruption to having rule of law, democracy, economic freedom, lack of corruption, efficient judicial systems with well-defined laws, who would benefit the most? They would, right? Because they own the capital. That's the paradox that he poses. The answer, of course, is—and you might have thought that when we were talking about the equity market, too—why wouldn't you, if you're the private individual who's part of this network, but working in the private sector, why wouldn't you benefit from joining the Novo Mercado? We talked about that yesterday. For some of you, you would benefit.
- 4.32 But if changing the rules allows new entrants to come in, that means new competition for you. You don't want that. But more generally, for this public governance problem, if changing the rules makes everyone better off, one of the things you wonder is, as John Wallis claims, the problem is, you can't credibly contract with the future. So if you're somebody like Vladimir Putin or Idi Amin or any of these people who's been engaged in

some pretty horrible practices, it was an amazing thing that in Chile, Pinochet, who had been the one in charge of some pretty awful things, that they actually pardoned him, and he stepped down.

4.33 That doesn't happen very much. It's usually considered pretty hard to step down, allow a new regime to be instituted, and then to have confidence that the new regime isn't going to come after you. So that's the problem. The problem is that this autocratic group, they're living a pretty good life. It's a little bit risky because they might have a revolution or somebody might kill them, and that does happen. But it's not so easy for them to contract their way out of this.

4.34 Now, Lee Alston has written a recent book called *Beliefs, Leadership, and Critical Transitions: The Case of Brazil, 1964 to 2014*, with a few other co-authors. And Lee Alston, in this book, is really focused in on this rare phenomenon of transitions, from this systematic corruption environment to a more favorable institutional environment, and trying to understand how it happens. In this book, I'll just recommend that if you're interested in this, you should read the book, he's focusing in on the Brazilian experience because it's one of the transitions that is most impressive.

4.35 I'm not saying, and they're not saying in the book, that Brazil is perfect now. They're not saying that there isn't some back and forth in terms of institutional regress. There has been in recent years. But what they're noticing is that when Henrique Cardoso, who was a socialist sociology professor, decided to become a reformer, he was very successful at creating a transition that nobody thought was possible. So let me just talk briefly about that.

4.36 Brazil has always been what I would describe, if I were going to characterize it, as a decentralized autocracy. So what I mean by that is, power in Brazil, the national government of Brazil, has never been an extremely powerful central government. This goes back to colonial history, where the coastal settlements were fairly autonomous. It has to do with everything from the way the rivers run in Brazil, the way the coastal waters are, the difference in the vast physical environment, and the difficulty in connecting it. So this was a very scattered political power environment.

4.37 The national government really only was able to exert power over one thing, the chartering of banks. The reason is because you could go into a place and see whether a bank had been created. So you can control the taxation. You can get tax revenues by having an army that can travel around and enforce the rules on the banks. Everybody else will say, "Oh, we had a really bad crop here. Sorry, we can't give you anything." But if you come into town and the bank is operating without a charter from you, you can shut them down. You can take their money because it's in the vault, too, right? Not a good strategy.

4.38 And so the point is, one of the only things that a weakly centralized government can do. It doesn't have a lot of tax resources. It's very fiscally weak. So what do you think it's

going to do? It's going to rely on financial repression and interest-free loans to get their resources. And how do you maximize the value of a loan? You make interest rates higher. How do you make interest rates higher? By having higher inflation. So believe it or not, that's a long story. I'll come back to it a little bit later.

- 4.39** The history of Brazil, Brazil is the most high-inflation country of the last 200 years. I would say nobody, not even in Latin America, is there anybody close. There are some countries that have had high inflation like Brazil, but not for 200 years, such high inflation. And so when you're trying to understand that, you understand that as weak fiscal situation, the national government only has one main source of revenue, and that's the inflation tax. And yes, the inflation tax, by which I mean the reserve tax, augmented more when inflation is higher, has been one of the main sources of government revenue.
- 4.40** That is, the government allows banks to be chartered, has the banks engage in pretty much the function of holding government bonds. This is not true anymore. But prior to 1990, holding government bonds, but then the government gets a huge amount of the value of the bank back by forcing the banks to hold zero-interest reserves. That's the equilibrium for a very long time. When you see what's happening with Cardoso, he comes into a situation after a military dictatorship, after a hyperinflation in a country that's had almost 200 years of just horrible fiscal history, complete lack of financial development.
- 4.41** He says, "I want you to believe in two things." Notice the title of this book, Beliefs, Leadership, and Critical Transitions. He says, "The first thing I want you to believe is that we can change," which is hard to imagine, right? How are you going to change it? By the way, they succeeded in the main thing. What is it? Inflation has been low in Brazil since the 1990s. It has never gotten back up to the levels it was previously. OK, so he said, "Two things I want you to believe. First, that we can do this."
- 4.42** Now, that's very hard because if you're living in a systematically corrupt environment, there's all sorts of stability coming from this, right? First of all, everybody believes you can't get out of corruption. If we all believe we can't, then we won't. You have to first believe somehow that it's possible to jump to a different regime. Then how do you believe that? Because someone exerts leadership, telling you some things that are credible and prioritizes those things. What were they?
- 4.43** He said, "You know from, every abuela can tell you," and of course, I don't know the word in Portuguese, "but every grandmother who's lived through the cycles of 20 or 30 years of bad economic policy can tell you that she doesn't believe what the government says." She doesn't believe what the candidates say. They always say they're going to cure poverty in four years. And he says, "You know that's not going to happen." So when a politician tells you, "I'm coming in and I'm going to cure poverty in four years," you know he's lying.

- 4.44 And what you want to believe is two things. First of all, we need to establish an inclusive society, one that's geared toward not just perpetuating this systematic corruption that favors the few. But we're going to do that by having a long time horizon. We're going to bring inflation down. We're going to reform our fiscal policy. We're going to depend on fiscal reform that makes the fiscal situation sustainable. And he said, "Those are my two policies, inclusion and fiscal sustainability." No more, don't believe anybody who says, I'm going to get you inclusion, I'm going to make you all rich immediately by spending, spending, spending. No, no, no. You know that that hasn't worked. It's never going to work.
- 4.45 And then what's one of the symbols of that is creating a policy that pays mothers to have their children go to school. Now, it takes a couple of decades for that to show itself. So if politicians, typically because of myopia, having to do with the electoral cycle, wouldn't want to do that, that wouldn't be their priority. They'd rather give away some money to some people to make themselves popular. And that's a problem, right? Because that's not part of sustainably increasing your inclusion.
- 4.46 But what he said is, "This is what we're talking about." And that policy became a metaphor for his leadership, for his saying, "You know not to believe what you've been hearing for two centuries. You know that's just a sham. Now, would you go with me into an environment of sustainable and responsible fiscal policy, combined with long-term planning for real inclusion?" Now, it hasn't worked out perfectly, but I agree with the authors of this book that it really was a transformation.
- 4.47 What made it possible? Part of what made it possible was the hyperinflation. I mentioned this yesterday, but it's quite important. Financial crises are often opportunities for someone to exert leadership. It happens very rarely. There has to be a possible consensus. But Cardoso, what was interesting about him is he took the moment of opportunity and he was able to articulate something that did resonate with people as a consensus because their life experience was consonant with what he was saying.
- 4.48 I just want to mention, we find very few of these credible transitions happening, but they do happen occasionally. Of course, progress is uneven after them. You see some ups and downs. But I think it's really interesting to note that in Brazil, they went from two centuries of very high inflation to so far about 30 years of very controlled inflation. We hope that it will persist.
- 4.49 When we're thinking about other markets, I'm going to talk mainly about debt markets. We've talked about equity markets. Before we leave this discussion of problems of systematic corruption, I also want to talk about two other related financial structure issues, and one of them has to do with foreign direct investment. Then I also want to talk a little bit about how systemic corruption in an economy can manifest itself differently in different sectors. We talked a little bit about that yesterday, but I want to emphasize that a little more.

- 4.50 OK, so first of all, foreign direct investment. Corruption, or any of these problems we've talked about, has been shown in empirical work to actually reduce foreign direct investment. People don't want to come to an economy and set up a home there as outsiders who aren't part of the protected network. And so if you have good rule of law and you have less corruption, my friend Shang-Jin Wei showed that getting your corruption level to fall or to rise, let's say, from the very low corrupt environment of Singapore to the highly corrupt environment of Mexico would be like taxing the foreign direct investors who are coming to your country. It would be like increasing the tax they have to pay by 20 percentage points.
- 4.51 So, if you're thinking, how do I attract companies to come to my economy to establish bases there, corruption turns out to be extremely important. Talking about the sectors and how corruption and systematic corruption matters differently for different sectors, remember, yesterday we talked about Russia and contrasting, let's say, oil and gas with mobile telephony. McKinsey used to have a group, and I think they still do it, that studies productivity sector by sector across the whole world, especially across emerging markets. And this is a really interesting and unique function that they have undertaken.
- 4.52 And there's a book that the McKinsey Institute published called The Power of Productivity, which I recommend to you if you're interested in this. So when they go into a country, let's say, like Russia, around the late 1990s, and they're asked to explain why are some sectors so unproductive, they do this pretty much by comparing all of the producers in that sector to producers in other similar countries. In the case of, for example, Russian supermarkets, what McKinsey did was they compared Russia to Poland and Brazil. And they said, why is it that Russia has so little productivity in dry goods sector? So in other words, supermarkets.
- 4.53 Now, this isn't rocket science. We're not talking about sending something to the moon. We're talking about whether you create supermarkets or whether instead you sell everything from kiosks. So guess which is more efficient? It's much more efficient to create supermarkets. So if you're the intermediary grocer, in the US, you get a very small spread. It's not a highly profitable business, but you run a supermarket and you're very competitive. You're able to price things very attractively for consumers because you pass on those efficiency gains. Grocers are very competitive with each other.
- 4.54 Why would a country like Russia decide to not do what Poland and Brazil had done and not have supermarkets? And if you read the McKinsey study, they said, well, the reason is because, when you're systematically corrupt, what that means is that nobody wants to run a big business because, let's say in Russia, when Boris the tax collector is deciding which two places to stop before he goes off to his dacha outside of Moscow, which two places to stop, what he's going to do is go to the big guys, stop there, and just basically extract a big bribe from them because he knows they're big.
- 4.55 And so corruption tends to manifest itself as having a big diseconomy of scale. Creating diseconomies of scale, you don't want to be the big fat supermarket that's attracting

the corrupt civil servant. The other part of that was that it was also hard not to enforce forgery. Everybody wanted forged goods, forged Stolichnaya vodka, forged Calvin Klein and similar designer products. Believe it or not, in emerging markets, forged designer goods are very much favored, because they're much cheaper.

- 4.56 But sometimes if you're doing that in a supermarket, you can get caught, or you can get a big bribe extracted from you not to have to be caught. And since forged goods was a big part of the retail business in Russia, that also favored kiosks because you knew that you could get away with selling forged goods in a kiosk, but not in a supermarket. So in other words, the lack of rule of law and the presence of corruption basically meant that dry goods distribution couldn't be done efficiently. When you're looking at failures, we're going to talk about financial system. But when you're looking at things in the real economy, you'll also find this.
- 4.57 What about the Russian steel industry? Well, the problem there was that there was a corrupt deal to get Gazprom—what's the main input to steel production? Natural gas. If you can politically control Gazprom, that means that you can operate a network of subsidies of giving some local manufacturers of steel free gas. When you're looking at the Russian steel industry, you see there's some steel manufacturers that are just as efficient as those in Poland and Brazil.
- 4.58 But then there's a very large number of them that look like holdovers from the Stalin era. How come they're surviving? They wouldn't survive in Poland or Brazil. How can they survive in Russia? Because they have free natural gas. Why do they have free natural gas? Because it's part of a political deal of the local government with the national government. Why does the local government want to keep those steel companies going? Because they employ a lot of people. I'm not making that up.
- 4.59 That's part of the McKinsey report on—so you go sector by sector and you basically unpack, what is the nature of how systematic corruption in this country is manifested in each sector? And they're not the same stories. They're all different stories. So more broadly, even though our focus is on finance, I want to just remind you that when you're thinking about economic performance, you have to have your mind open to thinking this way.
- 4.60 Now, a final point on this, in economics, we have something called the Coase theorem. I'm going to paraphrase the Coase theorem. It's a very widely discussed phenomenon. It basically says, from the standpoint of economic efficiency, it doesn't really matter where property rights are allocated so long as they're allocated to someone. The Coase theorem is sometimes used as a paradox. That is, well, then why is it that the development doesn't happen, because aren't property rights always established? The answer is, they're not.
- 4.61 In these soft environments where networks are defining effective economic control, property rights are not clearly established, explicitly governed by law, because it's not

in the interest of those in power to create explicit enforced laws. The Coase theorem is, if you're trying to understand the paradox of, well, if the Coase theorem is right, why isn't everyone able to develop? The answer is the Coase theorem is not right; or it's not that it's logically wrong, it's that there are places that choose not to have well-established property rights because it doesn't suit the people who currently have the effective control to establish those property rights. This is important, right? As you're puzzling over finance issues, I want you to also recognize that there's other related problems.

4.62 OK, now let's turn to debt markets. When we're thinking about the banking system in a country, there are two different sets of things that limit the operation of the banking sector. One of them is creditors' rights. So even if you're able to operate your banking system, that is, legally, you are allowed to operate freely, if there aren't good creditors' rights that are enforced, you can't get your money back. How are you going to lend in an environment where the courts are inefficient or the creditors' rights aren't well established?

4.63 We're going to talk about, we're going to really get into the weeds of, what are those creditors' rights? What are those court actions that tend to differentiate well-functioning banking systems from the standpoint of creditors' rights in emerging markets from those that aren't well established? But that's one set of problems, creditors' rights. But the other set of problems is financial repression. Financial repression is just happening because, from a fiscal standpoint, the government needs the banking system to perform the function of funding the state, which means the banking system can't perform the function of funding the private sector. Those are two completely different sets of constraints, and they're both very important. We have to keep that straight.

4.64 Then finally, as I've mentioned already, safety net policies are also a part of these political deals. They can distort bank lending, and they can do so by getting banks to take excessive risks and certain kinds of risks. I'm going to show you in particular that one of the things worldwide that we're seeing in the last couple of decades is that safety net generosity tends to be associated with increased real estate lending, and that's in developed as well as developing economies. I think it's quite plausible that there's an implicit political deal of expanding government safety net protection in terms of directing, through regulation, more credit to go to real estate, which is politically popular.

4.65 I'm just giving you some examples of ways in which, it's really this third thing is also a separate category, although it can be thought of as part of financial repression, part of that fiscal problem is a new dimension, the safety net policy. When we're thinking about the bond market, it's completely different. Let's think about the bond market operating outside. A little bit of historical background here. Prior to the 1990s, the

international bond market for emerging market instruments was really not very well developed. Let me take you back a little bit in time.

- 4.66** International lending from developed countries to emerging market countries, which weren't called emerging market countries so much then, they were called less developed countries, LDCs. That's happening first in the 1970s as part of a process called petrodollar recycling. What was happening there is, as OPEC was throwing up the price of oil and generating huge revenues for itself, it was parking those revenues in the American banking system. The American banking system at that time was actually undergoing something called a disintermediation problem that reflected its own problems, which were related to government policies in the US, that meant that it didn't have a lot of profitable ways of employing its funds within the US.
- 4.67** And so it was looking for profit opportunities outside of the US to, in the words of the time, recycle the petro dollars. So petro dollars are coming from OPEC into American banks, and American banks are lending them mainly to Latin American sovereigns and to very large corporations. So for example, the so-called grupos in Mexico are a great example of that. Grupo Alpha, for example, was a major group, a diversified Mexican conglomerate run by one of the six families, which also had big interest in a bank, one of the big banks in Mexico. So that's very typical.
- 4.68** In the 1970s, the way that the debt was structured was as bank loans from the largest American and some other developed country banks, but mainly American banks, to the sovereigns and to the large corporation conglomerates, not bonds. When the interest rates are elevated in the US in the mid- to late 1970s and early 1980s, we have a problem of repayment. Most of those loans were linked to the London Interbank Offered Rate. So when Paul Volcker jacks up US interest rates, it causes the London Interbank Offered Rate to skyrocket, and the debt service is no longer feasible. That's also happening in a time of recession because the tightening of monetary policy is causing a recession.
- 4.69** So it's the worst possible time to be able to try to repay those loans because the US market is undergoing a recession and your cost of debt service has skyrocketed because of the London Interbank Offered Rate. So you get a series of debt defaults of sovereigns and corporations who had been borrowing these bank loans. It's sometimes called the lost decade, because my first job, by the way, as an economist, was working in the summer at the IMF in 1981 in the middle of all this, and seeing how the US Banks, the US Treasury Department, the International Monetary Fund, and these sovereign borrowers had agreed to a postponement of working out this problem, which took the form of the IMF lent the sovereigns money, which the sovereigns then used to pay a little bit to the American banks.
- 4.70** And everybody pretended that the loans were going to be just fine. We had a phenomenon which is now called evergreening, which is you just keep rolling over and sometimes roll over the principal plus the debt service you haven't paid. So you have

this ballooning principle, right? So it was not sustainable. Everybody knew it wasn't sustainable. But it went on until the Brady Plan in 1989, which resolved this through a final agreement of writing down some of that debt. Most of us think that, in some countries, the debt wasn't written down enough, but it was a combination of writing down the debt and then collateralizing the remaining debt with US treasuries. That was the Brady Plan.

- 4.71 Why am I telling you all this? Because it's useful to understand that the debt markets were alive in the '70s in the form of bank lending. In the '80s, it was this lost decade of suspended access to debt while we resolved the problems of the LDC debt crisis. And then, once it's resolved in 1989 with the Brady Plan, you get a new birth of the international bond market. That's not bank loans, it's international bonds. Now, part of the reason that that was so favorable is, these international bonds were a class of debt that did not default during the 1980s. The reason for that is, it was too small. Why default on something that's trivial?
- 4.72 So people then said, well, it looks like sovereigns are very interested in maintaining their reputations in the international bond market. That was a mischaracterization of the facts. It was really just that the international bond market of the '70s and the '80s was so small that nobody needed to default on it. It wasn't very important. But then what you see is, all of a sudden people have a lot of faith in the international bond market as the new place for sovereigns to raise money and for corporations in emerging markets to raise money.
- 4.73 What is this international bond market? It's governed by the laws of foreign countries, and the debt is issued in foreign currencies. If you're a sovereign borrower or a corporate borrower, you're borrowing the money in a foreign currency, typically, that is not in the local emerging market currency, and your borrowing is enforced under the rules of another country. It could be the UK, it could be the US, it could be Luxembourg. Those are the most commonly used legal jurisdictions.
- 4.74 You might then say to yourself, well, is it true that by putting the legal jurisdiction outside of your country, you create a commitment to rule of law? The answer is yes and no. It is true that judges now in those other countries, you have imported the legal systems for adjudication of your disputes, but you still are located in the emerging market country. Those judges don't have the ability to reach into your country. They certainly can't do anything to the sovereigns of any great consequence. They can seize goods in shipment. They can seize some ships that belong to the sovereigns. They can do that, but really, it's not very important.
- 4.75 Furthermore, you can't really enforce the rules against corporations that are operating in another country. You can do some things that are inconvenient for them. But my main point to you is, this doesn't solve the problem of the lack of good creditors' rights in these countries. What you see is, in fact, the corporate bonds typically, with some exceptions— so, if you're, let's say, an energy company in Brazil, your sovereign yield

during some of this period might be lower than—I'm sorry, your bond yield might be lower than the sovereign yield. But for the most part, sovereign yields are a floor on the corporate yields in these countries.

- 4.76 And what that, I believe that really tells us is that the domestic situations of the countries and their domestic fiscal situations and governance issues are still driving the bus of the amount of risk that there is for all these companies. So their yields, which are reflecting risk, are typically seeing a floor at the sovereign risk. That means that we're all in the same boat. It is true that some corporations that are heavily accessing international markets for things like oil or gold or something like that might be able to get lower yields than the sovereign. But that's the exception, not the rule.
- 4.77 Typically, all the corporations are part of the same economic and legal environment effectively in their country. So I just want to emphasize, the international bond market is giving you access to foreign investors, but it's not a way to change your fundamental risk profile. That's dictated by the country you're operating in. The other thing that comes with accessing this new market, this new bond market, is just like the petrodollar market that I mentioned, you're now going to be subject to cyclical factors at the global level that are related to the dollar.
- 4.78 If you're borrowing in dollars, and most of it's in dollars, or if you're borrowing in euros, you are now subject to the global demand shocks that are related to that market. And we're going to talk about that as another institutional foundation, because you have to be cognizant of this as a source of variation, a source of risk, if you're now relying on those as part of original sin. So now in that post-1991, we're still in this environment of countries relying on the international bond market. We have had lots of defaults, the biggest one being the Argentine default, and we will go through the history of those defaults a little bit today, too.
- 4.79 OK, so that's an introduction to thinking about debt markets. Now we're going to go through them one at a time. Let's start with banking. First, we're only interested in things that matter for economic development. The reason why we are concerned about these threats to banking system development coming from, to being able to operate a credit system of doing arm's length credit transactions to the private sector. That's what we care about, governed by market factors. The reason we care about that is because there's a huge amount of evidence that this matters for economic development. Having this set of institutions in place is a key factor in promoting economic development.
- 4.80 It makes capital—I have a bunch of references here. I'm just going to say things that are obvious, but we have a huge literature that demonstrates it. It makes capital available to entrepreneurs. Young firms can't access the international bond market. They need banks. Banking systems tend to achieve better allocative efficiency, especially for people who aren't part of well-established political networks. What you see when you have private banks as opposed to government-owned banks, is that you see that the

parties that are able to access that credit are accessing it through their economic viability, not their political connections.

- 4.81** We can see very much that the private banks behave differently from the state-owned banks in that respect. State-owned banks make political decisions. Non-state-owned banks make economic and financial decisions. We see lots of entry of new firms when we create a viable banking system. That's one of the things that we see happening. By the way, that's true, both whether you are developing the domestic system through domestically controlled banks or foreign banks.
- 4.82** One of the things that happened since the 1990s is that foreign banks were allowed to come into emerging markets as brick-and-mortar establishments, setting up important percentages of emerging market banking under foreign control. And what was interesting about that is, while there's some evidence that initially, those banks lent to larger firms because they were the new guys on the block, they needed to lend to more established risks, that freed up capital from the domestic banks to focus more on the smaller firms.
- 4.83** So what we see is, economy-wide, any increase in banking presence is favorable for especially new entry and for smaller firms. And by the way, that entry of the foreign banks was especially good from the standpoint of political economy. Why? The political deals between the sovereigns and the domestic banks aren't the same as the political deals that could form between the sovereigns and the foreign banks. So in particular, can you imagine the Mexican government bailing out Citibank in Mexico? No. That's politically not going to happen.
- 4.84** If it's not politically going to happen, then that means that the nature of the political deal-making between the two is very different. So Citibank isn't going to be part of a corrupt political network. When I was working for the superintendency in Mexico on their banking system in 1999 and 2000, they were really tested in an important way by the two largest banks that were still in Mexico operating as independent, not linked to foreign banks at that time. Bancomer and Banamex announced they wanted to merge with each other. Neither had much net worth. They were both pretty much marginally insolvent.
- 4.85** Not as bad as some of the other ones that had recently been resolved, but they were not doing great. It was a moment of great courage for the government to say that they weren't going to allow that merger to disrupt the acquisitions of those banks that were already underway or about to happen by foreign banks. So, both of those banks were acquired by international global banks. I can tell you from having been part of that, that part of that was a political economy decision of the Mexican government, and I credit them with it, to decide to change the politics of banking once and for all, they hoped, by having foreigners acquire those banks so that the banks would no longer be part of the political economy network of corrupt acting.

- 4.86 And so that that would be a way through the credible commitment not to bail out these foreign banks, that you'd actually create a disciplined credit environment. I want to emphasize, foreign bank entry has been a big positive since the 1990s where it's been allowed. But getting back to our main theme, the point is that we're able to see small firms coming into existence, thriving, growing faster, as a result of making the banking system function for arm's length credit provision.
- 4.87 How bad is it in emerging markets? What I'm graphing here—it may be hard for you to see what the scale is—so the vertical scale is the ratio of private credit, provided by the banking system as a percentage of GDP. So that the top of this graph is 180%. Switzerland is the highest one on the far right. You can see the UK is the dark line that's almost at the highest point. Canada is the second from right dark line, and the US is the third from right dark line. The US is coming in at about 60% average during this period. Canada is coming in at almost 100%, and the UK is coming in actually above 100% of GDP.
- 4.88 The other dark lines in the second group toward the middle of the graph are Brazil and Mexico. You can see that they are much lower. So the ratio of credit to GDP is, even though you might think developing economies would need more bank credit relative to GDP because of the absence of securities markets, and because they have a lot more smaller firms, they're still in the process of economic development, nonetheless, they actually have less. And then you can see, for many of these countries, it's under 20%. 20% is the first height there. You can see all the way to the left. Democratic Republic of Congo is basically invisible. It's essentially zero.
- 4.89 So the average amount of credit provided by the banking system in the poorest countries—so these are grouped into four categories based on how poor they are in terms of income per capita. They basically have nonexistent banking systems. So historically, a banking system in a developing country, we would expect, based on 200 years ago or 100 years ago experience, we expect it to be more like 100% of GDP or more. But many of these countries have banking systems that are averaging 0%, 5%, 10%.
- 4.90 When we talk about a failure of institutions, I want you to understand, this is a colossal failure that prevents new firms, young firms, from coming into existence or being able to grow during their early phases of development. It makes economic development almost impossible. This is a very unhappy picture. What it's telling us is, systematic corruption, which is reflecting both the problem of lack of creditors' rights and also the fiscal use of regulation of the banking system, is undermining the ability of most developing countries to have effective private provision of credit. This is not a small thing. This is a fundamental impediment to economic development.
- 4.91 I'm going to show you that by showing you a bunch of studies of how much this matters for economic development. The discussion that we're going to talk about is, first of all, looking at this link, first between creditors' rights, which is one of the key factors, and

bank lending, and then how much that is mattering for economic development. We're also later going to talk, after we talk about the banking system, we're going to talk about global bond flows, and then we're going to talk about bank safety nets, which I've already mentioned.

- 4.92** But we're going to first just focus on this important link between bank credit and economic development and the foundations of having creditors' rights. Earlier in this class, I talked about this paper by Rousseau and Sylla that talked about foundational elements of financial institutions. And I just thought I would mention the list that we're usually thinking about, sound public finance being first and foremost. Because if you don't have, although they don't say this explicitly, if you don't have sound public finance, this is what I call Calomiris' law of banking, and I only have one. Here's Calomiris' law of banking.
- 4.93** If the sovereign has financial problems, the banking system will not be permitted to solve any problem other than the sovereign's problems. You won't get a banking system providing credit because the government controls the regulation of the banking system. The government that's struggling with its finances will never let a banking system develop to provide private credit if it doesn't have good finances. It will force the banks to do that.
- 4.94** By the way, what was the Bank of England? The first great chartered bank of the modern era, well, not the first. The Amsterdam Bank was maybe the first in roughly 1600. Bank of England, 1694. What was its purpose? It was a debt swap for the sovereign debt. That was its purpose. It wasn't to make loans. That function only came much later. Why were the British so worried about the sovereign finances? Because they were fighting a war with the French, which lasted until 1815.
- 4.95** After the French were finally defeated over that whole period, which spanned more than a century, after the French were finally defeated in 1815, then England and Wales were allowed to have a banking system that could provide private credit. Prior to that, the Bank of England operated with a monopoly, and it was mainly functioning to keep the sovereign going. So sound public finance is really a precondition, not just for the sovereign's credit, but for having a banking system.
- 4.96** And then stable money. Well, similarly, if you don't have sound public finance, you won't have stable money because inflation is one of the ways that an impecunious sovereign raises money through the reserve tax, as I've already pointed out. The reserve tax on the banking system is higher when the inflation rate is higher. So stable money is also fundamentally a fiscal problem, typically. And then thriving banking. Well, if you've solved the public finance problem, you will be allowed to have a banking system, if you don't have such a systematically corrupt environment that it won't permit it.

- 4.97** So let's think now about an example of that. So, Mexico, a hundred years ago, operating under, prior to 1912, the so-called Porfiriato, which is the dictatorship of Porfiria Díaz, who flees to Paris when the revolution displaces him in 1912. So Díaz is operating a banking system that, by historical standards in Mexico, it was supplying some credit, but it was still very scarce. It was like those examples in the graph I showed you a minute ago of developing countries with very small banking systems.
- 4.98** But why was it small? Because if you're an autocrat setting up this network, you have your group, and your group is getting control of a bank and getting control of the credit coming from the bank. The banks are all providing credit to the favored industrialists. Who's going to be willing to charter a bank in that autocracy? The expropriation risk is very high. And in fact, they were all expropriated when the revolution came in 1912. You knew you were in a very risky environment. How do you get people to invest their equity in a bank like that?
- 4.99** The way you do it is, you make it very profitable. How do you make it profitable? By giving them monopoly rights. How do you give them, what does that mean? It means that you keep credit very scarce. So if you're operating in an autocracy, there's also a logic there, because of the political instability, that you have to make the profitability of the favored few banks very high, and that was part of that equilibrium. So, again, thriving banking may be permitted, but you have to then ask the question, is the political economy going to favor it?
- 4.100** They mentioned central banking, and of course, that's great. You have to have an effective central bank. But again, when you're talking about stable money, there's this problem called fiscal dominance. You might say, well, the US central bank is very effective. The Fed has a great reputation. It's been around for 110 years. Maybe it doesn't have such a great reputation these days, but you might say, well, they do inflation targeting. They're all committed to good central banking. But what happened to the Fed when World War II came along and the national government needed huge amount of support to keep its debt service costs lower? Well, the Fed engineered a high inflation.
- 4.101** What would happen today, if we don't solve the fiscal problem in the US of Medicare and social security, that put us into an unsustainable fiscal situation? The Fed can't stop that inflation. So if it comes to a choice of, the bond market is not willing to fund the US government anymore because they believe that the US government is now at this inflection point of crossing over to the high inflation equilibrium, the Fed's choice is either buy US government debt, engineering the inflation as everyone expects they will, or refuse to buy it, in which case the US government has to default on its dollar-denominated debt.
- 4.102** Well, we know that's not going to happen. Congress can change the law, impeach the head of the Fed, whatever it takes. The point is, I'm going to keep emphasizing how political things, especially things having to do with the sovereign's finances, tend to

really be a fundamental precondition of all of these dimensions that Rousseau and Sylla are talking about. I mentioned yesterday that all of these institutional improvements really reflected incentives of the sovereign, especially the competition of the early modern period, but then also the rise of Germany and Japan, which both saw major financial institutional improvements, had an explicit military imperial purpose.

- 4.103** Basically, seeing that you couldn't be a player in modern warfare, late 19th-century warfare, without a big industrial base. Remember, Bismarck, iron and blood. Don't forget iron, and then later, steel. The need to be a modern economy, not because of what it would do for common people, was absolutely crucial to foundational elements of financial innovation. But by the way, they were also very interested in Germany, very interested in the welfare state. So you might say, well, gee, the German government seemed to really care about people. So they understood very early on the need for a welfare state.
- 4.104** And the view that I would propose to you about this to understand, that if we're asking young men to be willing to run in front of machine gun fire, having a welfare state is actually a really beneficial thing. That's the reason why many places, including Britain, developed a very generous welfare state as part of the change in the technology of warfare. There's a lot of things to understand about when it's in the interest of those in charge to create all sorts of things that are institutional innovations like democracy. Extending the voting franchise happens during this period, too, in Britain.
- 4.105** A little bit about the historical econometrics. Rousseau and Sylla also study these. Now, these countries are, all 17 countries, all very well developed. But during their development phase from 1815 until the early 20th century, they all are developing their financial systems at different times, growing their private credit availability at different times. That allows them to do regression analysis. What they find is that, during these formative years of the first banking systems in now-developed economies, that the formation of the banking system had huge effects on GDP growth.
- 4.106** Our first evidence about the developmental importance of bank credit comes from the histories of the now-developed economies. I'll just summarize what they find. They do this both with instrumental variables, which is, as we learned yesterday, a way to try to create more credible causation analysis. They also do it with just simple covariance, without really getting causation through OLS. What they're finding is, as the banking system is growing, and their proxy for that is broad money to GDP, because broad money is basically bank credit, a proxy for bank credit.
- 4.107** What they find is that over, and this is lumped into five-year periods, so the number 2.5 is telling you that, roughly speaking, every percentage growth in the banking system's credit translates into an annual percentage growth of GDP of about 0.5%. So if you grow banking credit by 1%, you'll grow GDP by 0.5 annually, 0.5%. When you look more at the later period, you don't find the same experience, that's 1945 to 1994. And of

course, the reason for that is that we don't think of—during the development stage, the growth of bank credit provided something that didn't exist before.

- 4.108** The organic growth of the banking system after it's developed is a different story. It's not having the same development implications. I think that's really an important point. There are now lots of other substitutes for firms, and banking systems are not being causally founded as institutions in the way they were during this early period. The crucial economic development phase is this banking system, once they come into existence and are allowed to thrive, play this role during the decades of foundational development of the economy. Then they continue to serve an important function for small firms on an ongoing basis. But in terms of their visible contribution to major phase of early development, it's not as apparent.
- 4.109** Another approach, econometrically, to measuring this contribution comes from Alan Taylor's work. What Alan Taylor did was, he said, let's write down a computable general equilibrium model. This is a structural model of different emerging market economies. In this model, we have a labor participation equation. We have an investment equation. We also have a relationship between the banking sector and investment. So he has a supply of capital through the banking system to investment.
- 4.110** Then he asked the question, if we're looking at the history of the 1980s and the early 1990s, at Asia and Latin America, we know that Asia experienced much more rapid growth annually than Latin America. I've already mentioned that was sometimes called the lost decade for Latin America. How much of that reflected the fact that Asia had more private availability of bank credit, that the Asian emerging market countries had greater availability of bank credit than the Latin American ones?
- 4.111** And in this computable general equilibrium model, what he finds is that that financial depth operating through the investment equation, access of funds for investment purposes, explains 5.4 of the 6.9% difference in the investment rate between Asia and Latin America, almost all of it. That translated into more than one percentage point annual growth difference in GDP between Asia and Latin America. The point is, this was also crucially important during the recent experience of emerging markets.
- 4.112** Another way to get at this comes from the work of King and Levine. What they do in their analysis is they say, suppose that we think that creditors' rights are a crucial part of the banking system. We also believe that creditors rights matter for economic development through making the banking system work better. That's a fairly plausible identification restriction. We're saying creditors' rights don't matter because they cause development out of thin air. They cause development through increasing the supply of bank lending.
- 4.113** If we do that, then we should be able to see that when creditors' rights are better, that means that the supply of loans is greater, and that translates into more economic development. Their paper, which is the last thing we'll talk about today, what they find

is, of course, financial depth varies greatly. This is credit over GDP, which is this variable. It's not here. This is looking just at liquid liabilities, but very similar. In poor countries, you see very little bank credit relative to GDP. That's the point of this summary statistics.

- 4.114** Now they're going to say, do we see that when the legal system is more developed, we get more bank credit to the private sector divided by GDP? That's going to be our core measure, which they call privy. They find that you need to include both the creditors' rights on the book, that is, the legal system's explicit creditors' rights, and we'll go into that in much more detail in a minute. Also, whether there's rule of law in the country that enforces the legal system. What they find is, include both of those variables, which they call creditor and enforce. And you explain more than half of the country variation in financial depth, the ratio of credit to GDP.
- 4.115** Creditors' rights and their enforcement are hugely important. Then they say, well, if I take that predicted value of bank credit coming from these exogenous variables, creditors' rights and their enforcement, does that also explain GDP growth or investment growth or productivity growth? The answer is, hugely. These are showing that the effect of creditors' rights operating through the private supply of credit have enormous effects on GDP growth, plant property and equipment growth, and labor productivity and total productivity growth.
- 4.116** So, what's really, this was a very influential study because it showed, like Alan Taylor's study, using completely different methodology, it showed two things. It showed that credit growth was very important. Exogenous influences on credit growth were very important for economic growth, using the credit system as a way to get that exogeneity. But then also it showed that creditor rights and their enforcement were both hugely important for allowing banking systems to develop. Take a break and come back and finish our discussion of the banking system. Thanks very much.

Lecture 5: Credit and Risk

At a glance — Credit and risk — banking-sector credit and financial depth.

- 5.1 Welcome back. When we left off, we were talking about evidence that banking sector credit, the depth of banking sector credit, let's say relative to GDP, was itself explicable to a large extent in terms of creditors' rights, and furthermore, that it had big effects on economic growth. And, in fact, in my own work that I haven't shown you, I found results that looked very similar to the results that I showed you from Rousseau and Sylla, which is that in many circumstances, when you're sort of building up a banking system and increasing it as a share of GDP, it's sort of, for every 1% of credit, you get kind of an annual GDP growth of an extra half a percent. So it's a very large effect.
- 5.2 But in addition to these effects on growth, we want to ask a few questions about that. How does this affect different sectors and how does it affect different people in the economy? So that's what we're going to turn to now. There's a lot of evidence that sectors are affected differently by availability to external credit. When you have a banking sector that's repressed for one reason or another, either because of the fiscal repression or because of the lack of creditors' rights, if you're in a sector that's more exogenously financial dependent, you suffer more.
- 5.3 Now, what does it mean to be exogenously financial dependent? It means that something about your technology in your sector, maybe for example, a larger minimum efficient scale of operations, means that it would be harder for you to fund yourself just out of retained earnings. So, suppose that your business is, you're making beads, necklaces out of beads, you might be able to, and you're putting those together with your own hands. That's a business that you can operate on a pretty small scale. In fact, scaling it up doesn't really change anything about the production process. It's just, hire more people to have the beads, right?
- 5.4 But then in manufacturing we tend to find higher minimum scale of operation generally. So across different sectors, some sectors tend to be more financially dependent exogenously because, to create a factory, to create an operation, requires an upfront investment that makes it harder to fund out of some sort of organic process of your savings and retained earnings from early operations of your business. And this study by Buera, Kaboski and Shin looks at this element, that is, higher minimum efficient scale of operations firms—do we see that they are more affected by the problems of a scarce credit banking system?
- 5.5 And the answer is, they are. And you see it not just in their growth, but also in their total factor productivity. So, we're going to come back to this a little bit later from a different perspective, but what we find is that, yes, in the sectors, if you're in a bank-scarce, bank credit-scarce environment, the sectors that are differentially affected tend to be those that are most exogenously financial dependent. And you can see from this

abstract of their paper that they're finding lower total factor productivity and that they can attribute this to the distorted allocation of capital, and also entrepreneurial talent, so that you need to be able to pay for talented entrepreneurs, as well as your minimum efficient scale investment in capital.

- 5.6** And so, if you don't have access to credit, you tend to have lesser entrepreneurs and you tend to have poorer investments in capital. We're going to come back to this because what do you think you do when you don't have access to credit in these minimum efficient scale operations? How do you actually proceed? And what I'm going to talk about a little bit later that echoes the findings of this, is that people in developed countries tend to buy used capital. So if you buy used capital, in fact, I might as well just talk about this now because it's so connected to this. I was going to bring it back later, but—
- 5.7** So in a capital-scarce environment, what people will do is, they only have limited resources. They need to start operations. How do you do it? Well, you buy used capital, sometimes very inferior capital. And when you do that, that inherently makes your productivity lower for a couple of reasons. One, just because it doesn't function as well, but secondly, because there's a connection between the cohort date of the capital and the cohort date of the things you're producing. So that, if you buy used capital, you're probably also producing older cohort merchandise. Older cohort merchandise won't command as high a price in the international market.
- 5.8** And so if you are destined to face financing constraints, one of the ways you'll deal with that is, of course, you won't be able to hire great managers. You might not be able to get into business at all. And if you can, you're buying used equipment and you're producing kind of some obsolete merchandise. When I say obsolete, I don't mean it can't be used, but it's just not going to command the same kind of price. If you've ever traveled through Mexico, as I did quite a bit, back in the, after the era where the Volkswagen Bug was no longer produced in the United States, you'd see Volkswagen Bugs everywhere in Mexico because the factory went to Mexico.
- 5.9** So used equipment has a new life in emerging markets because it's cheaper, but that has a cost which is that it's used equipment, it doesn't function as well, breaks down more easily, and it also is producing goods that tend to have lower value. What about the financial development's effect on different people? So, as part of this literature tracking the developmental consequences of inadequate banking systems, the literature, especially by World Bank researchers in the early 2000s, looked at how banking systems that were allowed to grow, how they also created more financial inclusion opportunities.
- 5.10** And so, the brief summary of this literature that I would give you is, growth of banking systems is very much correlated with financial inclusion. So banking, when banking credit's greater, it also means not just that there's more abundant credit, but there are more people involved in the banking system. So there are a couple of studies of this. So

first of all, just to show you some summary statistics, if we were looking at, by continents, different places, these diagrams tell you what the distribution by country is of the depth of the banking system. So we're looking at the percentage of households with a financial account. So this is having a banking account.

- 5.11** And what you can see is that in Europe and Central Asia, the median, which is that green dot, is around 30%, but the maximum goes all the way up to 80%. That's where the height of that black line is. And the 75th percentile is around 65%. The 25th percentile is around 16%. I'm just eyeballing it, but I think that's about right. So that's what this scatter graph is showing. It's showing sort of, by country, what's the difference. And so you can see in sub-Saharan Africa, the maximum is 60%. The 75th percentile is about 30%, a little below 30%. And the minimum is close to zero. That's probably Democratic Republic of Congo, which in my graph before was also kind of close to zero.
- 5.12** So the point of this is, there's a lot of variation across countries in financial inclusion. So when we look at different countries' experiences, we can also use that variation to see, how much does, is greater policies that produce more bank credit also producing more inclusion? Here's a graph of that. So all over the world, what you can see is, on the horizontal axis, we are measuring financial inclusion by the percentage of households that are using financial services. And on the vertical axis we're looking at this privy variable, private credit relative to GDP. And clearly, there's an upward slope.
- 5.13** So it tends to be the case that when you have more credit, you also have more financial inclusion. That's also true of developing countries as a class. This graph includes developed as well. But if you only looked at developing countries, you'd see a similar pattern. Now, from a policy perspective, we can analyze how financial inclusion is also covarying with different kinds of banking system policies. And what we've learned already in this course is that you can kind of see these banking system policies.
- 5.14** So if you were working, as I was, as an outside consultant with the World Bank and the IMF through most of my career, we all kind of understood that these policies were kind of correlated with those rule-of-law, systematic corruption kinds of things, too. And so I'm going to talk about them from that implicit perspective. So, as we look at a list of policies, when we see the policy, how does it co-vary with access to deposits and also access to credit? So we have household analysis of what percentage of households have access to some kind of loan? That's the access to loans. And we also, what percentage have access to deposits? So let's start with banking freedoms.
- 5.15** So if the banking, if you're able to organize a bank, remember, as I pointed out, many places have licensing requirements or chartering requirements in the case of banking that are very restrictive and that that's part of an environment to limit who can be a bank. Not surprisingly, when you have those kinds of policies, you tend to reduce financial inclusion. So they're a very big negative effect both on credit, percentage of households that have access to credit, and percentage with access to bank deposits.

- 5.16** Media freedom, again, not surprisingly, if you're limiting, this is if you're limiting, if you're in a place that limits media freedom, you tend to have less financial inclusion, because that's telling you something about the country, right? Official supervisory power has to do with the latitude of supervisors to decide things about banks. And one of the things that came out of a lot of the World Bank studies is, although this sounds like a productive, useful thing for supervisors to be able to have power to do things, if you believe in beneficial interventions, you might believe that. But it turns out that it tends to have the opposite effect, that the more supervisory power there is, that tends to be used in a way that's, if anything, a negative for financial inclusion.
- 5.17** Market-based supervision. Now what market-based supervision means is that you are using the supervisory function to force banks to be more disciplined. And what you find there is sort of the opposite effect from on financial inclusion. By the way, I should have mentioned, these two columns are the barriers. They're measured in terms of whether there are barriers, not sort of the, the flip side of inclusion, which is why the coefficients have the signs that they do. You may have been wondering about that, if I wasn't clear.
- 5.18** And so market-based supervision has a negative effect on barriers, while efficient supervisory power has a positive effect on barriers. So what does that mean? Well, places that have market-based supervision are places that are more interested in actual regulation on a meritocratic basis. So all of these are really, if you understand where they're coming from, these dimensions of policy are very much correlated with this overall theme of systematic corruption.
- 5.19** Government bank share. Now, what's interesting there is that a greater government bank share is actually a reduction in barriers for deposits, but an increase in barriers for loans. And the reason is that government banks tend to be operating very widely. They have a very wide distribution network and will go places that aren't necessarily profitable to set up depository gathering. But then from a credit standpoint, they're giving the credit away on a preferential political basis. So they're actually increasing barriers when it comes to lending.
- 5.20** Foreign bank share. That's interesting because from an access basis, foreign banks are associated with reduced barriers and creditors' rights, sort of statistically insignificant. And I don't think, I don't understand why that is, but it came out that way in their study. So the point broadly of this evidence is that financial inclusion, not surprisingly, reflects a lot of those same features of policy that we've been talking about.
- 5.21** So what kind of evidence, other evidence, do people have of the importance of creditors' rights? I'll just mention this study. So this is a study that looks at multinational firms and it asks the question, when multinational firms are deciding which countries to borrow in, do they borrow in countries—so they're operating in many different countries, but they have an internal capital market. They can borrow from one country

and transfer the resources to their operations in a different country. So given that, do they tend to borrow in the countries that have stronger creditors' rights or weaker?

- 5.22** Now, the reason I think this study is really useful is, if you listen to what some politicians in Washington say, and I won't name them, but they make it sound like if you weaken creditors' rights, you help borrowers. Yes. I mean, so if we have bankruptcy laws that allow you to default and declare bankruptcy more easily with less cost to yourself, that is a negative for creditors' rights. But why do we often see those policies being implemented? The argument is that they help borrowers.
- 5.23** Well, they may help the borrowers that go bankrupt, but they don't help the borrowers that don't go bankrupt. Because what happens? That means the interest rates go up for them, right, and the terms become less favorable. And, in fact, so what's nice about this study is that it does turn out that firms tend to pick jurisdictions to borrow in that are more creditor-friendly and less debtor-friendly. So, that's a pretty clear indication that so-called helping borrowers doesn't really help them.
- 5.24** I'm very proud of this study because it was done by Sujata Visaria, who was one of my PhD students. And Sujata looked at reforms that are occurring in India. What was interesting about these reforms that we're going to talk, I have another couple of PhD students who did some additional work with this, is that these were reforms that were occurring within the context of existing laws. They weren't adding new laws, but they were looking at ways to improve the operations of the courts. And they created these things called debt tribunal reforms.
- 5.25** And this happened, because this happened province by province, it didn't happen in India all simultaneously. Then what's great about that is, it allows you to go from, to have data about a place that just got the treatment effect, while other places didn't, and then have this scattered in time. As they're reforming the tribunals, we can see how the different practices are affecting the availability of credit. So what Sujata found was huge effects in reduced lags of enforcement in courts. I'll show you some data on that. And furthermore that it reduced the interest rates charged on borrowers by more than 2 percentage points.
- 5.26** So what's interesting is, of course, there's always opportunity for making things better. And in the case of these reforms, they were done kind of as partnering with the private sector to try to figure out, how can we streamline the legal process? And when you see how lengthy these enforcement procedures are, you're going to understand why there was a desire to streamline it. This gives you a list of which provinces are experiencing the reform and which dates.
- 5.27** So let's look at this. So this is a measure of, if you had treatment or didn't have treatment—so the top line are places that, before they got the treatment. And the bottom line, I'm sorry, the top line are those after they got treatment. The bottom line is before treatment. And so what's happening is, we're looking at the probability that

the invoice is paid, it's owed within 180 days. The probability goes from averaging about 0.47 to averaging about 0.63. So you can see that there's a lot of evidence that this is having a big effect on behavior of the borrowers.

- 5.28** What about the actual court adjudications when it goes to court? You can see that there's a—we have the average as the first line and then standard deviation as the second line. So, before the treatment, time to summons, 393 days. After the treatment, 136 days. You can still say that's a long time. Before we're really done, and we've sort of got the arguments starting in the defense and when the defendant is presenting the evidence in the court, yes, it went from almost 3,000 days to 700 days. 700 days is still two years, but 2,900. Like, you're not going to live that long, right? So it's almost complete.
- 5.29** If you think about how hard it is to get paid back, if you have to wait nine years, that's waiting a long time as opposed to two years. So these are very big effects on credit. And this kind of delay in emerging markets is when we're talking about judicial efficiency, I want you to be clear, it's not just a month or two here and there. It's the difference between sometimes nine or 10 years. And when reform happens, a couple.
- 5.30** This is the study by my other two PhD students that looked at the nature of the quality of capital. And what was interesting in the studies, they found that when you were experiencing, so they took advantage also of the debt tribunals and the fact that the debt tribunals increased the supply of credit when they were implemented. And what they found was a major effect on firms' investment. But what was so interesting is, the entirety of the effect of the increased credit supply that came from the credit reforms was reflected in the quality of the capital, rather than just the physical quantity. In other words, it wasn't so much that you bought more factories, you just bought newer ones.
- 5.31** And that was really interesting because it showed how much quality of capital was the dimension in which limited access to capital was expressed. There's another really interesting study that looks at what happened in Pakistan, the banking system, after 9/11. So Pakistan has a very repressed banking system. So, as of 2001, it's a banking system that's not doing very well. Private credit access is very limited. What was interesting, though, is that there's a kind of treatment effect here in exogenous shock, which is 9/11.
- 5.32** So at 9/11, believe it or not, there's a lot of international capital that chooses to go into Pakistan to house itself in the Pakistani banks. Why? Because after 9/11, Pakistan was a politically relatively safe place. They didn't think there'd be like, you know, a nuclear attack on Pakistan, because that was where a lot of the attacks were coming from, right? So the point is, this is a safe place, at least temporarily. And so we see this kind of exogenous, politically, geopolitically motivated movement of deposits.

- 5.33 And the point of the authors was, well, now that we see a big increase in supply, did we see an increase in credit? Now that these banks have more funds, are they using it to supply credit more broadly to a new group of people? And the answer was no. Why? Well, a combination. You could say it's a combination of the network, but you could also say it's a problem of creditors' rights. So what's interesting is, often the binding constraint is not just they don't have deposits, it's that they don't really have much to do with the deposits because of the limitations in creditor rights. That was the conclusion.
- 5.34 What about the reforms that happened after the fall of the Soviet Union? So what was interesting here is that it gave us a first glimpse of thinking about the two categories of creditors' rights. Because what this study was able to do for more than 100 banks that chose to be operating or were already operating in Eastern Europe, or foreign banks that were going into Eastern Europe, it allowed them to see, over the period after the 1990 collapse of communist control in Eastern Europe, there were reforms in creditors' rights implemented in the different countries, some more quickly than others. And reforms had two categories for creditors' rights.
- 5.35 One category is what I would call collateral-related reforms. So these are reforms that are changing the laws governing the disposition of collateral in a loan. By the way, we're going to look at another study focused just in on this question. And the other category is looking at bankruptcy rights. So bankruptcy rights have to do with what happens when you don't repay your loan. The bank has to go through a process. It has to try to enforce this collection, and the firm might declare bankruptcy.
- 5.36 And so, when we think about the sort of two large categories of reform, we can kind of divide them into new rules governing the disposition of collateral, like, oh, can the creditor seize the collateral without having to get a judicial order, just by virtue of the fact that I have a collateral right, or do I have to wait for a judicial order? Are there other limits to what I can do in seizing the collateral? So there are lots of issues about reform of collateral rights, and then there are issues having to do with bankruptcy law. Can the firm declare bankruptcy? What kind of a stay does that put on collections?
- 5.37 So you can see there are two kinds of categories, roughly, of creditors' rights that are the most relevant. And the authors divide these into two groups so that they can see, how do credit supply effects change as a result of creditors' rights changes, and which creditors' rights changes tend to be the most important for creditors' rights? And also, do foreign banks respond differently to domestic banks to these kinds of creditors' rights reforms? And this was also done by one of my former Ph.D. students, so I'm very proud of my Ph.D. students.
- 5.38 And so the basic conclusions of these, I'll take you through the regressions, but that collateral law changes are the ones that have the big effects. Bankruptcy law changes tend not to have big effects. Let me tell you how I see this from my experience in looking at bankruptcy law reforms all over the world. If you change a collateral law, it

has an immediate practical effect that's visible, because most of the reforms to collateral are of the form of allowing creditors to get access to the collateral immediately and more effectively.

- 5.39** Changing a bankruptcy law is kind of hard to predict what the effect is. You change the law, but how will it be enforced? This is as new to the courts when you change these laws as they are to the borrowers and the lenders. And often we know from experience, for example, in Thailand's bankruptcy reforms that happened after 1997, that they were not very effective. And in fact, there are lots of really interesting cases. When I teach this course, as a case course, I have a whole case about how the bankruptcy law reforms in Thailand didn't work because the courts didn't implement them the way they were written.
- 5.40** So, when you implement something like a bankruptcy law, you're relying on how a court's going to enforce something. And it's completely new. We don't really know how the court's going to enforce it. We don't know how it's going to play out. When you change collateral law, you're really changing something very practical and immediate. And in fact, what you're often doing is, you're taking it out of the discretion of the courts and putting it into immediate control of the creditors. And so that's how I interpret their finding.
- 5.41** And I will show you some more evidence about this in a minute. They find that the foreign bank entrants respond more to the legal changes in the existing banks. And I think that part of what's going on there is that, when you're part of, you've been there for a long time and you're dealing with what we call in banking soft information much more, you tend to be, therefore, have a broader sort of set of tools of knowing how to manage your credits' problems. Whereas if you're coming in from abroad, you're really much more reliant on seniority status through collateral and on making your credit decisions based on the enforcement of the collateral.
- 5.42** So this is reflecting in my mind the fact that foreign banks, especially in the first decade that they're entering, tend to be relatively more reliant on hard enforcement rather than soft information. And now I want to turn to a study that I did with some other co-authors looking at a particular multinational bank, which shall remain nameless because these data are confidential. It's a US bank that was operating in more than a dozen emerging markets. And so what we were able to get from that bank, when I say it was operating, I don't mean it was just lending long distance. I mean it had physical presence, brick-and-mortar presence, and it was making loans to medium-sized enterprises, not just to large corporations, small- and medium-sized enterprises.
- 5.43** And what we had access to was something that academics never have access to. We had access to the bank's own internal documents. So the bank was making collateralized loans and it was valuing the collateral, if it had to liquidate it. So you're not getting just some sort of phony number coming from what I value my plant on my books as a borrower. You're getting what the bank, having appraised the collateral,

thinks it's worth under a normal sale of the collateral, liquidating it after having possessed it.

- 5.44 And what we're looking at is, on all these collateralized loans, which were the only kinds of loans pretty much that the bank was making, how much was the bank willing to lend against this collateral? What percentage? And then we were asking the question, when creditors' rights were improved, did that make the percentage of the loan relative to the collateral go up? And the answer was hugely. And then we asked, because we had very granular kind of information about what was going on in these countries, we asked, what were the aspects of the creditors' rights that were most important?
- 5.45 What was the bank really depending on? What made the difference between the countries that they would lend, let's say, I'll make up a number, 50% and the countries they would lend 70 or 75% of the collateral? And that's kind of, that's the order of magnitude of difference. So what was it about the creditors' rights in the countries that could get really a lot more, where the small, medium-sized firms could get a lot more? And so we were able to pin that down to two particular things that were especially important. And what was great about that is, these are two things that really can be implemented very easily. They're really simple changes that could be made.
- 5.46 But you've been in this class long enough to know that that doesn't mean that the changes would be made. It was interesting that, when we prepared this paper, I presented it at the London office of an important multinational, multilateral organization that loved it. And they've been saying, "Oh, we've been trying to push this view. Yeah, this is really helpful to us." They were lawyers, we're economists. We're giving them fuel for their arguments about what kinds of things that they already believed were relatively effective.
- 5.47 But as you know, things that seem very mechanical and very easy to change are not necessarily things that people want to change. So we're going to show that these things have very big consequences. And then it might be puzzling to try to figure out, well, why, of course it's not puzzling to you all because you've learned to think about regulation from a political economy perspective. So, we're going to focus in on, we're going to divide assets into these two groups, which are called immovables and movables. So what's an immovable? An immovable is real estate. By the way, everything else is movable.
- 5.48 And you might say, well, wait a minute, what about machinery? Well, that's movable too. Sometimes in countries where rule of law doesn't work very well, you need to hire guards to be standing guard over your machinery, because over the weekend somebody might come and take the plant equipment away. And that's not an isolated occurrence. And it could be that sometimes this is even done in a corrupt way. So it's very interesting. So, yes, machinery is immovable. So are inventories, so are securities

or cash holdings, and so are accounts receivable. So all of these are assets. But what makes them distinct is, they're not immovable. Real estate is immovable.

- 5.49** So, why are we dividing these? Because we were already aware that collateral laws for these are distinct and that, actually across countries, even countries with fairly poor creditors' rights, collateralizing with immovables is relatively easy. Here's the thing about immovables. Remember when we were looking at the tribunal data, we saw that it could take several years to get your money back. So even if you had a very bad collateralization scheme against real estate, you'll eventually get it back. And if you're eventually getting it back, you've got it.
- 5.50** What if your collateral is inventories? That tends to be materials that are inputs used in production. If you can't get access to it, it's gone. It gets used, gets used or sold in many cases, too. And the same with equipment. So you can't wait, you can wait a few years. And almost all of the emerging market countries in the sample had systems that effectively worked reasonably well to give you, to be able to perfect eventually a collateral interest that was effective in your real estate assets, but not in movable assets. Movable assets are much harder to get to and you need to get to them quickly.
- 5.51** And so we figured to focus in on movable assets as the key dimension where the variation in creditors' rights was most relevant, that is funding machinery investments, accumulation of inventories, accounts receivable. Accounts receivable you get from selling goods to your customers, and you're acting as a kind of, all firms act as kind of little mini banks where they're selling things and then waiting for the 90 days to get paid back. So this is proprietary data from this bank I'm going to call Global Bank here, and it's unique.
- 5.52** And we're going to think about loan supply in terms of the ratio of the loan to the collateral. We're going to look at not just that loan supply consequence, but we're also going to look at how there's variation across sectors. So, one of the themes that we've talked about in our class is how all kinds of institutional failures in the financial sector and the legal sector, political problems, are reflected sometimes in the aggregate growth of the economy or the aggregate credit, but they're also reflected in sectoral differences. And so we're going to also explore that.
- 5.53** How? Well, the UNIDO data tell us the sector that's operating in the economy. And so we can see whether the shape of the credit is conforming to the sector's relative dependence on movables and nonmovables. Let me give you an example. What's the immovable in the production function where we have immovable capital and movable capital? Suppose that we have to be in a mining operation. Well, in a mining operation, it's inherent that immovables, capital, which is the mine itself and the operations of the mine, are inherently and exogenously a larger share of capital. That is, we can pledge the mine.

- 5.54 How about producing airplanes? Well, producing airplanes, immovables are not an important part. Movable assets are much more important. So there's a sector specificity to the exogenous reliance of the sector on movable versus immovables. And so, what that means is, we wanted to see whether it's the case that, when a country has a poor movable collateral law and we see less credit going to firms that are collateralizing their loans for movable assets, do we also see that there's a distortion that makes that country overproduce in the sectors that aren't movables-reliant, to overproduce in the sectors that are immovables-reliant? And the answer is yes, it's a quite big effect.
- 5.55 But what we're finding is that the distortions again are not just reducing the supply of credit, but making people do activities based on the supply of credit so that they're distorting their allocation of production. They're producing within manufacturing, which is what we're looking at, within manufacturing, in the parts of manufacturing that are more movable-dependent. If you're in a country that doesn't have good collateral laws for movables, you do a lot less production in that sector.
- 5.56 So this is all part of showing, what is it about creditors' rights that matters? First of all, does it matter? It matters hugely, which creditors' rights, and what are the effects? Reduced production also in a way that's affecting even the sectoral allocation of production. OK, makes sense. So it used to be that the World Bank, there was a scandal in recent years that unfortunately led to the demise of this data project. But they used to produce something called the Doing Business survey. And the Doing Business survey that was published talked about creditors' rights. And the unpublished version of it gave a breakdown of all of the components of creditors' rights. And so we were able to use that to detect which countries had which creditors' rights practices.
- 5.57 So, for example, one of them might be the existence of a collateral registry for security interests over movable property assets, unified geographically and by asset type, with an electronic database indexed by debtors' names. And if you remember, when I was talking about my experience in Mexico in 1995, that was the thing I was talking about. I was talking about telling them to create that. And what they said at the time was "Hahaha, of course we know that's a good idea, and of course we're not going to do it."
- 5.58 So the point of, what's the point of having a collateral registry? If you don't have a collateral registry—now I know you're going to come out of this and think I'm training you to be criminals—if you don't have a collateral registry, borrowers double-pledge their collateral or triple-pledge it. If you have a collateral registry, then when you're a lender, you can go to the registry and see whether the collateral that's being offered to you has already been pledged. That's a pretty important thing to be able to see. That's going to, by the way, turn out to be one of the two most important dimensions of improving creditors' rights.
- 5.59 Notice, it's not a legal thing. It's the existence of a registry. These were sponsored typically by the central bank and by the main bank regulator. They could be created privately, but they were often sponsored by the central banks to try to improve

creditors' rights. And they were very effective, as we'll see. And the second one on this list is that the creditors are able to get immediate possession over the collateralized assets without waiting for a court ruling. And if you're worried about movable assets, which can disappear, that's a really good right. So I'm telling you in advance, those turn out to be driving the bus.

5.60 Most of these other things are legalistic. Does the creditor have seniority? So if you're a collateral, if you have collateral, you should be paid first. Yes, everybody's got that. The real differentiator in movable rights are these two that I talked about. They're the ones that really tend to be absent when creditors' rights for movable assets aren't being taken seriously. And they have huge effects. What effect do they have? About 20%. So when you create, when you have those two things present, you'll increase your lending relative to collateral by about 20 percentage points. So yes, instead of lending 50%, you'd lend 70.

5.61 That's a big effect. We group these, just for purposes of exposition, into weak law and strong law. And that's just that they have weak along all those dimensions. On average, they're weak. Later we're going to show you, there's those two elements that are really driving the bus. And you kind of group the countries into two points above median and below median in terms of the strength of their legal systems. I'm not going to talk a lot about how we constructed this index, but you can read the paper if you're interested.

5.62 And the thing that you see from the beginning is, if you look at immovable assets, you can see that the difference between weak-law countries and strong-law countries is pretty small. That is, it's about 10 percentage points in terms of collateralization. On average, the difference is, at the bottom it says 0.815 for this ratio of loans to collateral for immovable assets. In the strong-law countries, it's 0.91. But this is not a regression. This is just an average. And it's not a perfect measure, but it's smaller.

5.63 And then when you look at the difference in movable assets, movable assets, it's 83% versus 44%. So that's a very big difference. And if you take the difference in the difference, that is, you take the roughly 40% minus the 10%, you end up with a difference of about 30%. So what we're expecting is that we're going to find very large difference. And then in the regressions we find things that are quite similar to that. This is just the regression specification. What it's doing is, it's allowing this interaction between, we're going to show that specific to movables assets.

5.64 So we're holding constant the immovables and saying, whatever the immovables' credit is, we're going to say that the movables is going to be, after taking that level of credit into account, we're expecting to find that's where the real difference is between the countries, which is what we do find. And that we're expecting that difference is really resulting in the interaction between the fact that it's a movable collateralized loan and the extent of creditors' rights. And that's what this delta coefficient is going to be estimating. So that's the thing we're really interested in, that little delta.

- 5.65** And so what you see is, the number is, for law times movable, that's what we're interested in. That's the delta. It's 0.21. And you can see it's depending on different kinds of things, like whether we use fixed effects for countries and different kinds of things, it still doesn't matter. You get 0.21 no matter what you do. So what that means is that the difference in credit as a percentage of the collateral being offered in a strong-law country goes up by about 21%. What we also did was, we looked at a particular case. So, in Slovakia they changed the law to allow, to improve creditors' rights substantially in 2002.
- 5.66** So what we're able to do is, we're able to see whether what we've estimated is called a cross-country effect because we're looking at differences across countries. And we're finding that the same bank that's lending to a dozen countries in the countries that have strong movables collateral rights and registry, that in those countries the amount of lending is 21% higher, but that's based on a cross-country comparison. Slovakia is a unique case where we can see a within-country difference because they change that, their practice, in the middle of our sample period.
- 5.67** So do we find in Slovakia that there's a similar change? And the answer is, reassuringly, yes. We find that, for movables assets, it goes up by about 21%, so that the within-country variation and the cross-country variation are confirming each other. And that's reassuring. So we do a bunch of Slovakian reform regressions and pre and post. You can see now it's the post times movable and you can see that's about 20%, same as the cross-country.
- 5.68** What about the sectoral effect? And you can remember what we're talking about here is, is it the case that exogenously movables intensive capital industries tend to be more affected and tend to therefore produce less as a share of GDP and as a share of manufacturing, if we don't have good movables collateral laws? And the answer is yes. So the way we get at this is, we have data from the US on the capital intensity. So the idea here is, the research strategy is, you take a country where you know that the collateral laws are just as good for movables as for nonmovables, and the US is that case.
- 5.69** And so then we can see what the sectoral use of movables and immovables is in this particular setting. So, this is what we're going to call the exogenous—sort of think of it as technologically determined—intensity of each sector in movables and nonmovables. That allows us to order sectors into sectors that are exogenously movables-intensive and those that aren't. So what are these? Air transport is exogenously the most movables-intensive. It's on the far left there. Actually, this is immovables intensity that we're defining here. It's the ratio of land and buildings divided by total assets.
- 5.70** So you can see that if you're in the air transport business, or financial intermediation, or water transport or construction, you have a lot of movable assets. But if you're in the real estate-owning business, or you're in the mining business or the tobacco products

business, you have a lot of immovable, exogenously immovable high reliance. OK, so now what we can do is, with that in hand, we can then look at, let's just look at the data first of all. So when we looked at this, what is the share within these manufacturing sectors?

- 5.71** If you're in a weak-law country, what's the average sort of intensity for the weak-law and the strong-law countries? And so what you can see is that in output, the weak-law countries have the number 0.7 and the strong-law countries have a 0.52. So what that means is that just looking at an average, weak-law countries are much more immovable-intensive. Now, that's not very strong evidence. So let's use a regression. And in the regression, what we can do is, we can look much more clearly to see whether there's a sectoral reallocation effect, that when you are in a country that has movable collateral possibility, that you see that you're much, that is specifically linked, other things being held constant, to specializing more in movable assets.
- 5.72** And before I show you the regression, let me show you the Slovakia example. So here, Slovakia, we have the immovable sectoral share divided by the movable sectoral intensive share. And you can see, after 2002, what happens? It falls dramatically from 57% to about 52, 53% over a very short period of time, which is really remarkable, that you see that kind of sectoral reallocation happening as a result of just introducing improvements into the legal system's ability to collateralize and to register collateral interest properly. And then we see that also similarly in the regression results.
- 5.73** So I've already told you what the findings say. But what's really interesting is, this is using an actual database with the kinds of data that we don't usually get, which is allowing us to see what the true liquidation value of collateral is, dividing loans into movables and immovables types, knowing a lot of detail about the countries, and even having an example of the country that changed its behavior.
- 5.74** I want to move on now to talk about international bonds, especially to point out that once you've tied yourself to international bonds denominated in dollars as a result of this original sin problem, basically you are a fiscally weak country, so you have to access —and also, you may have other reasons, like a small banking system, that make you want to be relatively dependent on international markets for your credit, even for your sovereign credit, as well as your corporate credit. So you've decided to tie your trailer to the global bond market.
- 5.75** And what I want to emphasize about this is, this brings with it a cost of a certain kind of cyclical instability, which is inherent to relying on borrowing in another country's currency. So let me show you. Of course, I've already told you about this in the 1970s, because what we had in the 1970s was, when countries borrowed from the banks before the international bond market was relevant, we had this increase in LIBOR that happened during a Fed tightening that drove a lot of countries along with the US recession. That rise in interest rates drove them into default.

- 5.76** But now what I want to do is point out that this is something that is also true for the international bond market. So let's look here at the early 1990s. When long-term interest rates are coming down in the US, that's reflecting monetary policy changes. What happens when long-term interest rates are declining to the volume of international bond issues? Do you see a connection? Yes, pretty big one, right? So notice that those are a mirror image of each other. When interest rates are low, we tend to see international bond issues being higher.
- 5.77** You might say, well, that's pretty obvious. Shouldn't that be true? Actually, it's not so obvious to me as a finance professor why it should be true. And I'm going to show you that it has an explanation, at least in the current era. We think that it has a lot to do with the supply of institutional investors' credit to emerging markets, that when US interest rates are lower, institutional investors are much more willing to supply risk in emerging markets.
- 5.78** I'll tell you a little bit about this literature because it's developed a lot in the US. When risk-free interest rates, that is the ones that monetary policy controls, the Fed funds rate or the repo rate, when those rates are very low, we also tend to find risk premia are very low. So that the first thing to notice is, if you're investing in a portfolio of US securities, when the Fed loosens, and monetary policy is looser and nominal interest rates in the US are falling, that is causing both the yield on risk-free assets to go down, but it's also causing the risk premium on risky assets to go down.
- 5.79** So when nominal interest rates on risk-free assets fall, bond spreads, credit spreads also decline. And by the way, so do required returns on equity so that the risk premium in the equity market also falls cyclically as US risk-free interest rates decline. So at the compensation that you get, if you're an institutional investor investing in US bonds and stocks, the compensation you get is falling, both because the floor rate, the risk-free rate's declining, and also the risk premium is falling.
- 5.80** Now, it isn't obvious to me as a financial economist why that would make, should make, institutional investors decide to take more risks and new risks, but we have this expression in financial markets called search for yield. And what search for yield says is, when you are experiencing lower yields through the decline in the risk-free rate and the decline in the risk premium, you tend to look for ways to boost your yields.
- 5.81** Now that's a behavioral kind of fact or hypothesis. It's not one that's easy to generate in finance theory. So I'm just pointing out, this is something that many people have shown and I'm going to show you some evidence for it myself, but it isn't obvious that it should be true, but it is true, and that's probably what's driving this. So when interest rates fall in the US on dollars, we tend to see suppliers of capital, especially institutional investors, be much more interested in investing in emerging market bonds, both corporate and sovereign, emerging market dollar-denominated bonds.

- 5.82 That is not just true of this period, but let me remind you that we had this little thing called the global financial crisis in 2008. When we had the global financial crisis, the Federal Reserve lowered interest rates and kept them very low for a very long time. In fact, it's only really after 2021 that we started seeing real big increases in those interest rates. So we had a long period from 2008 to 2020 where interest rates were pretty much at their nominal floor of zero for almost that entire period.
- 5.83 Now, if we know that periods of low US interest rates tend to be periods where investors become more interested in finding risks outside the US, what would you expect that would have produced for emerging market bond issuers? I hope the answer in your head is a big boom in the issuance of emerging market debt, both for corporations and for sovereigns. And it did. And what's especially worrisome right now to me, as US interest rates are going up, as those debts, those bonds, which tend to have longer maturity as they mature, the borrowers are going to have to roll those debts over in much higher interest rate bonds, which they might not be able to afford.
- 5.84 Remember the 1970s going into the 1980s? So I think right now we're in a position where there's a lot of concern that sovereigns have borrowed too much. And when they were borrowing, and corporations, and when they were borrowing, the Fed itself was believing, and many macro-economists were arguing in the US, that there was a permanent lowering of real interest rates in the US. A lowering of those rates from the historic average of a real interest rate of about 2% on short-term interest rates averaging about 2% real, which means the nominal interest rate minus the expected rate of inflation.
- 5.85 But many people were arguing two years ago that, and certainly four or five years ago, that there had been a global change, that those rates had fallen to maybe only 0.5%. My view is that that was an exaggeration and that, in fact, currently, the long term is probably much more likely to be around 1 to 1.5% and maybe even higher. But what that means is, people were believing, hearing and believing, that there was a permanent low interest rate environment. So they were really goosing it up in terms of their leverage. And now, as those debts are going to mature and have to be refinanced, they're going to be experiencing much higher interest rates going forward.
- 5.86 There's also a problem of fiscal, potential fiscal dominance in the US, which I alluded to before, which is that we haven't fixed Medicare and social security. And if we have a big run-up in risk, in our inflation risk, before we actually have the inflation, as the markets start to become fearful of that, US interest rates will rise because of inflation risk fear. That will also be a very big negative potentially for emerging market sovereigns and corporations. So I'm currently very worried about this one-two punch of low interest rates for 12 years, which were advertised as part of a permanent lowering of real interest rates, followed by a steep increase in interest rates and probably a permanent realization of higher real interest rates.

- 5.87 So, apart from what we're going to talk about in this paper, I just want to say, I'm extremely concerned about the ongoing risks. So when we talk about the fact that when you tie your wagon to the dollar, you're tying your wagon to a particular dollar-related interest rate risk, which we saw in the LDC debt crisis of the 80s, we saw in this graph about the early 90s, what we're potentially going to see in really a vast and unprecedented scale in the international bond market in potential defaults, both corporate and sovereign defaults, if US interest rates remain very high and if it turns out that I'm right, that real interest rates are going to be higher, and if it turns out that I'm right, that within five to 10 years, unless corrected, there's going to be a major inflation risk fear that's going to be reflected in very high real interest rates on dollar-denominated assets.
- 5.88 So this hooking your wagon to this risk, coming from the original sin problem of the weak fiscal situations of these emerging markets, is really a big risk right now going forward, not just something in the past. In this paper that I'm going to talk about, my co-authors and I wanted to look at a particular phenomenon related to this search for yield in this big growth in debt. And that was that we saw this unusual, not only increase in debt, but increase in particularly large debts. The corporations that didn't seem like they should be borrowing this large amount of debt, given their size, were also participating in very big accumulations of debt. And we wanted to try to understand what was driving this.
- 5.89 And so we're going to now get into the mechanics a little bit of how the international bond market works and how bonds are traded and why some bonds are more liquid than others. And you're going to see that part of the magnification of this potential risk, coming from people borrowing unusually large size debts when they did choose to borrow, reflects the fact that debts of 500 million or more are more liquid than other debts. And that means that they have a much lower yield, which means that, from the issuer's perspective, they're much cheaper.
- 5.90 So when you have, all of a sudden, a lot of foreign investors who are interested in buying debts out of the search for yield, it turns out that they're willing to buy them at much higher prices, lower yields, if the debts are particularly large, because those debts are more liquid. Why are they more liquid? Because they're included, by virtue of their size, they're included in the major market indexes. Debts that are included in the index, because passive funds invest in the index, passive funds therefore want to hold the debts that are included in the index. So those debts, as the funds lose or gain investment size, they tend to trade in those bonds. So those bonds are exogenously more liquid.
- 5.91 And because they're more liquid, they have lower yields. And because they have lower yields, issuers, even issuers that wouldn't normally be buying a \$500-million, be selling a \$500-million debt, choose to do that because the yield's so low. And then they just accumulate a large amount in cash, dollar cash, because they want to take advantage

of the market timing, because they don't know when the Fed might decide to raise rates, the market environment changes, and they won't have that same opportunity of borrowing cheap.

- 5.92** I'm going to show you that all that's true, that it's especially, that what's going on is that corporations that are borrowing from emerging markets in this dollar-denominated market are especially attracted to debts that are \$500 million or bigger because they're included in the important index. They have much lower, it turns out, yields as a result of that inclusion in the index. And many of those borrowers, the ones that are doing, sort of, really borrowing much more than they can use are accumulating as cash.
- 5.93** Now what we don't know from the financial information that we can observe is, we don't know whether they're holding it in dollars or in their local currency, cash. We have reason to believe, through some indirect regression analysis we've done, that a large percentage of it is being held in local currency. And that's a problem because when the dollar interest rates go up, the local currency is also going to depreciate relative to the dollar, which means that they're going to have to pay back this debt, not only with higher interest rates, but in a depreciated—they'll have to pay back an appreciated amount of debt.
- 5.94** And we're finding that a lot of these borrowers are from the non-tradable sector, meaning that they don't have an inherent hedge. So it looks like they're not maintaining their holdings of cash. At least there's some indirect evidence that they're holding a lot of local currency rather than dollars. And it also looks like this is true for non-tradables firms, not just tradables firms. So that means that we're seeing a new kind of risk because they're borrowing a huge amount relative to their operations that's potentially at risk through a rise in interest rates, which will tend to also produce a depreciation on their currencies.
- 5.95** So I'm very scared about what could be happening if I'm right, that over the next several years we're going to see the kinds of global changes that I'm expecting. I'm not Nostradamus, I could easily be wrong. I'm not in the forecasting business. But we have seen these kinds of things before and I think it's certainly within the realm of possibility.
- 5.96** OK, so the key thing, I'm not going to go through all this, is that trading in these, there's a lot of cost to trading the bond market. It's all still done, not all, but mostly done through telephone. You want to sell a bond, you have to find someone who wants to buy it. If you have a bond that's part of the index, it's going to be much easier to find someone who wants to buy it because you know that there's regular trading of the passive funds in these index eligible bonds, unlike the stock market. So in the stock market, how do you get into the S&P 500 or the Dow? You have to be invited. But in the bond market, that's not true. If you issue bonds of a certain magnitude, as long as they're not exotic in their terms of payment features, they are automatically included in the index.

- 5.97** And so you know that you can, simply by virtue of the amount that you choose to borrow, you can get yourself included in the index, and therefore, get the advantage of a lower yield because the bonds you're issuing are more liquid. So there are a couple of indexes. I'm not going to explain all of this, but it's the CEMBI narrow index that has the \$500-million cutoff. There's also a CEMBI that is emerging market corporate. That's what CEM means, corporate emerging market bond index. There's also a broader one, with a cutoff of only \$300 million. Both of those cutoffs seem to matter a little bit, but the 500 matters a lot more.
- 5.98** These graphs really tell the story. So you can see, in 2008, after the problems of 2008, which is a global financial crisis, you see a rebuilding of sovereign debt. And this is just looking at corporate debt. But I could show you the same graphs for sovereign debt in terms of the value of bonds and the number of bonds that exceeds its prior peak in 2006. Remember, 2006 was also when interest rates were very low. From 2002 to 2005, they were unusually low. So that was also driven by the cycle of US monetary policy. So the point is, monetary policy is driving the bus in the magnitudes of these markets. And you can see the number of bonds and the value of bonds is going up.
- 5.99** The gray area here is telling the whole story though. So the gray area is the volume of bonds of a denomination of \$500 million or greater. And the red area is \$300 million or greater. So what you can see is, for bonds that are less than \$300 million, they're actually shrinking in number of issuers and in size, whereas it's the ones that are index relevant that are really accounting for all of the increase. This is basically showing you that there's, relative to prior experience, that is prior to 2008, that the jump in the frequency of issuers at the 500 is unusually large.
- 5.100** So what's happening is, it was always the case that if you were at 500, if you were at 490, you would issue 500. Right? Nobody's issuing 490 because, but you wouldn't if you were at 4, you wouldn't issue at 5. The advantage of the issuance premium of price and reduction discount in yield wasn't so great. And so what this is showing you, with the red line being at that vertical distance at 500 being steeper, is showing you that the effect in terms of the number of people who choose to, who would have been below 500, who choose to be at exactly 500 or above, is the step function is unusually large.
- 5.101** And in fact, here's something really interesting. Suppose we just looked at the total number that are issuing exactly at 500. So if you're issuing exactly at 500, it means you're really stretching to reach that 500. It's not that you need 500. There's a sort of discrete thing that's making you reach up to 500. And what we're finding is, institutional investors' flows of funds into emerging market bond funds looks a lot like the total number of issuances of 500, right? Don't they look pretty similar? Those graphs, there's no logical reason why those need to be the same. The cumulative flows into emerging market debt funds by institutional investors are very much obviously driving the bus of the decision to issue exactly at 500. And that's very consistent with the story that I was telling you.

- 5.102** Also, when we compare developed versus emerging market at the 500, the developed country corporate bonds have the same cutoffs, 500, but we don't see the same behavior. And the reason is because, if you're doing a search for yield because US bonds and US monetary policy rates and everything has unusually low rates, you're not going to search in other developed economies; you're going to be searching in emerging markets to get that little goosing from having riskier investments at higher yield.
- 5.103** So what you can also see here is that, while it was always true that at that 500 cutoff, there was a little bit of a yield discount, that what the red graph shows you is that that yield discount became much larger after 2008, so the premium could be really large. Now I'm going to, unfortunately I'm not going to be able to statistically distinguish very tightly in our estimation of that premium. We're going to find it's statistically significant. But I'm not going to be able to tell you whether it's a number like 0.2%, which is really big, or 0.9%. So that's almost crazy, that you could get like almost a whole percentage point lower yield going from 490 to 500 in your issuance. But we're not able to rule that out.
- 5.104** My own feeling is, it's probably a lot closer to 0.2, just because I just have a prior that says it can't be crazy big, but it's a big yield and it's driving the bus, as you'll see. And again, we don't see, we see something similar happening in developed markets, but it's not as big. So the point is, doing this sort of graphical analysis, you're able to see what you'll see in the regressions more exactly, which is, this is a phenomenon that's pretty much restricted to emerging markets. It shows up both in the quantities, much more happening at 500 and above, but especially at 500, and much bigger discount, because the search for yield that's happening as a result of this persistently low interest rate environment in the US, people know the Fed's lowering interest rates and it's keeping them there for a very long time.
- 5.105** That is a behavioral change that's making people want to reach funds, wanting to reach into emerging markets in an unprecedented way. This is just showing you some nuances in terms of the distribution. And the bottom line point here is that we're seeing the effect of what we're going to call reaching firms, firms that typically would have been too small to reach for a 500 issue in prior years deciding to now reach much higher. So the percentage of small firms doing large issues is going up. It's not just percentage of firms.
- 5.106** We're also finding that this is reflected in the uses of funds, that what you see here is that, for smaller firms, we're seeing that a lot of small firms are exhibiting a behavior that they didn't exhibit before. If you look in the right side of the graph, what that says is that even two years after the offering, firms that would typically not have been borrowing \$500 million and above are, even two years after the offering, they're still holding 81% of the proceeds they raised in the bond market as cash. So they weren't

borrowing for immediate use, they were borrowing for market opportunity coming from this Fed policy.

- 5.107** So I hope that you find this interesting. The point is, we're getting into a little bit of more institutional details. But the point from the emerging market standpoint is, when you play in this field, this field has its own rules. And you're subjecting yourselves to a new instrument called international bonds. And you, the sovereigns and the corporations, are therefore subjecting themselves to different kinds of determinants of supply and different kinds of potential fluctuations with new kinds of risks. It's all those things that we see.
- 5.108** I didn't show you the data on one more thing which I want to mention. The other thing we're expecting to see is, if we looked at the mutual funds in the US, what we're expecting to see is that funds that didn't used to invest in emerging markets are the ones that are driving the bus here. Why? If you're an emerging market fund, you're investing all along. Who's the ones reaching for yield? The ones who were really not involved in emerging markets before and are now deciding to dip their toe into the emerging markets. What we find is that that's true.
- 5.109** And why is it that they're especially willing to pay a higher price for these 500 plus bonds? It's because they're inexperienced in the emerging markets, so they really prize liquidity. The reason why the liquidity discount went up so much is, it's these crossover funds who don't usually invest in emerging markets, who are reaching for yield and are willing to pay a lot for liquidity because of their inexperience and their desire to be able to get out of whatever they're buying. And so we are finding exactly that, too.
- 5.110** So the story is pretty consistent across all of those different pieces. It's time for our break. When we come back, we're going to talk about sovereign debt and about a few other, sovereign debt in the international bond market and about a few other specialized topics.

Lecture 6: Emerging Market Dynamics

At a glance — Emerging market dynamics and sovereign debt risk.

- 6.1** Welcome back. We're turning now to talk about sovereign debt risk, which of course, is a fiscal problem, like a lot of the problems, or at least some aspects of many of the problems we're talking about. I want to start by showing you what it feels like to be tracking an emerging market sovereign that's getting into trouble. The most recent one that I thought I'd share with you is Venezuela in 2018.
- 6.2** When you're tracking this in the market, you're typically watching a couple different investment services that are sending you weekly, or sometimes more frequently, information about what's going on. And I want to really emphasize, the theme of political risk that we've been emphasizing through all of our discussion, just comes out constantly in these kinds of communications. And you learn a lot about, you can learn a lot about the function and structure of these markets by just subscribing to one of these services so that you're kind of getting the weekly news coming from emerging markets.
- 6.3** So, in January 2018, one of my favorite services decides it's time to start talking about Venezuela more. And I'm just going to read you a few sentences from this longer thing. It goes on for several pages of my slides here. "In 2017, Venezuela proved that the region's most dysfunctional government"—that's saying something—"with the longest running recession could still find ways to make the situation worse." So you can see there's some sort of gallows humor whenever you're tracking emerging markets.
- 6.4** "The political scene will continue to dominate 2018. The PSUV's stance of hostility towards the private sector, scapegoating for mismanagement and rampant seigniorage-fueled inflation." Seigniorage-fueled inflation means using the reserve tax that I talked about. And you notice seigniorage-fueled inflation is all about trying to make forced revenue from inflation coming from your reserve tax. "Scapegoating for mismanagement, rampant seigniorage-fueled inflation will all continue in 2018. This will result in the fifth straight year of deep economic contraction." So we're seeing financial repression happening in real time increasing during this discussion, on the banking side too.
- 6.5** "Venezuela's problems stem from chronic mismanagement with economic policy rife with price controls, inconsistent, sometimes arbitrary enforcement of laws, nepotism and cronyism, and apparent lack of understanding of basic market principles." How bad does it get? I'll show you a graph. "But Venezuela's benchmark 2027 US dollar sovereign bond is yielding an astounding 40.7% at the time of writing, although sovereign and quasi-sovereign bond payments have begun experiencing delays, triggering a CDS payout and a default designation by ratings agencies. So far, the Venezuelan government continues to service the debt."

- 6.6** So this is sort of the beginning. You can see they're going into the beginning of default. What you see happening as things start to really go south, you see that who's holding the debt changes. So normal bond funds were holding the debt until this transition moment. And then the debt gets sold to specialists who are sometimes called vulture investors, who specialize in the negotiating with a defaulted sovereign creditor, a debtor. So you're, that's an interesting aspect, too. So one group of people, you know, different people have different training. Some people are trained to construct and maintain diversified portfolios where they vary the country weighting of their portfolio based on political risks and other things.
- 6.7** But then other people are trained to do the hard work of negotiating with defaulted debt issuers to try to—and sometimes that can go on for more than 10 years, more than 20 years. And so you can see here they're talking about how it's now, if you read the bottom here, it says that "our core scenario is that Venezuela will likely eke out enough oil revenue in 2018 to continue making late payments. But we believe that legal complexity will and should ward off all but a handful of specialized distressed debt investors." So it's now moved to a new debt class with a new set of specialized investors.
- 6.8** This is a graph of sort of what that looks like, going from October of 2017 into January of 2018. So you can see, yields are already very high in 2017, and then they skyrocket on some of them. Not surprisingly, this is a sovereign problem, and it's going to be reflected not just in the sovereign yields on dollar-denominated debt, but also in the exchange rate. One of the most predictable things is that because it's reflecting a fiscal problem, the government will try to increase the inflation tax before the sovereign default.
- 6.9** So you'll see in an analysis I'm going to show you in a minute that when people are looking at the scoring of sovereign debts in regression analysis, inflation has a lot of predictive value, inflation in the country, for the sovereign debt rating. And that's because it's sort of the canary in the coal mine, that before we get explicit sovereign default, you were usually seeing some extreme policy changes to try to make, print money very dramatically, try to increase the taxation of the banks through forced reserves.
- 6.10** In this case, the Venezuelan government is also announcing that the banks should be more patriotic and buy more government debt and help resolve the problem. And of course, we're your regulator. You'd better be patriotic or else we can do things to you. So that's how it feels as you're watching this happen. It's not a coincidence that since Venezuela's main source of external revenue is coming from the oil market, that there was an oil market downturn that's happening in 2015. And you can see it persistently very low compared to prior years, oil prices in a couple years preceding that.
- 6.11** And we could think of that as contributing to the fiscal problem that the government's facing. It also, because it's a sort of outlaw regime, it also has serious problems in terms

of the age of its infrastructure in the oil industry, a lot of not-well-functioning. So it's not just the price, but it's also things that are happening that are specific to Venezuela. So when we talk about, and when these analysts talk about, long-term deterioration and dysfunction in the public governance, that sort of thing turns out to be very relevant for them.

- 6.12** Another thing that's worth mentioning is, and now we'll turn to talk about the pricing of this, that the market is, and I'm going to go back all the way to the 1990s here, that the market has a way to look at all the factors of the country, kind of divide them into things you might call capacity to pay or willingness to pay. Now they're not very easily divisible because, of course, you could be more willing to pay. You could say that some people, creditors say, it's just that you're not willing to pay. And countries say, well, if paying you means that I have to cancel all welfare payments to the population, that's more of a capacity to pay problem, not a willingness to pay. So there's not a sharp distinction between the two.
- 6.13** But when I think about it from a regression standpoint, let's turn to this. This is a study that was done, I think in the early 1990s. So this is, we're just going to focus on the first column here. And what these researchers did was, they said, let's look at all the factors that go into where the credit rating is of a sovereign. So some of that is GNP per capita. So if you have more GNP per capita, holding constant everything else, you'll have a better sovereign rating. If you have more GDP growth per capita, holding everything else constant, you'll have a better sovereign rating.
- 6.14** And when you look at a regression coefficient like this, an ordinary least squares regression, the correct interpretation is, this is telling you the partial covariance of the rating with this variable, holding all the other things in the regression constant. So when we say that holding GNP per capita per constant—I'm sorry, when we say GNP per capita and GDP growth are both favorable in terms of ratings, it's holding all the other factors in this regression constant. What are they? Well, the amount of debt that's issued is here, total external debt. And you'll see that variable has a very negative coefficient. The more promises you make, the more promises you will likely break. Right?
- 6.15** So holding the amount of debt constant but holding your GNP per capita and your GDP growth, more debt makes your rating worse. Well, that makes sense, right? Inflation is very negative. I already mentioned why. Inflation is an indicator of times of fiscal stress. So if you're having higher inflation, people in the sovereign debt market understand that that puts them at risk. And so the sovereign debt rating will deteriorate as a result of that.
- 6.16** What's interesting is, external balance and fiscal balance are not really included here as significant variables. You might say, well, I expect the fiscal balance, whether we're running deficits, to be very important and I expect the external balance, whether we are, have a big current account deficit, to be important. And you'll hear that a lot. The

reason that they don't come in is a few things. First of all, they're both highly endogenous variables. So the fiscal deficit is itself largely reflecting momentary cyclical changes, not just long-term policies that matter for creditworthiness.

- 6.17** And secondly, it's often mismeasured. So often there's accounting fraud, effectively. I could talk about many examples of that. So fiscal balance is not a very reliable indicator of fiscal stress, both because it's highly cyclically sensitive and because it often, it doesn't really count the numbers correctly. External balance is a highly endogenous variable. Am I running a current account deficit because I've just done a reform, my exports are growing really fast, and everybody in the world wants to lend money to my country? Because that can make your current account deficit very big.
- 6.18** Or am I running it because exports are shrinking, and imports haven't changed? So if I'm running it because I just did a major reform, so it's Argentina in 1992, big current account deficit, I'm importing machinery. Everybody wants to lend me money so I can import machinery. My exports are growing, have increased their growth. So you can't really, I don't think, always reliably tell. So external, the current position of your external balance doesn't come in as very important. Highly endogenous variable. Ambiguous meanings.
- 6.19** Industrial. Well, what that's telling us is, are you a fairly diversified economy? Because emerging markets tend to be the commodity producers. So if you also have an industrial base, it's putting you in a particular category. And then finally, default. Notice, very big negative number. Default is telling me whether in the past you've had defaults. So it's, you know, if you're Argentina, we know that you're riskier. Even though you've done this reform and you're making lots of promises and everything looks very good, we know you're riskier because you've had a history of having defaults over many, many decades.
- 6.20** So the market doesn't forget. I'm going to get into that in a little more detail, but it shows up in your rating after controlling for GDP per capita, GDP growth, the amount of outstanding debt, right? So it's an incremental effect after controlling for all that. One thing that's special about emerging markets compared to developed economies is, the ratio of government debt to GDP takes leaps. So if you're analyzing Germany's ratio of government debt to GDP, or the US, it doesn't take leaps. And the reason is, because the things that make emerging market debt to GDP ratios take leaps aren't present in developed economies.
- 6.21** So what makes leaps happen? Well, first of all, suppose you have a lot of debt that's in the form of dollar-denominated debt, and all of a sudden your exchange rate depreciates. Then the dollars relative to GDP go way up. So because you're borrowing in hard currency, you have this inherent risk to have leaps in your debt, that people who borrow, countries that borrow in their own currency, don't experience. Secondly, the other thing that makes leaps happen are banking crises where, when the banking system has an explosion, this sort of deferred maintenance, and then all of a sudden

we find out that there's a huge amount of bad loans in the banking system. How big is the cost of that?

- 6.22** The median of the hundred countries that had a banking crisis, according to the IMF database, which is, I think a good database, the median in terms of the cost of resolving it, because governments nowadays bail out failed banks and pay the depositors of those failed banks, avoid depositor loss typically, not always, but typically. And so that means that the resolution cost becomes a fiscal cost of the government, the median resolution cost for the 100, which is itself an astronomical figure. So let me just back up.
- 6.23** We're living through a global pandemic of banking crises, the likes of which we've never seen before in terms of the number of them since 1980, more than 100 now. And the median resolution cost of those is 16% of GDP. To give you a sense of how big that is, for the US during the Great Depression, the negative net worth of failed banks was about 3% of GDP. I don't know of any historical episode—there were only two I know of that were 10% of GDP all around the world from 1873 to 1914. The median is 16%.
- 6.24** But then if you take out the developed economies, that median goes up. So it's the typical experience for a developing economy is to have a resolution cost of 20 to 30% of GDP. Indonesia, it was double that. So when you think about, momentarily, those tend to also coincide with big exchange rate depreciations. So what we see is big leaps. This is from a study by Reinhart and Rogoff, and they wrote a book called *This Time Is Different*, all about how people are often in denial about crises. And what they're seeing here is that the historical average for these big bank, post-banking crises, the historical average is that there's an 86% increase in the sovereign debt.
- 6.25** So if you think of that relative to GDP, maybe before the crisis you thought your debt to GDP ratio was 40 or 50%. Didn't sound too bad. But then the crisis happens, you've had a big exchange rate depreciation, so your sovereign debt denominated in dollars just went way up relative to GDP, and you've had a big resolution cost. And so your sovereign debt relative to GDP has doubled. So this is a risk and a problem relating, so the point I'm making is, the fiscal weakness in the US, when we had our financial crisis, it was in 2008, or if you look at the, in Europe, because those are developed economies, they have much bigger GDP, much smaller ratio of financial system to GDP in terms of the risks.
- 6.26** And so what's happening there is, the cost, fiscal cost is a couple percentage points of GDP and there's no exchange rate depreciation magnifying all of that. But in emerging markets, it's a very different story. And so it's not just that they are weak fiscally, but because of a combination of the fragility of their banking systems and the denomination of their sovereign debt in dollars, that weak position is compounded by those additional risks of big jumps in their debt ratio.

- 6.27** Now, an interesting question to study is, we've already seen that the sovereign debts reflect the institutional memory of the market, knowing that you've defaulted before. How long do you get punished for a default? So what typically happens in a default is, you default, you start negotiating with your creditors. It can take, depending on the country, a few months or many years to resolve it. The defaults are also of different characters. Sometimes countries just default on their debt service. They just postpone paying debt, but they're able to, in the renegotiation, pay off a somewhat diminished interest rate, but without any principal reductions. Sometimes they make big principal reductions, 50%, 80%. So defaults aren't all the same.
- 6.28** Sometimes they involve just a short period of a reworking of the interest rate. Sometimes they revolve a much longer period and a major sort of reduction of principal. And so one of the questions is, how does the market punish haircuts? And what we're going to see is the market punishes haircuts depending on how big the haircut is. I'm sorry, haircut means how much you write down your debt. And what we're also seeing is, in recent years, haircuts have gotten bigger. That is, as the amount of debt has trended up, that also, when a default happens, it tends to incur bigger haircuts. And that also means bigger penalties.
- 6.29** What's a penalty? A penalty has two different pieces to it. One of them is how long before you can issue anything in the sovereign debt market. So, notice Argentina defaults in 2002. The sovereign debt market is still not taking new Argentine debt placements in the international bond market. So sovereign defaults can be very long-lived. And then when you are coming back, what's the yield premium relative to what we would expect or what other countries are paying? How much?
- 6.30** So in a regression analysis, Crucis and Trebis are going to give us two kinds of penalties. They're going to tell us how, as a function of the haircut, you got penalized by longer delay in being able to get back to the market and a larger yield requirement once you did come back to the market. And this is a path-breaking study because it really was able to systematically analyze and quantify those two penalties. And so the answer is, the penalties are big. So the market does—there's a big literature in economics theoretically trying to figure out whether, can the market penalize you? How can the market penalize sovereigns?
- 6.31** And this paper sort of resolves it. It says the market remembers, just like we saw in our regression analysis, that the market remembers in terms of its rating for the debt, the market remembers in terms of yield, which is, of course, related to rating, and also in terms of delay in getting back. So this graph gives you a sense of how big the debt offerings are. And so the size of the debt offering is indicated by the size of the circle. So you can see that there are some—this is in deal volume. You can see there are some deal volumes in the '80s that are pretty large, but they do tend to trend higher over time.

- 6.32 This gives you a sense of, in the restructurings, blue is telling you that it's a restructuring of debt servicing without principal reduction, whereas the red dot is telling you that there's a principal reduction. So what you can see is, as the debt offerings are getting bigger over time, when the defaults happen, they tend to involve bigger write-downs, which include principal reduction. And so the authors are going to take advantage of this differentiation to say, does the market penalize you differently if you do a smaller haircut?
- 6.33 So you might think it would make sense, because if you default, if you're trying very hard to minimize your penalty on creditors, then you sort of try just to do a rescheduling without writing down principle. And why would you work so hard at that? Because presumably you're hoping that people remember that you did a small haircut. And so what they show is, this is kind of the point of their empirical exercise, so let's just, for sake of convenience, divide the haircuts into greater than 37% of present value or less than 37% of present value of the debt.
- 6.34 And what you can see is that the post-restructuring spreads, that is, once the restructuring happens, the spread on the debt is much greater for the bigger haircuts. In other words, the yield penalty is much higher. It's higher by a couple hundred basis points, 2 percentage points. So, everyone's not treated the same because you default. If you default and you keep the haircut low, it has a very favorable effect on your post-restructuring yield.
- 6.35 And then they say, how long does it take you to get back into being able to issue in the market? When the haircut's less than 30%, you typically are readmitted to do new issues in about—so the probability of remaining excluded is only 25% about three years after the offering. So there's a three-quarters chance that, within three years of your default, you're going to be reissuing. But if your haircut's greater than 60%, there is a 50% chance that you won't be back in the market even after a decade. So then all defaults are not equal in terms of the way they're penalized. And this is just a fact paper that allowed us to see that. So markets definitely keep track of what you've done in the past and they penalize you.
- 6.36 One of the issues that has come up in this emerging market literature is, given especially the risks associated with foreigners changing their level of interest in your economy, deciding, as we saw, sometimes capital inflows are positive, sometimes they're negative—how do countries, emerging markets, try to manage those risks and how should we think about it? So I just wanted to talk about that briefly, too. We've already seen, from looking at our equity markets, and I could have also spent some time on foreign direct investment, if you look at foreign direct investment and equity, and I would say the same about long-term bonds, the value of being able to access capital is great.
- 6.37 And emerging market issuers in the equity market, of course, are taking great advantage of that. The concern people have is with a phenomenon called sudden stops.

And what sudden stops are all about is big reversals in capital inflows as a fraction of the size of your economy that might be very disruptive, that cause typically big depreciations of the exchange rate, and that basically withdraw savings, which can lead to a collapse of investment. So let's look at, first of all, what are these things? So from this seminal paper by Calvo and Reinhart in 2000, on sudden stops, you can see that there are lots of reversals here that are 10 or 20% of GDP. So what is a reversal?

- 6.38** If it was 15% and then it falls, it becomes negative 5% of GDP, that's a 20% reversal. Or if it's positive 10% and it becomes negative 10%, that's a 20% reversal. So some of these are pretty big reversals. As a fraction of GDP, you can see Thailand in 1997, it's a 26% reversal. And you can think of that as withdrawing from investable savings 26% of GDP, because you were getting the positive, now you're actually getting a negative, that's withdrawing funds. So you can see why this is a concern.
- 6.39** But what I want to emphasize is, as we talked about before when we talked about equity, you can't reverse equity. Once you've invested in equity, the firm doesn't give it back, doesn't give you the money back, when you want to withdraw it. Foreign direct investment similarly is a long-term investment. Once you've built the factory, yes, it's true that you can repatriate funds and you can exit. Sometimes that happens, but it's not, it's not what the sudden stops phenomenon is about.
- 6.40** Similarly, long-term debt is not a sudden stop phenomenon. The debt is issued, it matures many years in the future. The creditor can't ask for the money back. So what is the sudden stop phenomenon? It's short-term dollar-denominated debt, typically that's coming from developed country banks to emerging market country banks. And that is the toxic short-term dollar-denominated debt that's really driving a lot of these stories. It's risk money, it's money that's coming—so you might say, well, why would anyone lend that way instead of equity, foreign direct investment, or long-term bonds? And the answer is, it's opportunistic short-term investments that have to do with something you might call a hot money market.
- 6.41** And especially when the recipient banking systems are in trouble, they will bid up the price to try to get some of that hot money. And then the banks in the developed countries that were lending it, were lending it because they were getting high interest rates for doing something very risky. Often those loans ended up being a big part of the resolution cost of the banking crisis, because that meant you had to pay back the developed country banks. And of course, the developed country banks have secretaries of the treasury or ministers of finance who are sitting at the IMF because they run the IMF. I don't mean it metaphorically, I mean literally, they run the IMF.
- 6.42** And so if their banks have lent, let's say, a lot of money to Chile, they will make sure that the IMF lends the sovereign the dollars so that they can bail out the banks with the dollars to then send back to the developed country banks. We call that a moral hazard problem because then the developed country banks, of course, know they're going to

be made whole by their own ministers of finance through this little circuit of IMF money going to the sovereigns, going back to the developed country.

- 6.43** So when you think about the sudden stop problem, it's really a very specific kind of capital flow problem related to hot money markets of interbank short-term dollar-denominated lending, which is magnified in its moral hazard problem by the fact that the IMF participates in the exposed bailout. Now, knowing that, that's not the only reason why the hot money will go there. And I'm going to talk about another one in just a minute, because we know there's also search for yield, and the banks can themselves be part of, the developed country banks can themselves be part of a search for yield phenomenon. And I'll talk about that in the case of Colombia. That's coming up just in a minute.
- 6.44** But what I really want to emphasize is, when we look at the capital flows even during the 1990s, these crises that are happening in Mexico, with some spillover effects in 1994-95, and again in '97 and '98 in Asia and Russia, what you can see is portfolio investment on net is still positive into emerging markets from developed countries in 1994, 1995, 1996, '97, '98. Portfolio investment, that means equity and long-term bonds, foreign direct investment, very positive. Where it gets negative is, what? The short-term interbank dollar-denominated lending.
- 6.45** And so the answer, of course, is, "Doctor, it hurts when I do that. Stop doing that." And a well-governed, well-regulated banking system is one that's going to try to discourage those kinds of capital flows from happening. But of course, if your banking system's insolvent and your banks are taking big bets and they need some form of dollars, they get desperate and they start borrowing that. And then when they explode it, it just makes the resolution costs all the bigger.
- 6.46** So one of the things that has happened is, well-functioning emerging market regulatory systems for banks have developed this understanding and they tried to penalize these. And the way they penalize them is with special reserve requirements on hot money flows. So that's sometimes called the encaje. So if your banking system gets a lot of short-term dollar-denominated inflows, then you're required to hold those dollars in reserve, maybe 20%, 30%, it can vary, 40%, as a way to discourage the inflows of making you have to hold them as cash.
- 6.47** And I guess another way to think about this is that the financial crises people often implicate the capital flows, but the capital flows are really reflecting the fact that you have bad prudential regulation in the first place that's creating these insolvent banks, like in Mexico in 1994 or in East Asia in 1997, who then go to the developed country banks because they're desperate for funds, they need those dollars, they get them, and then their bet doesn't pay off because they were too far gone to begin with. And then that just increases resolution costs.

- 6.48** You might say, well, why do they do it? Well, if you already are insolvent and you're the bank manager and the equity holder in the bank and you're already insolvent, what have you got to lose? Nothing. So you might as well double down on your bet and hope that somehow things turn around. And that's the kind of mentality that drives a lot of these hot money flows. So, emerging markets, I would say, on balance, since the 1990s, have been moving in a positive direction.
- 6.49** And this is my bright sort of spot in this course. There does seem to be a recognition that these things, as I told you, the average cost for these emerging markets of many of these catastrophes, which are concentrated especially in the 1990s and the early 2000s, are so large that they become transformational events and very disruptive events for the economy. And so you're starting to see countries as they progress, starting to take their regulation more seriously.
- 6.50** Let me give you a case. Now, it's very sad to talk about Colombia nowadays because it's become a narco state and it's just really imploded. I mean, it's been a very hard thing to watch. Similar thing in Mexico, to a lesser extent in Brazil. A lot of countries in Latin America right now are just really in terrible shape from a political economy standpoint. And it's very hard to understand it. As of, let's say 2002, Colombia was just doing great. As of that date, Mexico was also doing quite well. They had come out of long histories of institutional backwardness and depressing stories of one-party rule. And they were, Mexico was looking much better.
- 6.51** So everything, if you, you know, as of 2002, 2004, 2006, even 2008, I was really a Latin American optimist. And strangely, so in the period I'm talking about here in Colombia, 2006 to 2008, Colombia is really at the top of its game. And its central bank is run in a very smart way by my friend José Uribe. And he is really the pioneer of some of this anti-hot money policy and showing how well it can be eliminated by the right kinds of regulations. So even though Colombia right now has kind of returned, unfortunately, to being a basket case and a narco state, it was actually looking great in 2008.
- 6.52** And so what happened was, as interest rates in the US were lowered, if you were a fairly good, institutionally progressing emerging market country, people really wanted to send money to you. So Colombia was progressing. That meant that it was attracting, as interest rates were very low in the US, let's say circa 2005, 2006, it was experiencing huge capital inflows. And so, to give you a sense of how large those are, financial system loans, so what happens when capital flows in? It makes the banking system, the money is deposited effectively in the banking system, makes the banking system grow.
- 6.53** Well, when the banking system's funding grows, it starts lending money out. So the banking system went from lending 10% growth, as of December 2005, to triple that by December 2006. When a banking system, we tend to know, when a banking system accelerates its growth rate from 10% to 30%, that means that credit standards are not being applied the same way. That is an unsustainable and very questionable rate of

growth. But if you're flush with cash, often that drives the bus. And your lending officers are empowered to make a lot more loans.

- 6.54** Because of the expansion that was happening domestically, because of this expansion in credit, you were getting also expansion of aggregate demand beyond what the supply of the economy could maintain, which means you got the consumer price index rising from 3.5% in April 2006 to 5% by April 2007. Real GDP growth was 8% in 2007. Because everybody wanted to lend Colombia money, as I'm saying, the current account deficit rose from 1.8% of GDP to 3.6% of GDP. So that's a very huge.
- 6.55** Remember, we're talking about sudden stop risk, right? So they're now seeing a big increase, and, of course, they're worried about sudden stops. So what happens here is really interesting. The monetary authority gets together with all of the other financial regulators. Initially, the monetary authority thinks they're going to solve the problem by increasing interest rates. Because you're used to thinking contractionary monetary policy is increasing interest rates, right?
- 6.56** But if the source of the increase in the money and credit supply is incoming foreign investment based on search for yield, what happens when you increase the interest rate? You get bigger capital inflows. That's what happened. They tried to solve the problem with an increased interest rate, and all it did was compound the problem because they got increasing foreign capital inflows. So then they decided, we have to use something called macro-prudential regulation. We have to. So they did everything. They increased capital requirements. They increased provisioning requirements for loans, which is sort of a momentary increased capital requirement because of suspicions about loan quality.
- 6.57** They increased also reserve requirements, especially reserve requirements relating to the encaje. And so doing all this actually got them a soft landing. And they didn't have a recession in 2008. They didn't have problems coming from this big surge in capital inflows because they actually stopped the acceleration in credit before it became a systemic problem. So I think that this is a good example of how there's been a lot of learning in emerging markets. We'll come back to that theme at the very end.
- 6.58** Everything isn't bleak. Although, of course, in Colombia it is right now, because they decided to give up their good institutions for reasons I can't explain. One of the phenomena that has been worldwide, which has also engulfed the emerging markets, is the growth of real estate lending. And real estate lending, often with a political purpose of expanding access to housing credit. So if you're in Brazil, it's called Minha Casa, Minha Vida, right? Something like that. So you get the idea.
- 6.59** Housing finance is not just some sort of plain vanilla category. It has to do with essential and highly politically charged consumer preferences. And that's not just true in emerging markets. It's also true in developed economies. In fact, there's been a worldwide global boom in bank mortgage lending. This is actually just for developed

economies, because of the database that these authors had, but it's true in emerging markets, too. You can see from 1970, mortgages as a share of total bank loans rose from about 40% to about 60%. So that's a pretty big rise.

- 6.60** And it's definitely reflecting in part the political preferences that regulation gives to mortgage credit. And those political preferences are, of course, not coincidentally driven by voters' preferences. So this is a problem of democracies, not just a problem of autocracies. What's worse, during the same period, I would say the other big change is that the whole world became enamored of very generous bank safety nets. And in a paper that I wrote, was published a few years ago with Sophia Chen, what we found is that there's a connection between these. That is, when deposit insurance becomes more generous, covering more of the deposit base of a country, we find that banks tend to increase their real estate lending.
- 6.61** And there's every reason to believe, from a political economy standpoint, that that's intentional, that part of the role of expanding deposit insurance coverage from banks is to make them more willing to do real estate lending. Deposit insurance reduces liquidity risk because depositors don't have as much of an incentive to worry about bank solvency issues and liquidity issues. And so, depositors are much more willing to be passive, which allows banks to invest in more illiquid kinds of instruments like mortgages.
- 6.62** So I could go through a lot of evidence about this, but I just want to kind of quickly show you that, just like the graph that I showed you about mortgage lending, deposit insurance is sort of one of the major phenomena of the last 30 years. So if you look at this graph, what it says is, as of 1960, there's only one country in the world that has deposit insurance. I think that the US is alone then. Canada follows in the early '60s, I think. And then we had a few countries by 1970, only about 20 countries, a little less than 20 by 1980 have deposit insurance. But by the time we get to 2000, it's almost 100.
- 6.63** So this is a really sudden phenomenon. How did it happen? Well, it was a combination of things. But one of the things that we see econometrically is encouragement for creating deposit insurance by the International Monetary Fund, the World Bank, and the EU on their members was a big part of the story. And for some reason, all those multilateral agencies decided in the 1990s to either require or to strongly encourage the enacting of deposit insurance.
- 6.64** My own view is that, when it comes to the IMF and the World Bank, this was the result of myopia. They knew as a matter of research that instituting more generous deposit insurance would, or in many countries, just creating deposit insurance, would have long-term negative consequences. And I'm going to show you that evidence. But the IMF was struggling with some very big country crises and it lacked the resources to actually bail out all those countries. In fact, remember, in 1995, the US treasury had to come up and sort of partner with the IMF with an additional \$50 billion for Mexico.

- 6.65 And so I think that the IMF did this because the reduction in liquidity risk short-term kind of put some oil on the, to calm the seas a little bit. But unfortunately that oil also can catch fire later. And it did. But so it's hard to know exactly what the motivations were. But what we know econometrically is, the IMF, the World Bank, and the EU all decided to decide all of a sudden in the 1990s that they wanted to promote deposit insurance. And that had a big positive effect. We're going to use it as instruments because we want to make sure that we're looking at exogenous and external pressures on countries. And so econometrically we're going to use that.
- 6.66 And what we find is that, in our analysis—and I'm proud to say this paper won the best paper award in the Journal of Financial Intermediation—what we show is that, when the generosity of deposit insurance went up, banks increased their risk-taking in three ways. They held less cash and made more loans, that is. That's way number one. They increased their leverage, held less capital relative to their assets, and they increased substantially their proportion of mortgage, real estate kinds of loans. All three of those have been implicated in making systemic risk bigger.
- 6.67 And we also find, although here, depending on the econometric methodology, the results are either significant or not significant. But prior research has also found this. We also find that this substantially increased the probability of banking crises. But of course, it has to, because if all your levers of risk, those are the only levers we can think of, if all the levers of risk are going up, then it's not surprising that you're going to get more banking crises.
- 6.68 So this brings us back, of course, to the political economy lesson that we've had in this class, which is, we've known for a long time that banking collapses are more likely as a result of increasing deposit insurance. In fact, the literature has been uniform in this. There are scores of articles now published in journals, all reaching the same conclusion. And yet, when we look over time, we see that deposit insurance has actually increased. So if you're an economist and you think that efficiency and wise judgment should be where you look to understand policy, you have a hard time understanding why everyone is saying something's a bad idea and it's happening all over the world.
- 6.69 And so something about the political economy of both the behavior of the multilateral institutions and the national governments in expanding the safety net has to be the answer. Having discussed the problem of excessively generous deposit insurance in incentivizing banks to take on big risks leading to major banking crises that are hugely disruptive, both in terms of causing major recessions and also having big fiscal costs for governments, the question arises, well, what would be the right way to approach prudential regulation in emerging markets? And we have lots of examples.
- 6.70 Interestingly, one of the first country disasters of a banking crisis, which was Chile, 1982-83, then was the leader in terms of reform. And Chile came up with new approaches to prudential regulation starting in the mid-1980s, which actually led the way for the developed world, as well as the developing countries. And many developing

countries that are interested in reform hire the Chilean experts to come in and advise them about how to do it. Very limited deposit insurance protection, very ambitious and honest accounting of banks. And interestingly, Chile has not had a banking crisis since 1982-83.

- 6.71** What is the formula that we should be thinking about? And I also want to talk about how, believe it or not, Argentina managed to develop a very successful bank regulatory regime in the 1990s, which I was involved with the central bank in designing. So the key to successful risk management in banks is that you actually manage your default risk. Your default risk is the result of three things. If leverage is higher, that increases default risk. If your cash holdings relative to your risky assets are lower, that makes default risk higher. And if the types of loans you're making are riskier, that makes your default risk higher.
- 6.72** That's pretty much all you need to know. OK, we could get more complicated, but that's really it. More leverage, more lending, and riskier lending. That's really the three things you have to control. And so how do you control them? How do you know when enough is enough? And part of the answer is, develop a regulatory system that learns from market information. Markets are actually providing us information all the time. If the deposits are not protected, one of the things that the market is giving us information about is the riskiness of the deposits.
- 6.73** Because if the deposits aren't protected, they have interest rates, and those interest rates are reflecting risk. Believe it or not, deposit interest rates in unprotected banking systems can vary hugely in ways that are very much related to risk. We also have similar evidence from a lot of other sources historically. And so we're going to learn from looking at those deposit interest rates, but we're also going to have a system that learns in other ways. How about loan interest rates? So in some countries, if you see a higher interest rate on loans, guess what?
- 6.74** You assume those loans are riskier, and you require the capital that the bank maintains, sort of the degree that it's going to leverage its funding of that loan, to vary with the interest rate that's charged on the loan. So you can use both interest rates on the liabilities of the bank and interest rates on the assets of the bank, as a supervisor, if you want to, to get market information. Of course, you can only do that if you don't have everything on the deposit side perfectly protected. You can't use the deposit interest rates very well then. But you can still use the loan interest rates.
- 6.75** You can also require, as the Argentines did, you can require the banks to have to target particularly savvy investors for some fraction of their funding. That's sometimes called a subordinated debt requirement. There are different versions of this subordinated debt requirement, but what it does is, it says, you're going to go out and find some institutional investors to hold your debt, longer-term debt, and we're going to see what the yields are on that debt. And we're going to require that you have a certain proportion of your funding in that debt.

- 6.76** The Argentines did that in the 1990s, and in fact, that was a suggestion that I gave them. What's some of the evidence that markets do, market pricing is useful? Well, one piece of evidence would be, in the Argentine banking crisis that happened in 1995, some banks were allowed to fail. The banks that were allowed to fail were not bailed out. The depositors were not bailed out. If you look at the interest rate that those banks were paying, those uninsured banks were paying in 1993, compared to the interest rate that the banks that didn't fail in 1995 were paying in 1993, guess what?
- 6.77** The interest rate was 3.5 percentage points higher on the deposits of the banks that failed in 1995. Remember, two years prior to the failure, the interest rates were three and a half percentage points higher. And that evidence was actually quite convincing. To the central bank, it looks like the markets were kind of aware that these banks were riskier. And then they also noticed that loan interest rates, and by the way, we've seen this for the US as well and for many other countries, that the best way to gauge the riskiness of loans, we don't use it.
- 6.78** Despite the fact that many researchers, including researchers at the New York Fed, myself, and many others have recommended it, much research has found it, use the interest rate, the all-in interest rate cost on a loan, as a gauge of its risk. Makes sense, right? It's what the bank's charging. Regulators in the West don't do that. The Argentines did it in terms of the capital charge. So there were some really great examples of this, of the use of market information and trying to get markets involved in monitoring banks more and investing in information about banks more through a subordinated debt requirement.
- 6.79** By the way, in Mexico, after their crisis of 1995, there was a lot of interest in the banks. They were insolvent, they had to be bailed out, their assets had to be resolved. And because there was a change in government, the PRI ceased to be in control of the lower house of the Mexican parliament. All of a sudden it became a question whether deposit insurance— because remember my story, in the early 90s, 100% of liabilities were protected by the government. However, people started wondering, given how crooked everything seemed to have been going in the regulatory process and the lending process of those banks, in 1996, people started wondering whether maybe we weren't going to bail out all the liabilities of these banks. Maybe the government was going to renege on its deposit insurance promise.
- 6.80** And when that became a real political possibility, you started to see interest rates on deposits of Mexican banks, all de jure, 100% insured, varying dramatically. So, for example, Banamex, that was kind of marginally insolvent, was paying 17% interest rates on its funds, Whereas Bank Serfin, that was deeply insolvent, was paying 30%. Now, those were all insured funds. They shouldn't have differed. But it's because the insurance became a question that you saw that happening.
- 6.81** So there's a lot of really interesting information about how to use market information and what we know, that a few of the lessons that we learned, and there's a lot of World

Bank research that came out in the early 2000s that kind of confirmed all this. I'll read you a quote from the head of market research at the World Bank on emerging market banking system regulation.

- 6.82** "Across the different statistical approaches, we find that empowering direct official supervision of banks and strengthening book capital standards of banks do not boost bank development, improve bank efficiency, reduce corruption in lending, or lower banking system fragility. Indeed, the evidence suggests that fortifying official supervisory oversight," which we already discussed earlier, "and powers actually impedes the efficient operation of banks, increases corruption in lending. When the supervisors have more discretion, they use it to extract bribes and therefore hurts the effectiveness of capital allocation. In contrast to these findings, banks' supervisors ran regulatory policies that facilitate private sector monitoring of banks."
- 6.83** And they mean limiting deposit insurance coverage, requiring certain kinds of debt, as I mentioned, improve bank operations and reduce bank fragility. And they have a bunch of regression results that confirm that, specifically something they call the private monitoring index, which is really driven by the percentage of the deposits that aren't insured, is the most robust predictor of banking system stability of all of their measures.
- 6.84** In the Argentine case, it was really an interesting combination of many different reforms. But to me, the thing that was really distinctive about it was the focus on market information. And they developed something they called the BASIC system, B-A-S-I-C, because it included all of those different pieces each, each of those letters fits in as one piece of the acronym. And the system worked extremely well. And you could see in, some banks weren't able to comply fully with it. But those banks also had other signals of weakness. So failing to be able to comply with the system, failing to be able to raise the subordinated debt that was required, was actually a really good indication that the banks had big non-performing loan problems and other things.
- 6.85** So I think that we know how to implement effective regulation. And it starts with limiting the protection of the banks. In the Argentine data, I did a study that also showed that all of the key ingredients of their regulatory system came together to create deposit market discipline. And I showed that in two ways. First, looking at the growth rate of deposits, whenever the system got weaker based on fundamentals, we saw uninsured deposits leaving banks in a continuous way. Not bank runs, but just withdrawing. And that stabilizes the bank because, when deposits withdraw, it reduces the bank's leverage.
- 6.86** And so actually, there was this nice stabilizing system. When banks got riskier, there was an immediate market reaction, reallocating deposits from risky banks to less risky banks, which created a strong incentive for the banks to maintain their risk. And then we also did a similar analysis looking at the interest rate on deposits. And what we found there was really startling. We found that, when banks experienced increases in

their risk, their deposit interest rates went up. How quickly, though, did those deposit interest rates revert back to the mean?

- 6.87** Within a quarter, because the deposit outflows were creating strong incentives for banks to manage themselves and also were themselves helping to stabilize the banks by reducing leverage. And all of these results, I'm telling you, the reason that of course I thought of all these things and ran these kinds of regressions is that I had been finding them in historical unprotected banking systems in my research for decades.
- 6.88** When I was hired first by the Colombian central bank to do an analysis to see whether their prudential regulatory system—this was in, I think, 2009, something like that—to see whether their prudential regulatory system was, was behaving effectively, I said, "I can tell you how we can tell. We're going to, please run this regression. I want to see whether deposits leave the banking system when your measure of capital falls, your measure of—" I gave them these three measures, capital cash ratios and also riskiness of loans through looking at non-performing loan ratios. And I said, "Do those three variables, on the right-hand side of a regression, with deposit flow on the left-hand side, do they predict deposit outflows?"
- 6.89** And they said, "Professor, it's amazing. We've never run that regression before, but it works really well." And I even told them what I guessed the magnitude of the coefficients were, and they— I was right. Of course, they thought I was Nostradamus. Because here I had told them to run a regression on data I'd never used before and that the signs of the coefficients would be statistically significant in these directions of roughly these magnitudes. But the reason is because depositors all over the world, kind of unprotected depositors, kind of behave similarly. They monitor banks. Now, not all depositors monitor banks, just big institutional depositors. That's good enough.
- 6.90** And they are the ones that typically move when they see risk happening. Why? Because unprotected depositors are not investing in the stock market. Right? You put some of your wealth into the stock market, you don't worry if it goes up by 2% one day, down by 1% the next day. It's fine. But when people put some fraction of their wealth into a banking account, they don't want it to be risky. And when, if they're savvy investors and they're monitoring that, when they see the markers of risk going up, they put their bank deposit elsewhere. That's what we call deposit market discipline.
- 6.91** And what I found in decades of research is that it's quite reliable. Interestingly, I'm finding that the audience for that kind of thing is much greater in emerging markets than it is in developed economies for this kind of lesson. And I think part of it is because the emerging markets experience such big losses from the disruptions of these banking crises that they have stronger incentives to try to avoid them.
- 6.92** So a list of specific ideas that I think is sort of best practice in emerging markets. Free entry of banks. Allowing people to organize banks, not restricting it to your favored friends. Allowing foreigners in particular to open up new banks. Using market

information to alert supervisors. Baking the loan interest rate into a measure of risk that's used in capital standards for banks.

- 6.93** I also believe that there's a role for a kind of subordinated debt requirement in the form of what I call contingent capital that would allow us to require banks to have somebody out there in the market who's very savvy, who's holding some debt in the bank, that we know the yield on that debt is going to be very informative because those investors are good at measuring risk. And most importantly, limiting very sharply the amount of bank protection.
- 6.94** OK, moving on to our last topic, I want to end on a hopeful note. I want to show you that, in liberalizations of emerging markets, there's evidence that learning is going on, and that it shouldn't surprise us that when you change everything as part of these massive liberalizations that we were talking about, that you don't get everything right initially, and you have to learn. So the basic evidence of this comes from a paper written by Kaminsky and Schmukler in 2003. And what they did was, they looked at 28 liberalizations. Some of these included developed economies, but mainly emerging market economies.
- 6.95** These were liberalizations that had to do with the three ones that we've talked about before. Capital flows, foreign stock ownership, and domestic banking system deregulation. And their measure of systemic problems is collapses of equity markets. And that's a good measure. If you're looking for times when financial systems really implode, a nice measure of it is equity markets fall a lot. And so they looked at equity busts to try to identify crisis risk. What they did was, they said, after these periods of liberalization, do we see, let's say, three years after the liberalization, let's call that the short term, do we see that the risk of a major financial crisis, which we're proxying with this collapse of equity markets, do we see that that risk is higher or lower?
- 6.96** And what we see is that, in the fourth year after you have done your liberalization, if we measure it then, the risk of a big financial problem is actually higher. But then they look out a few more years and they see that that risk is actually lower. And the inference is, as they say, "short-term pain, long-term gain" is the title of their paper because they think that there's a learning process that these data are exhibiting. And that's an optimistic note.
- 6.97** Here's what the booms and busts look like. You can see that, in emerging markets, because they're riskier places, the booms in the stock markets tend to be steeper and the busts also tend to be steeper. But what's interesting is, as they show here, so you can see that the pre-liberalization sort of probability of a financial crisis is given by the white vertical distance. I'm sorry, that's the short-term post-liberalization probability of crisis. The pre-liberalization is the striped line. The short-term post-liberalization is the white box and then the far right sort of gray shaded one is the longer-term liberalization.

- 6.98** So what they're showing for emerging markets and for developed economies, both, they find that actually liberalization is associated with a declining risk of a financial crisis, but not immediately. So on that note, that hopeful note, that it is possible for learning to take place, although there's a huge amount of political resistance to institutional improvement. I'd like to end and maybe take some of your questions.
- 6.99** So are there any examples of situations in which there's a large influx of, say, poor decisions by investing groups or maybe angel investors coming into an emerging market, investing, and then that propelling the institutions to change for the better, sort of putting the cart before the horse, and then do those changes tend to last, or do those just exist during the duration of the investment?
- 6.100** Good question. So first of all, there were some studies after the Asian financial crisis to try to see whether the precipitating events of the withdrawal of funds were originating in institutional investors coming from developed economies. And the answer basically was not. And that was also true in Mexico, that the beginning of the Mexican collapse in the end of 1994 was actually a run on the peso and on the tesobonos, which were a short-term instrument that were actually originating within Mexico.
- 6.101** So what we know is that, in terms of judgment about when crises are risky, we know not to blame foreign investors for precipitating the crisis. What about their investment during boom periods? So there are some indications of excessive optimism by investment groups, and unfortunately for them, they're the ones who paid the cost. And the most obvious example of that was Citibank's investments in Mexican banks, which did not turn out very well for Citibank, but was great for the Mexican taxpayers because what that did was it meant that these banks, foreigners, were so interested in acquiring them, anticipating major legal reforms that would make bank lending more profitable.
- 6.102** And that meant that they were willing to bear some of the fiscal burden by buying banks that were insolvent and for a premium, therefore saving the taxpayers some money. Unfortunately, the legal reforms didn't happen fast enough or thoroughly enough to really make those investments worthwhile. So that there are lots of cases of, where I could think of, and I point to, of foreigners being a little too optimistic during boom periods in emerging markets. But they tend to be, I would say, the ones that are bearing the costs of that.
- 6.103** Now, where, on the hot money side, that's a different story. For example, I was telling the story of search for yield, especially with short-term investments in Colombia, which wouldn't have been such a positive thing if the government hadn't put an end to it. What we're seeing right now, though, we don't know. So the protracted period from 2008 to 2020 and the huge amounts of bond investments, as I mentioned before, could turn out to be a case where the losses to the bond investors will be even, also losses in terms of defaults, recessions, and all sorts of problems for the issuers, both sovereigns and corporates.

- 6.104** So there are, when you're taking big risks on a system-wide basis, both the borrowers and the lenders are subjecting themselves to some major systemic potential loss. So we have to wait and see. And I guess you could say that the recycling of petrodollars into the Latin American economies by the banks in the 1970s didn't end so well for them. By the way, again, Citibank was a big loser in Mexico. Chase got most of its money back, got all of its principal back. I think Mexico got about half of its money back. So there have been some pretty large losses.
- 6.105** And you know, your question also was kind of getting at, and how does that affect the recipient country? And of course it's a, it's a complicated question because how does it affect Mexico, if the Mexican government's spendthrift behavior in the late '70s and early '80s, and the power of the sort of autocrat alliance with the six families, is magnified by access to credit? So I'm not sure that, you know, access to credit and the role of the banks in the '70s, especially given how badly it ended, at a big cost to the economy in terms of recession and all the rest of it. So tying your horse to these international debt market little engines can be a highly risky thing, not just for the lenders, but also for the borrowers.
- 6.106** In this last lecture, you spoke in depth about Argentina in the 1990s and early 2000s. In my limited knowledge and perspective, post-2020, Argentina has drastically shifted their economic policy. Have these shifts in economic policy made Argentina a unique economic experiment? And if so, what do we stand to learn?
- 6.107** I would say that it's been a lot of ups and downs in the past four years. We have a new president in Argentina, but he doesn't have, because of the Argentine political system and the way it's structured, he doesn't have support. He doesn't have a party. He doesn't have support in the legislature. The Senate is different from the lower house and has this kind of rigged system of keeping old dinosaurs in power.
- 6.108** So it's kind of hard to—I wouldn't characterize Argentina as having committed, really, as a country, to a new path of reform. And whatever commitment that is, it would have just occurred in the last year. Now, of course, there were other things going on in the prior years, ups and downs, I would say. I would say the last time we really had a consensus in Argentina in favor of a reform agenda of emergence was from starting in 1991 and going through until 2002. And when the Argentine crisis hits, that consensus evaporates, and it's really never been rebuilt.
- 6.109** So Argentina is a bit of a unique case in the sense that it's one of the most, sort of depressing, long-lived rejections of emergence, where we had a consensus for it. And part of the reason why that consensus disappeared in the 1990s was kind of a very rough period for Argentina. I don't want to exonerate them. There were lots of domestic political things that they didn't do right as part of their liberalization. They never succeeded in really fully implementing free trade reform. There were labor reforms that were never fully implemented. There was fiscal reform that was never implemented.

- 6.110** When Menem was elected, all of these things were prospects. There was no sort of real program of how they were going to be implemented. And then, while they were all hoping to implement them, they had constructed this kind of sword of Damocles in the form of this convertibility law that meant that, if everything didn't go just as promised, they were going to have a real destruction of their dollarized economy, which did happen.
- 6.111** Part of the reason why things were hard to do is, you had a crisis in Mexico in 94-95, a crisis in East Asia in 1997, a crisis in Russia in 1998, a Brazilian devaluation in January 1999. And if you're Argentina, you're living through all of that, and all of those are affecting your access to capital. Let me explain that. It'll just take 30 seconds, but it's really important. Funding of emerging markets is being accomplished largely by specialist emerging market funds. If they have a loss in one country, for example, Russia, in 1998, they will actually reduce the amount of investment they're doing in other emerging market countries.
- 6.112** The reason for that, the reason that that happens systematically in the world is, and this is a very unlikely explanation, but I'm telling you, I believe it very much. It's a matter of human capital. So the number of people who really are experts in structuring funds that specialize in emerging markets is fairly limited, and they're fully invested, that is, they have structured, they've put their own wealth or a substantial chunk of it into organizing these funds. When these funds have losses, they're not easily recapitalized suddenly.
- 6.113** So they can't just—so what do you sell? If you're experiencing, let's say, a default in Venezuela, you can't sell the Venezuelan debt so easily right away. So what you sell is all the good ones because you have to reduce your risk. Because there's the same kind of risk discipline on the hedge funds that are investing in emerging markets, as I described in the deposit market, there's a similar kind of risk. Those hedge funds might get some repo funding or short-term debt funding that's subject to that same kind of risk intolerance.
- 6.114** So what that means is, when they have losses, what you'll see is, when Russia has its crisis, Brazilian sovereign debt spreads go way up. And Brazil ends up having, by the way, a devaluation not long thereafter. Why are there linkages across these emerging markets that don't have to do with competition for import markets, in their exports, that is. They're not competing to have their exports imported by particular countries. That's not the whole story. It's not just contagion in some irrational sense.
- 6.115** It's that the capital base that can provide funding to emerging markets is fairly specialized. And if they experience losses coming from one country, there's an immediate transmission of reduced tolerance for taking risk in the other countries. So believe it or not, the scarcity of global human capital that's knowledgeable about managing these risks, which I hope we've, we've illustrated in this class, are fairly

complicated, is actually, I think, part of the story for why you don't immediately recapitalize these funds when they suffer a loss.

- 6.116** So, Argentina was dealing with the Mexican crisis of 1994-95, the Asian crisis in '97, the 1998 Russian crisis, and the 1999 Brazilian devaluation, which did have an effect because they're both competing commodity exporter countries. So that was a lot of shocks. I'm not trying to say that they did all their domestic policy right. They didn't. But it was a particularly unlucky experience and it really did sour the population to the idea of hooking your little trailer onto the engine of global progress, which is too bad.
- 6.117** Thank you, Professor. I want to reference the case of Brazil in 1994. Fernando Henrique Cardoso spearheaded currency change. He created a nuclear currency that at the time was tied to the dollar and 1:1 ratio. That was very controversial at the time. It's positive and negative effects. Bringing that kind of policy in light to what we just discussed in sovereign debts and possible defaults, and how exchange rates can actually really make the whole deal much worse, would that kind of policy and currency changes be valid?
- 6.118** It's a really interesting question and it's very helpful in answering that question to contrast the exchange rate policies of, and more generally, the financial systems of Argentina and Brazil in the 1990s. So Argentina adopts in 1991 a convertibility law, and beyond that, it also allows its banking system to issue dollar-denominated deposits. And by the time of 2002, all the deposits in the banking system pretty much are dollar-denominated. So they're all in with a dollarized financial system and with a very strict convertibility one-for-one exchange rate, which they call a currency board.
- 6.119** Some people might question whether it was true currency board, but basically it was a one-for-one hard convertibility constraint that couldn't be relaxed until it collapsed. Brazil is a different country. So one of the places in the world that is developing index financial contracts earliest is Brazil in the 1970s. And it's also developing all sorts of instruments to insulate domestic financial contracts from inflation risk, overnight repos. So in fact, in the 1970s, Brazil and the US are the two countries doing most of the sort of overnight repo business.
- 6.120** So the point I'm making is, Brazil did not dollarize. In fact, Brazil never dollarized and they kept their financial system and the risk that comes from— because if you dollarize, you have a depreciation. Everybody who owes money now owes a lot more money and then everyone's insolvent. That's the problem in a country like Argentina, of course, then they have to deal with that by relaxing the law and not having everybody have to repay and it's a catastrophe, right?
- 6.121** Brazil, even though they did the real plan, as you correctly said, with a one-for-one dollar-to-real exchange rate, that plan was not a hard and fast rule. And in fact they didn't keep it. Now I would say that it was a success partly because it was flexible. So, OK, when Cardoso promised he'd keep one for one, but even though he didn't succeed, and Brazil didn't succeed in keeping it one for one, they had this, as I mentioned, they

had little devaluations all along the way, moving from one for one. And then in 1999 they had a bigger one, about 20% devaluation.

- 6.122** But that, if you look at the inflation experience, sort of, even though the letter of the law of that commitment wasn't maintained, the spirit of the law was, because the whole point was to use the real commitment as a credible commitment to limit fiscal spending, so that inflation would change. The inflation process would fall from hyperinflation to inflation in single digits. And that's been a success. So in fact it was the fact that Brazil was so different from Argentina. So I would say that what Brazil did was, they created a credible commitment. I'll explain why in just a second.
- 6.123** They created a credible commitment out of seemingly nothing, but also a flexible one. And it was really, I would say, more the leadership of the transition toward believing that fiscal discipline was an important ingredient of the economy. And in fact it was maintained, not perfectly, but compared to the previous 200 years, it was really big progress. But not tying himself to the mast of a convertibility law and having a dollarized financial system, instead, the financial system remained real-denominated and, in fact, the banking system had real limits on dollar risk exposure.
- 6.124** So that when the devaluation happened, reporters called me on the phone the day of the devaluation in Brazil and they said, "Are we going to have a banking crisis in Brazil?" Because all of the devaluations they had seen in 1995 in Mexico, 1997 in East Asia and in Russia had produced bank failures. And I said, "No, you're not going to see a bank failure." And the reason is because there is no connection between the banking system and the exchange rate, because it's not a dollarized banking system.
- 6.125** So what was the commitment of the real plan? There was no—when he announced the real plan, he didn't announce anything very credible about his plans for fiscal policy, although he did some things, ultimately. What he did was, he put, I think it was about, maybe it was \$30 billion, I can't remember, of actual dollar reserves on the table. And that was the nature of the commitment. "I'm keeping the exchange rate at one-for-one. And if you don't believe it, you can come and ask for your reals back. Here's, I think it was 30 billion sitting on the table." That was about the only real concrete commitment that you could point to.
- 6.126** It was something about his leadership and about the way he was expressing these two pillars of inclusion and sustainability that somehow were convincing. It's a little hard to put your finger on what makes leadership, right, but there was something about that moment that made it convincing and, I would say, successful. Even though ultimately, what's the current exchange rate between the real and the dollar? It's one to five, right?
- 6.127** So you could say terrible failure, but I would say no, it was exactly that flexibility that was necessary to have a flexible exchange rate system that would deliver single-digit inflation, not zero inflation, but at least single-digit. So I think this is another lesson. We

haven't in this discussion, because of our focus on financial institutions, rather than macroeconomic policies like exchange rate policies. That would be another course. I'm thinking of offering that course in this same suite at the Peterson Academy because it's completely complementary, but it's a different set of topics, right.

6.128 But I'm really appreciating the question because I think that's the lesson I draw from it. At the time I was wrong. In the 1990s, I was actually believing that the real plan wasn't going to be a success and that the convertibility would because it was a stronger commitment. Right. But that turned out to be wrong. Young professors.